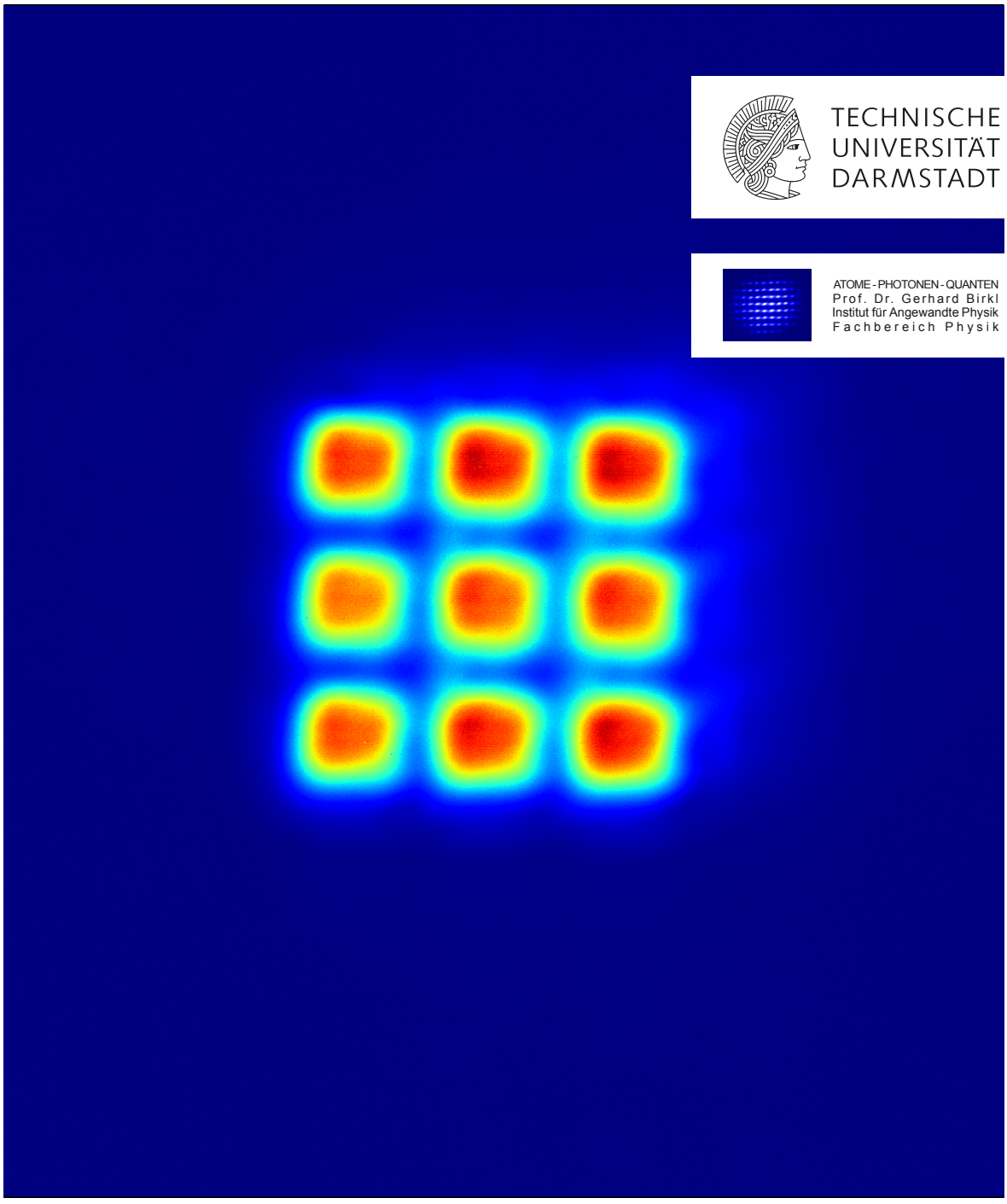
Shaping of uniform intensity profiles for single-site addressing of ^{85}Rb atoms

Strahlformung von homogenen Intensitätsprofilen zur Einzelplatzadressierung von ^{85}Rb-Atomen

Master thesis by Daniel Hagn

03.11.2026



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Cover image: A 3×3 grid of multi-tone Top Hat profiles generated using two orthogonally oriented Acousto-Optic Deflectors (AODs). Each profile is formed by the superposition of 3×4 individual Airy profiles, each corresponding to a beam diffracted by the AODs.

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List of Abbreviations

AOD Acousto-Optic Deflector

AOM Acousto-Optic Modulator

ARC Alkali-Rydberg-Calculator

CNOT Controlled-NOT

DM Dichroic Mirror

DOE Diffractive Optical Element

EOM Electro-Optic Modulator

FBS Focal Beam Shaper

LCM Least Common Multiple

PBS Polarizing Beam Splitter

PV Peak-to-Valley

QUIPS Quantum Information Processing System

RF Radio Frequency

RMS Root Mean Square

1 Introduction

The technological advances of recent decades have led to a substantial increase in classical computing power. Nevertheless, certain computational problems, such as those of a combinatorial or logistical nature, remain intractable for classical computers within a reasonable amount of time. Quantum computers have the potential to solve specific classes of these problems more efficiently than classical computers. A prominent example is Shor's algorithm for the factorization of large numbers, which illustrates the computational potential of quantum computers [1].

One approach to realizing a quantum processor involves neutral atoms that are trapped in registers of optical dipole traps and can be coherently manipulated using laser beams. The Atoms-Photons-Quanta group led by Prof. Birkl is pursuing this approach with the Quantum Information Processing System (QUIPS) experiment, which uses ^{85}Rb [2]. Using spatial light modulators and Acousto-Optic Deflectors (AODs), large two-dimensional atomic arrays containing up to a thousand atoms can be generated [3]. The Talbot effect enables the realization of three-dimensional structures, providing a pathway toward further scaling [4]. In this system, the hyperfine ground states $|F{=}2\rangle$ and $|F{=}3\rangle$ of $5\,^2S_{1/2}$ in ^{85}Rb are used to encode the qubit states $|0\rangle$ and $|1\rangle$, respectively.

Transitions between these states form the basis for the logical gate operations performed in this system. Of particular relevance to this thesis is the Raman laser system used to drive these transitions, which was developed and characterized in previous work [5] and subsequently integrated into the experimental setup [6]. The system is designed to provide the required light for two applications. In the first, the entire atomic plane is illuminated at once. In the current implementation, sufficiently uniform illumination is achieved for a 32×32 grid of ^{85}Rb atoms [7]. This setup is currently being commissioned, and first tests are being conducted in preparation for experimental use. The second application is single-site addressing of individual atoms, enabling optical transitions and, consequently, gate operations to be performed locally. A displacement of the atom relative to the focal point of the addressing beam, for example due to beam-pointing errors of the dipole trap or the addressing laser, can negatively affect the optical excitation and reduce the fidelity of the transition between the two qubit states. To counter this issue, previously constructed optical systems have successfully demonstrated the use of uniform intensity profiles generated with an optical Beam Shaper for addressing the entire atomic plane [7, 8]. Building on these results, several approaches for generating locally uniform beam profiles are investigated and simulated in this work with the aim of improving single-site addressing for use in a new Raman laser setup.

The first approach uses an optical Beam Shaper [9]. Since the uniform illumination of the entire atomic grid has already shown promising results [7, 8], it is investigated whether a similar concept can be applied locally to achieve uniform intensity profiles close to the diffraction limit. Another possibility is the superposition of multiple individually deflected Gaussian beams to form a uniform beam profile. This method is based on a multi-tone AOD setup, which was first implemented for the movable optical tweezer setup [10]. For this new approach, the influence of stationary interference and beating is simulated in this work prior to experimental implementation.

2 Theoretical Background

This chapter introduces the theoretical background required for the generation of uniform intensity profiles. It begins with the fundamentals of quantum computation as the underlying motivation, followed by an introduction to the relevant atomic physics and the optical components used in the experimental setup. Finally, the principles of optical beam shaping and the interference effects arising from the superposition of multiple optical fields are discussed.

2.1 Basics of Quantum Computation

To perform information processing based on quantum states, a system has to fulfill a certain kind of criteria. In 2000, David P. DiVincenzo defined five such criteria that a physical system must fulfill in order to a realize quantum computer [11].
The fundamental unit of information in a classical computer is the bit, which can either take the value zero or one. Its counterpart in a quantum computer is the so-called quantum bit, or qubit. Qubits are physical two-level systems with the possible states $|0\rangle$ and $|1\rangle$, which exist in a superposition $|\Psi\rangle = a\,|0\rangle + b\,|1\rangle$ of both states and are subject to the normalization condition $|a|^2 + |b|^2 = 1$. While the states of a single-qubit form a two-dimensional complex vector, N qubits result in a 2^N-dimensional vector. A physical system consisting of such qubits must be adequately characterized, which means that the Hamiltonian operator $\hat{H}$ and all energy levels must be sufficiently well characterized with respect to possible couplings between internal states, other qubits and external fields. If such couplings are unknown, errors may occur that reduce the fidelity of information processing. To achieve sufficient computational power, the physical system must be scalable, making scalability a central focus of research.
As in classical computing, a quantum computation requires a well-defined initial state. Therefore, all qubits must be initialized in a known reference state, which can be achieved, for example, through laser cooling and optical pumping.
Decoherence characterizes the dynamics of qubits under environmental influences. It is the loss of phase information required for quantum interference and, consequently, can reduce the computational advantage of a quantum computer over a classical computer when solving certain problems. If the gate operation time is significantly shorter than the decoherence time, the quantum coherence can be maintained throughout the computation, allowing the computational advantages of quantum information processing to be preserved.
Gate operations form the central building block of quantum computation and consist of sequences of unitary transformations $\hat{U}_i$. A distinction is made between single-qubit gate operations, which are performed on only one qubit and multi-qubit gate operations, which act on multiple qubits simultaneously. A quantum computer requires a universal set of quantum gates, consisting of a complete set of single-qubit gate operations and one two-qubit gate operation, such as the Controlled-NOT (CNOT) gate [12]. A CNOT gate flips the state of the target qubit depending on the state of the control qubit according to:

$$|00\rangle \rightarrow |00\rangle\,, \qquad |01\rangle \rightarrow |01\rangle\,,$$
$$|10\rangle \rightarrow |11\rangle\,, \qquad |11\rangle \rightarrow |10\rangle\,.$$

Gate operations are only supposed to act on qubits for a specific amount of time t, which can be achieved using focused laser pulses of appropriate duration.
At the end of a calculation the result has to be read out. Assuming that the density matrix of a qubit has the form $\rho = p\,|0\rangle\langle 0| + (1-p)\,|1\rangle\langle 1| + \alpha\,|0\rangle\langle 1| + \alpha^*\,|1\rangle\langle 0|$, the measurement should yield $|0\rangle$ with probability p and $|1\rangle$ with probability $1-p$. This measurement destroys the superposition of the two states of the measured qubit and projects it onto one of the two eigenstates. A method for measuring the qubit state is fluorescence imaging. By applying additional laser light, a transition is driven for one of the two qubit states, causing it to emit photons through spontaneous emission, while the other state remains dark.
In addition to these five criteria, two further criteria have been defined that address the ability to exchange quantum information between different computing units, thereby enabling the realization of quantum networks [11]. Atom transport within an atomic grid can be achieved using movable optical tweezers, as demonstrated, for example, in [10]. Transporting these stationary qubits, used for the actual computation, over long distances while preserving their coherence is challenging. The ability to convert stationary qubits into “flying” qubits and vice versa would enable the transmission of quantum information over such distances. A possible realization of a flying qubit is a photon, which can encode quantum information in properties such as its phase or polarization.

2.2 Manipulation of Atomic Qubits

One physical system suitable for the implementation of a quantum processor consists of neutral atoms trapped in optical dipole traps and manipulated by laser light. Atoms are identical physical systems and can therefore be cooled and addressed collectively. Coherence times of neutral atoms in the range of seconds are possible and result from the long life times of stable atomic energy levels [13].

2.2.1 Two-Photon Process

For ^{85}Rb, the basis of every logical operation is a two-photon process. The coupling of the two qubit states $|0\rangle := |g\rangle$ and $|1\rangle := |e\rangle$ is achieved in a two-photon process involving a third intermediate state $|i\rangle$ in a λ-scheme as shown in Figure 1. Here, each level $|k\rangle$, with its energy $\hbar\omega_k$, interacts with the incoming light described by a classical electric field of two frequency components $\omega_{1,2}$, which are stable in their relative phases $\phi_{1,2}$:

$$\mathbf{E} = \mathbf{E_1}\cos(\omega_1 t + \phi_1) + \mathbf{E_2}\cos(\omega_2 t + \phi_2).$$

The light matter interaction is described with the dipole operator $\hat{\mathbf{d}}$ resulting in the following Hamiltonian for the three level system:

$$\hat{H} = \hbar\omega_e |e\rangle\langle e| + \hbar\omega_i |i\rangle\langle i| + \hbar\omega_g |g\rangle\langle g| - \hat{\mathbf{d}}\cdot\mathbf{E}. \tag{2.1}$$

The frequency difference between the two qubit states $\omega_{eg} = \omega_1 - \omega_2 - \delta$ for ^{85}Rb is approximately 3.035 GHz. A laser setup for generating this difference with an Electro-Optic Modulator (EOM) was built [5] and further developed [6] and can now be used to address 32×32 atoms in the atomic plane at once [7]. The Rabi frequency for each level is given by:

$$\Omega_g = \frac{\langle g|\,\mathbf{d}\cdot\mathbf{E_1}\,|i\rangle}{\hbar}, \tag{2.2a}$$

$$\Omega_e = \frac{\langle e|\,\mathbf{d}\cdot\mathbf{E_2}\,|i\rangle}{\hbar}. \tag{2.2b}$$

If the detuning Δ relative to the intermediate state $|i\rangle$ is much larger than the individual frequencies ω_e and ω_g, then no population gathers in the intermediate state $|i\rangle$ and it can be adiabatically eliminated [12]. The system can therefore be approximated as a two-level system described by the new Hamiltonian:

$$\hat{H} = -\hbar\begin{pmatrix} \frac{|\Omega_g|^2}{4\Delta} & \frac{\Omega_{\text{eff}}^*}{2} \\ \frac{\Omega_{\text{eff}}}{2} & \frac{|\Omega_e|^2}{4\Delta} \end{pmatrix}.$$

The effective Rabi frequency for this two-photon process is then given by:

$$\Omega_{\text{eff}} = \frac{\Omega_e^*\Omega_g}{2\Delta}, \tag{2.3}$$

and is therefore dependent on the detuning of the laser light and the Rabi frequencies of the single-photon processes. The single-photon Rabi frequencies are themselves proportional to the electric field and thus proportional to the square root of the laser intensity.

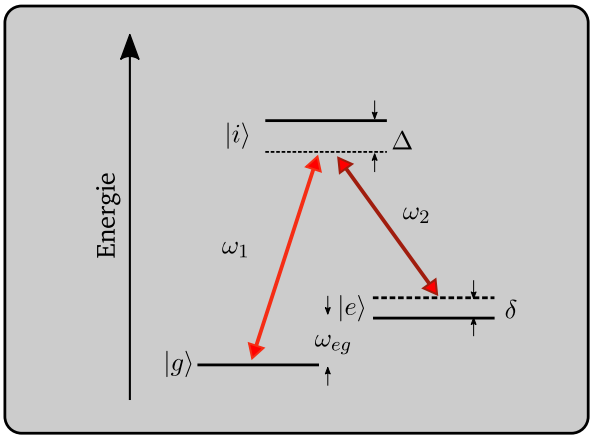


Fig. 1: Schematic of a two-photon process in a λ scheme used for the coupling of the qubit states [5].

2.2.2 Bloch Sphere and Single-Qubit Gate Operations

To realize a Quantum Computer, a universal set of quantum gate operations is needed. Such a universal set consists of a complete set of single-qubit gate operations and one entangling two-qubit gate, such as a CNOT [12] gate. Important single-qubit gate operations are the following:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{2.4}$$

To visualize single-qubit gate operations in a two-level quantum system, the model of the Bloch sphere, including the Bloch vector, can be used. A quantum state $|\Psi\rangle$ is represented as a Bloch vector by the spherical coordinates r, Θ, and φ. The length of the Bloch vector r is conserved in undamped systems and, due to the normalization of quantum states, satisfies $|\Psi|^2 = r^2 = 1$. The polar angle Θ specifies the relative proportions of the ground states $|0\rangle$ and $|1\rangle$ and depends directly on the duration of the incident laser radiation, which causes the population of the ground states to change. For $\Theta = 0$, the state is $|\Psi\rangle = |0\rangle$, and for $\Theta = \pi$, it is $|\Psi\rangle = |1\rangle$. The azimuth angle φ specifies the axis of rotation of the Bloch vector as the population in the two ground states changes. φ can be adjusted via the phase of the two photons involved in the two-photon process [12]. A visualization of the Bloch sphere is shown in Figure 2. The general form of a Bloch vector is:

$$|\Psi\rangle = \cos(\Theta/2)\,|0\rangle + e^{i\varphi} \sin(\Theta/2)\,|1\rangle\,. \tag{2.5}$$

Single-qubit gate operations can generally be represented as rotations of the Bloch vector. In a given basis, $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the gate operations can be written as:

$$\hat{R}(\Theta, \varphi) = \begin{pmatrix} \cos(\Theta/2) & -ie^{-i\varphi} \sin(\Theta/2) \\ -ie^{i\varphi} \sin(\Theta/2) & \cos(\Theta/2) \end{pmatrix}. \tag{2.6}$$

For $\Theta = \pi$, the initial qubit state is flipped and gathers additional phase depending on the phase φ of the laser:

$$\hat{R}(\pi, \varphi)\,|0\rangle = -ie^{-i\varphi}\,|1\rangle \quad \hat{R}(\pi, \varphi)\,|1\rangle = -ie^{i\varphi}\,|0\rangle\,.$$

For $\varphi = 0$ this is effectively the realization of an X gate with an additional phase factor: $\hat{R}(\pi, 0) = -iX$, while for $\varphi = \pi/2$ the Y gate is created: $\hat{R}(\pi, \pi/2) = -iY$. A Z gate can be realized by a combination of different laser pulses:

$$\hat{R}_z(\pi) = -iZ = \hat{R}(\pi/2, 0) \cdot \hat{R}(\pi, \pi/2) \cdot \hat{R}(-\pi/2, 0). \tag{2.7}$$

A 2π-pulse with $\Theta = 2\pi$ returns the Bloch vector to its initial position but introduces a global phase in the quantum state with a shifted sign:

$$\hat{R}(2\pi, \varphi)\,|0\rangle = -\,|0\rangle \quad \hat{R}(2\pi, \varphi)\,|1\rangle = -\,|1\rangle\,.$$

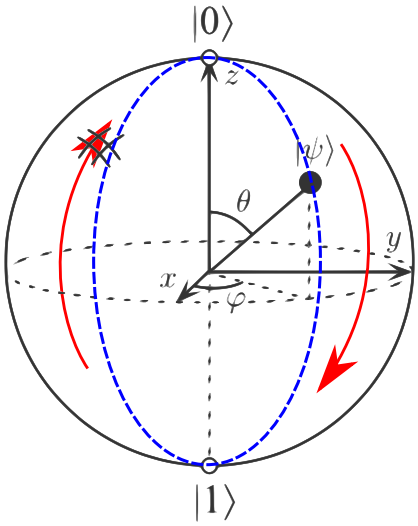


Fig. 2: Schematic representation of the Bloch sphere. Single-qubit gate operations result in a rotation (red) along paths (blue) on the surface of the sphere and allow to create superpositions of both states $|0\rangle$ and $|1\rangle$ [14].

2.2.3 Atomic Properties and Thermal Motion of ^{85}Rb

The Quantum Information Processing System (QUIPS) experiment uses ^{85}Rb atoms trapped in optical dipole potentials for quantum information processing. Its energy level structure [2] is particularly well-suited for the implementation of a quantum processor, since the required transition wavelengths can be generated using conventional laser systems. The hyperfine ground state $|F{=}2, m_F{=}0\rangle$ of the ground state $5^2S_{1/2}$ is the qubit state $|0\rangle$, while the state $|1\rangle$ is represented by the state $|F{=}3, m_F{=}0\rangle$. A CNOT gate with ^{85}Rb can be realized with a two-photon process using the wavelengths 420 nm and 1012 nm through a process called Rydberg Blockade, as demonstrated in [15]. Single-qubit gate operations need a wavelength of 795 nm or 780 nm in a two-photon process depending on whether the D_1 or D_2 line is used for the transition. The D_1 line of ^{85}Rb is shown in Figure 3. According to the frequency splitting, the two photons participating in the logical operation must have a frequency difference of approximately 3.035 GHz.

To describe one atom inside an optical dipole trap, the model of a thermal harmonic oscillator [12] can be used. The wavefunction Ψ_n for a level n of the harmonic oscillator is given by:

$$\Psi_n(x) = \frac{1}{\sqrt{2^n n!}\pi^{\frac{1}{4}}\sqrt{a}} H_n\left(\frac{x}{a}\right) e^{\frac{-x^2}{2a^2}}. \tag{2.8}$$

$a = \sqrt{\hbar/m\omega}$ is the characteristic length of the harmonic oscillator and $H_n\left(\frac{x}{a}\right)$ are the Hermite Polynomials. The temperature T of the thermal state in combination with the frequency ν_r of the dipole trap ($\omega_r = 2\pi\nu_r$) determines the average quadratic distance of the atom to the center of the trap:

$$\sigma^2 = \frac{\hbar}{2m\omega_r} \coth\left(\frac{\hbar\omega_r}{2k_BT}\right). \tag{2.9}$$

The spatial probability density $Pd = |\Psi|^2$ of the atom is given by a Gaussian distribution:

$$Pd(x,y,z) = \frac{1}{(2\pi)^{\frac{3}{2}}\sigma_x\sigma_y\sigma_z} e^{-\frac{x^2}{2\sigma_x^2}-\frac{y^2}{2\sigma_y^2}-\frac{z^2}{2\sigma_z^2}}. \tag{2.10}$$

For a symmetric dipole trap in the two-dimensional atomic plane, the equation simplifies to:

$$Pd_{2D}(r,\varphi) = \frac{1}{2\pi\sigma^2} e^{\frac{-r^2}{2\sigma^2}}. \tag{2.11}$$

With this probability density, it is possible to effectively calculate the region of interest in which a single atom can be found. This is of further interest for single-site addressing when measuring whether an atom is uniformly illuminated with a laser beam profile. Taking the atomic positions into account for single-site addressing also enables simulations of optical beam pointing relative to the dipole trap holding the atom, allowing the properties of different beam profiles to be evaluated.

By integrating over the area, the probability $P(r \leq R)$ of finding the atom within a radius R is given by:

$$P(r \leq R) = \int_0^{2\pi}\int_0^R Pd_{2D}(r,\varphi) r dr d\varphi = 1 - e^{-\frac{R^2}{2\sigma^2}}. \tag{2.12}$$

Within a radius of $R = 3\sigma$, the probability to find the atom is $P(r \leq 3\sigma) = 98.89\%$.

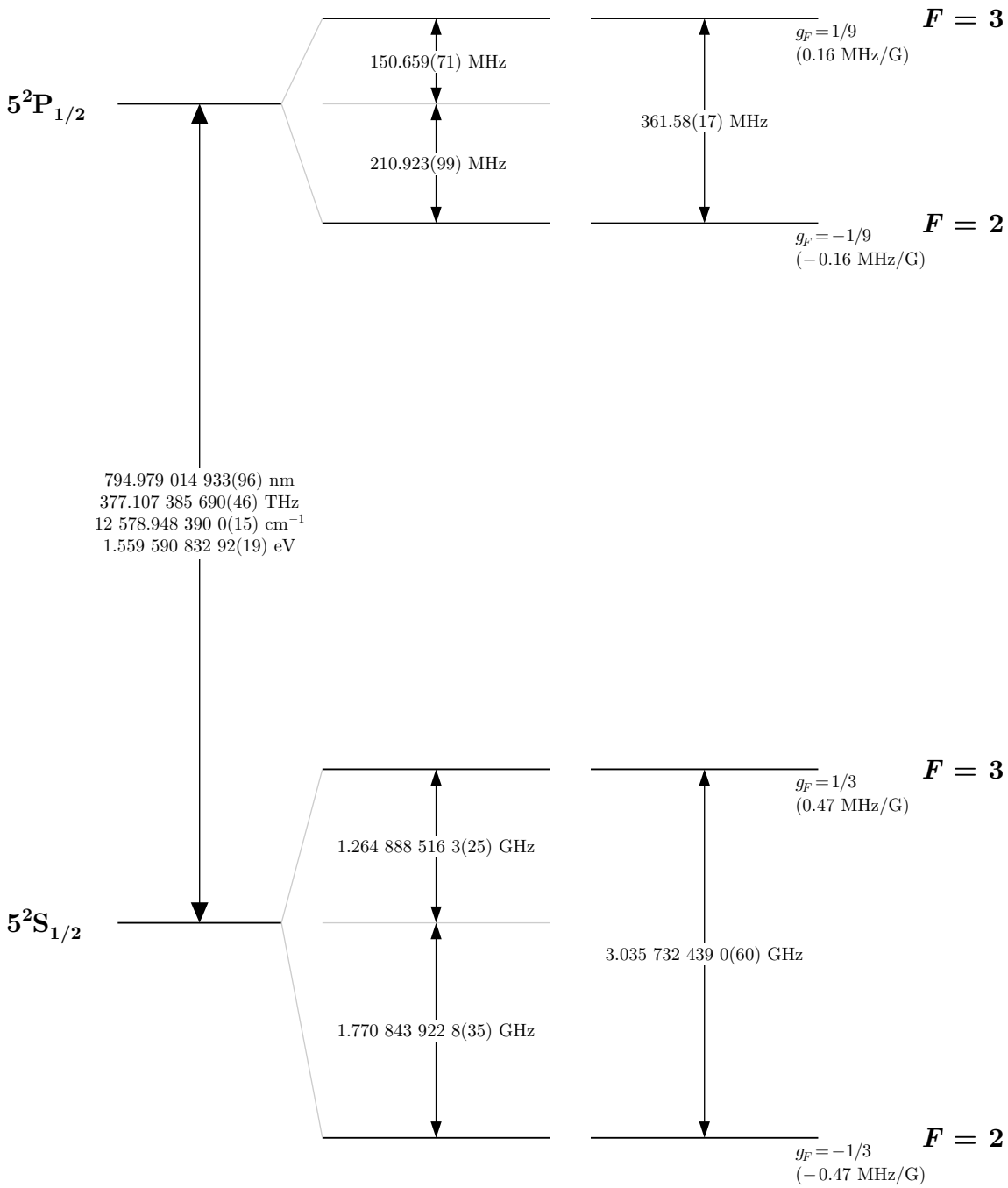


Fig. 3: Hyperfine structure of the D_1 line of ^{85}Rb with the frequency splittings between the hyperfine energy levels in different units and the g-factor for each level. The ground state $5^2S_{1/2}$ with its splitting is used for quantum information processing [2].

2.3 Acousto-Optic Deflection

In this work, a new optical setup for the single-site addressing of ^{85}Rb atoms for single-qubit gate operations, referred to as Raman tweezers, is to be simulated, designed and tested. Afterwards, an optical system combining this new local addressing scheme with a global Raman laser system is to be built. While the global setup reproduces the system developed in [7], using a Beam Shaper as its defining optical component, the local setup includes an Acousto-Optic Deflector (AOD) for single-site addressing, as demonstrated in [10].
When a Radio Frequency (RF) signal is applied to an AOD crystal, it induces a sound wave and, consequently, a periodic variation of the refractive index Δn [16]. The wavelength $\Lambda = \frac{v}{f_s}$ of this acoustic wave is determined by the speed of sound v in the medium and the RF frequency f_s of the applied signal. This sound wave acts like a static diffraction grating on radiation propagating through the crystal, since the variations in the refractive index have a significantly longer wavelength than the light $\Lambda \gg \lambda$. Diffraction at an angle Θ_B occurs if the Bragg condition is fulfilled [17]:

$$\sin(\Theta_B) = \frac{\lambda}{2\Lambda} = \frac{\lambda f_s}{2v}. \tag{2.13}$$

Thus, selecting a specific frequency results in a corresponding deflection angle, allowing different regions of the target plane to be selectively illuminated. A schematic representation of this condition is shown in Figure 4. The diffraction efficiency of an AOD is defined as the ratio if the diffracted intensity I_d and the incident intensity I as $\eta = {}^{I_d}/_{I}$. The first-order diffracted laser beam experiences a frequency and wave-vector shift by the corresponding frequency Ω and wave vector q of the sound wave due to energy and momentum conservation:

$$\omega_j = \omega + \Omega_j, \quad k_j = k + q_j. \tag{2.14}$$

In a multi-tone setup, multiple acoustic waves are generated inside the AOD crystal, resulting in different gratings and therefore diffracted beams. The variation of the refractive index in this case can be written as a sum over all N sound waves:

$$\Delta n = -\Delta n_0 \sum_{j=1}^{N} \sqrt{\kappa_j} \exp\left(i(\Omega_j t - \vec{q}_j \cdot \vec{r})\right) + \text{c.c} \tag{2.15}$$

For all N waves, a normalization condition must hold for the power ratios κ_j:

$$\sum_j \kappa_j = 1.$$

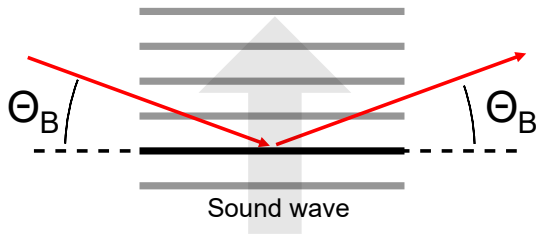


Fig. 4: Schematic drawing of Bragg interference in an AOD [18]. The incident light beam is diffracted at the sound wave with the Bragg angle Θ_B, resulting in a deflection angle of $2\Theta_B$.

2.4 Top Hat Profile Generation

The fidelity of quantum gate operations can be increased by using a uniform laser beam profile. In practice, the Gaussian intensity profile of the Raman tweezers can be transformed into a Top Hat profile, providing a more uniform intensity across the addressed atomic array. This reduces the sensitivity to beam-pointing errors between the dipole traps and the Raman tweezers, thereby reducing power fluctuations and, consequently, Rabi-frequency fluctuations.
To quantify the flatness of a Top Hat intensity profile, the uniformity U is defined as:

$$U = \frac{\Delta I}{\langle I \rangle}, \tag{2.16}$$

where ΔI denotes the standard deviation of the intensity and $\langle I \rangle$ its mean value. A smaller value of U indicates a more uniform intensity distribution, with $U = 0$ corresponding to an ideally uniform profile. The aim of optical beam shaping is therefore to minimize ΔI while keeping the mean intensity $\langle I \rangle$ approximately constant.
To estimate the extent to which laser radiation from neighboring positions in the grid is emitted into the central position and potentially reduces the fidelity of gate operations, the crosstalk η can be defined. It is given as the ratio of the power of the nearest neighbors P_{NN} in the target area to the power of the central profile P_0 in the target area:

$$\eta = \frac{P_{\mathrm{NN}}}{P_0}. \tag{2.17}$$

The power of the central profile P_0 represents the desired laser power of the central beam profile at the atom's position, while P_{NN} denotes the unwanted light illuminating the atom. The number of considered nearest neighbors when evaluating the crosstalk depends on the structure of the atomic grid. For a Cartesian grid, the eight nearest neighbors are included in the ratio η, whereas a hexagonal grid would include six nearest neighbors. Considering all possible nearest neighbors provides an upper bound for the crosstalk and thus represents the worst-case scenario for unwanted light. In practice, the measured crosstalk is expected to be lower than the simulated values. A key challenge for generating Top Hat profiles for effective single-site addressing of individual qubits is to minimize the uniformity U while keeping the crosstalk η low. In this work, two methods are examined and analyzed for generating local Top Hat profiles in regards of their compatibility with the already running experimental setup.

2.4.1 Beam Shaper

A Beam Shaper is an optical component that can be used to transform a laser beam profile into a desired form based on the incident beam profile and the properties of the Beam Shaper itself. A refractive Beam Shaper modifies the intensity distribution of a beam by changing its propagation direction through refraction. Specially designed optical surfaces redistribute the laser beam such that regions with high intensity are expanded, while regions with lower intensity are compressed. Since the transformation is achieved through the geometry of the optical surfaces, most of the incident optical power remains in the desired beam.
A diffractive Beam Shaper uses a spatially varying optical phase to modify the propagation of the incident beam. Such a Beam Shaper can be implemented as a Diffractive Optical Element (DOE) with a structured surface that determines the local phase shift. To generate a uniform intensity profile, a focusing lens must additionally be placed behind the diffractive Beam Shaper

[19], forming the Focal Beam Shaper (FBS). The resulting field $E_f(x_f, y_f)$ in the focal plane of the focusing lens can be paraxial approximated as the Fourier transform of the incident field $E_0(x, y)$ modulated by the transfer function of the Beam Shaper T_{FBS} [20]:

$$E_f(x_f, y_f) = \mathcal{FT}[E_0(x, y) \cdot T_{\text{FBS}}(x, y)]. \tag{2.18}$$

The transfer function for the FBS is given by the phase modulation $T_{\text{FBS}} = e^{i\phi_{\text{FBS}}(x,y)}$. A schematic of the beam path of an FBS is shown in Figure 5.
In previous works [7, 8], a Beam Shaper in combination with a focusing lens was used to create a rectangular Top Hat profile in the atomic plane capable of addressing the whole ^{85}Rb array, consisting of more than 32×32 atoms with a pitch of 5.288 µm. In this work, such an approach is tested near the diffraction limit, in which a beam profile resembling a Super Gaussian function is created with the goal to minimize the uniformity and the crosstalk. For a local approach, the created Top Hat profile has to be projected onto every atom site in the atomic plane. Thus, in addition to generating the Top Hat profile with the desired uniformity and crosstalk, the deflection process must also preserve the beam quality. As already discussed, the method of single-site addressing is realized with an AOD. The deflection of a Gaussian beam through the sound wave within the crystal of the AOD changes the frequency ω and the k-vector $\vec{k}$. The acousto-optic crystal additionally imposes a spatially dependent phase on the Gaussian beam because the acoustic wave takes a finite time to propagate through the crystal and therefore interacts with different parts of the beam at different times. Additionally, these crystals are cut such that different parts of a wide beam experience different optical path lengths and therefore different phases. For these reasons, the AOD potentially introduces a new, unknown transmission function $T_{\text{AOD}}(x, y)$. The resulting field $E_f(x_f, y_f)$ is then given by:

$$E_f(x_f, y_f) = \mathcal{FT}[E_0(x, y) \cdot T_{\text{AOD}}(x, y) \cdot T_{BS}(x, y)].$$

In the case of $T_{\text{AOD}}(x, y) \neq 1$, the Top Hat profile can be distorted or even destroyed entirely, resulting in Airy spots in the atomic plane and thus eliminating the advantage over previously developed setups using Gaussian spots [18].

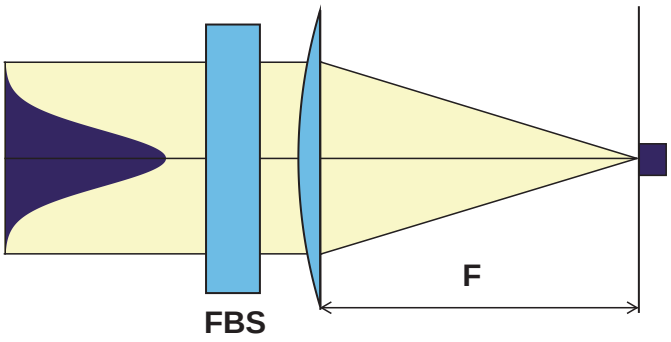


Fig. 5: Schematic beam path of an FBS. The Gaussian input beam, shown on the left, passes through the FBS, which imprints a phase modulation, and is then focused by a lens of focal length F. In the focal plane, a Top Hat intensity profile forms on the screen. Taken from Fig. 2 of [19] and upscaled using AI tools (Claude Opus 5, Anthropic).

2.4.2 Multi-Tone AOD

Another approach for the generation of uniform intensity profiles uses an AOD. The basic concept consists of superimposing multiple deflected laser beams to generate a uniform laser intensity profile. Each spot is generated by the AOD in a multi-tone configuration, where the RF signal consists of multiple frequency components f_s, with each component corresponding to one spot. The incident laser beam in front of the AOD and the spots within the atomic plane can be described as Gaussian beams:

$$I_g(r) = I_0 \exp\left[-2(r - r_c)^2/(\omega^2)\right]. \tag{2.19}$$

Here, ω is the beam waist, defined as the radial distance from the beam center r_c where the intensity has decreased to $I_0/e^2 \approx 13.5\,\% I_0$. The parameters for optimizing the uniformity of a Top Hat profile are the beam waist in the atomic plane and the spacing between neighboring spots, both determined by the optical setup. The waist within the atomic plane is a projection of the beam waist in front of the AOD. Thus, varying the magnifying optics used is a way to change the uniformity. The spacing between neighboring spots is determined by the angle of deflection created by the AOD and thus by the applied frequency as well as by the subsequent imaging optics. In this work, another type of beam profile has to be considered for beam sizes approaching the diffraction limit. An Airy profile, described by a first-order Bessel function J_1, is used to model the individual spots in the atomic plane, as it reproduces the side structures observed in [10], which do not appear for a Gaussian beam profile:

$$I_a(r) = I_0 \left[\frac{2J_1\,(kr)}{kr}\right]^2. \tag{2.20}$$

The Bessel function is parametrized such that its central peak has the same waist ω as the Gaussian function. Here, $k \cdot r_0 \approx 3.8317$ is the first zero of the first-order Bessel function. Matching the waists of the two profiles yields $r_0 \approx 1.483\omega$. A visualization of this parametrization is shown in Figure 6.

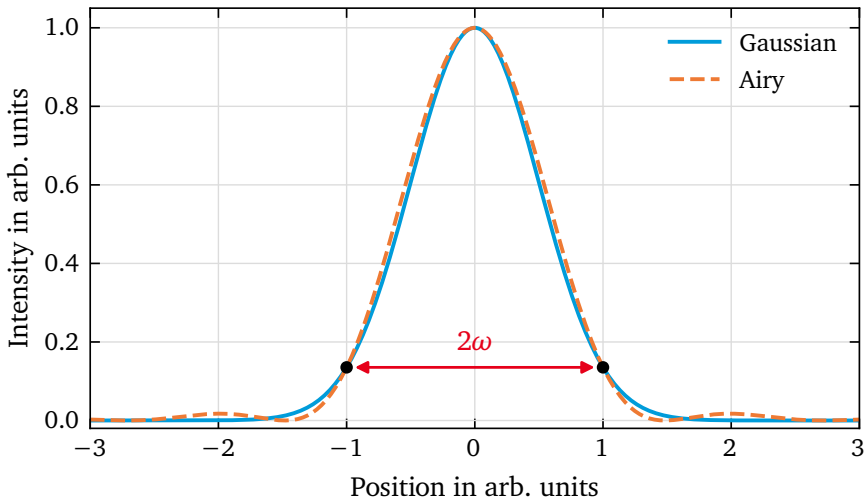


Fig. 6: Plot of a Gaussian and an Airy function used for the parametrization of a single spot profile in the atomic plane. The Airy profile is parametrized with the same waist ω for its central peak as the Gaussian profile.

When multiple coherent laser beams are superimposed, interference occurs. Because of the imposed frequency shift of the AOD, vertically or horizontally neighboring overlapping spots do not spatially interfere. Along the anti-diagonals of the grid, the frequency shifts are identical, resulting in spatial interference. For example, a base frequency of 100 MHz is chosen with a frequency span of 1 MHz in each dimension. For eleven spots per dimension, this results in a frequency spacing of 0.1 MHz between adjacent spots. The frequency 200.3 MHz can be obtained from different combinations of frequency shifts along the two deflection axes, such as 0.1 MHz + 0.2 MHz or 0.2 MHz + 0.1 MHz. A visual representation of the interfering spots is shown in Figure 7 on the left. For intensity profiles with different offset frequencies, such as 100 MHz along the x-axis and 105 MHz along the y-axis, this problem does not occur on the anti-diagonals of the profile. One method to avoid stationary spatial interference is to choose a different number of tones for each axis while keeping the same frequency width.
One approach is the use of $N_x = N$ and $N_y = N + 1$ tones. In this case, the resulting profile remains symmetric, and only one pair of spots on opposing edges of the profile experiences the same frequency shift, as illustrated on the right in Figure 7. The condition for the selection of the waist ω and the profile width W of such a profile that limits stationary spatial interference is given by:

$$\sqrt{d^2 + d^2} = \sqrt{2}d > 2\omega. \tag{2.21}$$

Once no overlap of these spots is ensured, the matter of unwanted spatial interference poses no limiting problem for the generation of uniform intensity profiles. A more general discussion of projections of two-dimensional light patterns with an AOD regarding interference can be found in [21].
Since the profile generation involves the superposition of coherent laser beam profiles with closely spaced frequencies, time-dependent interference, also known as beating, must be taken into account. Beating can be a limiting factor for efficient single-site addressing by disrupting Rabi oscillations due to strong intensity oscillations and therefore distorting single-qubit gate operations. The beating frequency can be estimated from the frequency difference between neighboring spots, which typically lies in the kHz to MHz range.

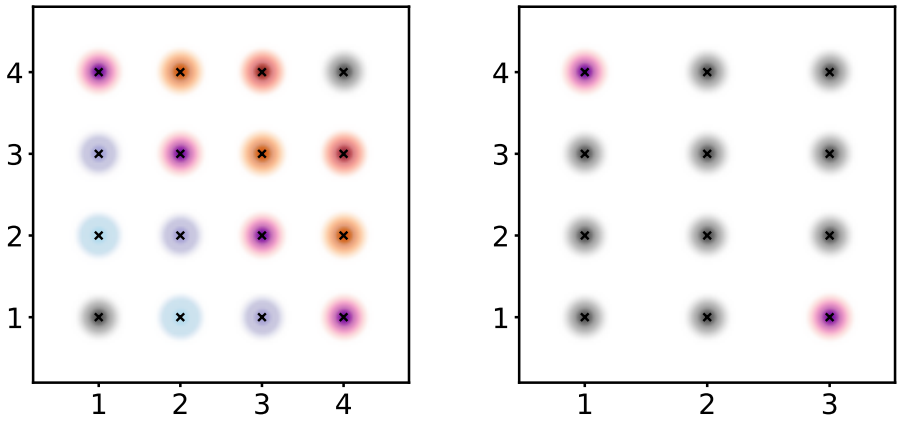


Fig. 7: Left: Representation of a 4×4 spot grid. Spots with the same frequency shift that cause stationary interference are marked with the same color. Right: Using different numbers of tones (here 3×4) leaves just one pair of spots with the same frequency shift and reduces interference. Left figure adapted from Fig. 3 of [22], with labels and annotations removed using AI tools (Claude Opus 5, Anthropic). Right figure created using AI tools (Claude Opus 5, Anthropic).

2.4.3 Time-Dependent Interference

In this work, a simulation is conducted to determine the magnitude of the time-dependent interference for the multi-tone AOD setup. The optical fields in the simulation are treated using a standard complex-field description of coherent light, in which the total field is calculated as a superposition of the individual fields and the intensity is proportional to the squared absolute value of the resulting field [23].
Both AODs shift the optical frequency. A spot $s = (i, j)$, produced by tone i of the x-AOD and tone j of the y-AOD, thus has the frequency:

$$f_s = f_{x,i} + f_{y,j}, \qquad f_{x,i} = f_{\mathrm{x},0} + W_f \frac{i}{N_x - 1}, \quad f_{y,j} = f_{\mathrm{y},0} + W_f \frac{j}{N_y - 1}. \tag{2.22}$$

Here, W_f denotes the frequency width of the generated spots, which is equal for both dimensions as discussed above. $f_{(\mathrm{x,y}),0}$ is the offset frequency of the respective AOD dimension and $N_{x,y}$ denotes the corresponding number of spots. To take interference effects into account, the electric fields of the spots are superimposed rather than their intensities. For each spot s, the spatial field amplitude is given by:

$$g_s(\mathbf{r}) = \sqrt{a_s}\, u(\mathbf{r} - \mathbf{c}_s), \qquad a_s = a_{x,i}\, a_{y,j}, \tag{2.23}$$

where u is the real mode in the focal plane, normalized to $u(\mathbf{0}) = 1$ (Gaussian e^{-r^2/w^2} or Airy $2J_1(kr)/(kr)$), $\mathbf{c}_s$ is the position of spot s in the focal plane and $a_{x,i}$, $a_{y,j}$ are the intensity weights. In the linear regime of the AOD, the diffraction efficiency is proportional to the RF power, so a is an RF power ratio and $\sqrt{a}$ the corresponding RF voltage ratio. The spot phase is the sum of the phases of the two tones, one from each AOD:

$$\phi_s = \phi_{x,i} + \phi_{y,j}, \tag{2.24}$$

so only $N_x + N_y$ of the $S = N_x \cdot N_y$ spot phases are free parameters. Only the differences between spot frequencies are relevant for the analysis of the time-dependent interference. If all differences are integer multiples of a fundamental frequency f_0 (their greatest common divisor), every spot frequency can be written as

$$f_s = f_{\min} + k_s f_0, \qquad k_s = \frac{f_s - f_{\min}}{f_0} \in \mathbb{N}_0. \tag{2.25}$$

Here, $f_{\min} = f_{x,0} + f_{y,0}$ holds. The fundamental frequency can be calculated with the Least Common Multiple (LCM) of tone indices through:

$$f_0 = \frac{W_f}{\mathrm{LCM}(N_x - 1,\, N_y - 1)}, \qquad T_0 = \frac{1}{f_0}. \tag{2.26}$$

For a 3×4 grid this results in $f_0 = W/6$. The index s, as introduced above, labels a spot and k_s its so called frequency order. Spots with the same order form the group:

$$\mathcal{S}_m = \{\, s : k_s = m \,\}. \tag{2.27}$$

The superposition of all spot profiles in $\mathcal{S}_m$ with their respective phases ϕ_s defines the time-independent electric field, referred to as the group field:

$$H_m(\mathbf{r}) = \sum_{s \in \mathcal{S}_m} g_s(\mathbf{r})\, e^{i\phi_s}. \tag{2.28}$$

The time-dependent electric field of a single spot, on the other hand, is given by:

$$E_s(\mathbf{r},t) = g_s(\mathbf{r})e^{i\phi_s}e^{i2\pi f_s t} = e^{i2\pi f_{\min} t} g_s(\mathbf{r})e^{i\phi_s}e^{i2\pi k_s f_0 t}. \tag{2.29}$$

For the total field, obtained by summing over all spots, the definition of the group field can be used:

$$E(\mathbf{r},t) = \sum_s E_s(\mathbf{r},t) = e^{i2\pi f_{\min} t} \sum_m H_m(\mathbf{r})\, e^{i2\pi m f_0 t}. \tag{2.30}$$

Since only the squared absolute value $|E(\mathbf{r},t)|^2$ is of interest, the common phase factor $e^{i2\pi f_{\min} t}$ can be neglected. The time-dependent field can therefore be expressed as a superposition of monochromatic frequency components, following the standard Fourier description of polychromatic optical fields [23]:

$$E(\mathbf{r},t) = \sum_m H_m(\mathbf{r})\, e^{i2\pi m f_0 t}. \tag{2.31}$$

The intensity is then given by:

$$I(\mathbf{r},t) = |E(\mathbf{r},t)|^2 = \sum_d D_d(\mathbf{r})\, e^{i2\pi d f_0 t}, \tag{2.32}$$

which is a trigonometric polynomial with the Fourier coefficients $D_d(\mathbf{r})$ and the beat frequencies $f_{\text{beat}} = |d| f_0$. The parameter d is called the beating order and is defined as the difference in the spot orders $k_{s,s'}$ of the overlapping spots with $s = (i,j)$, $s' = (i',j')$ and $\Delta i = i - i'$, $\Delta j = j - j'$:

$$d = k_{i,j} - k'_{i',j'} = 3\,\Delta i + 2\,\Delta j. \tag{2.33}$$

The Fourier coefficients are calculated in terms of the group fields and the individual spot fields as:

$$D_d(\mathbf{r}) = \sum_{m-n=d} H_m(\mathbf{r})\, H_n^*(\mathbf{r}) = \sum_{k_s - k_{s'} = d} g_s(\mathbf{r})\, g_{s'}(\mathbf{r})\, e^{i(\phi_s - \phi_{s'})}. \tag{2.34}$$

Because $I^*(t) = I(t)$ holds true, the Fourier coefficients fulfill:

$$D_{-d} = D_d^*, \qquad D_0 \in \mathbb{R}. \tag{2.35}$$

Calculating the time average of the intensity over one period T_0 leaves only the term $d = 0$:

$$\langle I(\mathbf{r},t)\rangle_t = \frac{1}{T_0}\int_0^{T_0} \sum_d D_d(\mathbf{r})\, e^{i2\pi d f_0 t} dt = D_0(\mathbf{r}) = \sum_m |H_m(\mathbf{r})|^2\,. \tag{2.36}$$

This result can be expressed in terms of the spatial field amplitudes:

$$\langle I(\mathbf{r},t)\rangle_t = \underbrace{\sum_s g_s^2(\mathbf{r})}_{\text{incoherent sum}} + 2 \sum_{\substack{s<s' \\ k_s = k_{s'}}} g_s(\mathbf{r})\, g_{s'}(\mathbf{r}) \cos(\phi_s - \phi_{s'}). \tag{2.37}$$

The time average equals the incoherent sum of the individual spot intensities plus a phase-dependent interference term between spots of the same frequency order. With $f_s = f_{x,i} + f_{y,j}$, stationary spatial interference occurs for $k_{x,i} + k_{y,j} = k_{x,i'} + k_{y,j'}$. For a 3×4 grid with equal frequency width, $k_{x,i} = 3i$ and $k_{y,j} = 2j$, exactly one pair of spots has the same frequency. The

two diagonally opposite corner spots, as already shown in Figure 7, cause stationary spatial interference with their contribution to the group field:

$$k_{(0,3)} = k_{(2,0)} = 6, \qquad H_6 = g_{(0,3)}\, e^{i\phi_{(0,3)}} + g_{(2,0)}\, e^{i\phi_{(2,0)}}. \tag{2.38}$$

For $N_x = N_y$, all spots along a given anti-diagonal ($i + j = \text{const}$) share the same frequency and thus contribute to the spatial interference. For a single pair of spots, the stationary spatial interference term in Equation 2.37 vanishes when their phase difference satisfies $\phi_s - \phi_{s'} = \pm\pi/2$. Calculating the variance of the intensity with $\langle I^2\rangle_t = \sum_d |D_d|^2$ results in:

$$\mathrm{Var}_t[I(\mathbf{r},t)] = \langle I^2\rangle_t - \langle I\rangle_t^2 = \sum_d |D_d(\mathbf{r})|^2 - |D_0(\mathbf{r})|^2 = \sum_{d\neq 0} |D_d(\mathbf{r})|^2 = 2\sum_{d>0} |D_d(\mathbf{r})|^2\,. \tag{2.39}$$

To characterize the beating, a time-dependent uniformity $U_t(\mathbf{r})$ is defined as the ratio of the temporal standard deviation to the temporal mean intensity at a given position $\mathbf{r}$:

$$U_t(\mathbf{r}) = \frac{\sigma_I(\mathbf{r})}{\langle I(\mathbf{r})\rangle_t} = \frac{\sqrt{\mathrm{Var}_t[I(\mathbf{r},t)]}}{D_0(\mathbf{r})} = \frac{\sqrt{2\sum_{d>0} |D_d(\mathbf{r})|^2}}{D_0(\mathbf{r})}. \tag{2.40}$$

This quantity captures the temporal variations of the intensity at each position. The Root Mean Square (RMS) time-dependent uniformity over a region of interest R is then given by:

$$U_t = \left\langle U_t(\mathbf{r})^2 \right\rangle_R^{1/2}. \tag{2.41}$$

Using the RMS value weights regions with strong beating more heavily than a simple arithmetic mean would. The uniformity can be expressed as the quadratic sum of the contributions from the individual beating orders with $\sigma_d^2(\mathbf{r}) = 2|D_d(\mathbf{r})|^2$ as:

$$\left\langle U_t(\mathbf{r})^2 \right\rangle_R^{1/2} = \left\langle \frac{\sigma_I(\mathbf{r})^2}{\langle I(\mathbf{r})\rangle_t^2} \right\rangle_R^{1/2} = \left(\sum_{d>0} \left\langle \frac{\sigma_d^2(\mathbf{r})}{\langle I(\mathbf{r})\rangle_t^2} \right\rangle_R \right)^{1/2} \equiv \left(\sum_{d>0} (U_{t,d})^2 \right)^{1/2}. \tag{2.42}$$

The quantity U_t thus provides a measure of the temporal intensity fluctuations averaged over the region of interest. The smaller U_t is, the more temporally stable the profile is.

3 Test Setup for Generating Top Hat Profiles with a Beam Shaper

To generate beam profiles with uniform intensity, the Beam Shaper FBSR-800-3.5 manufactured by the company TOPAG is used. The incoming Gaussian beam requires a specific $1/e^2$ beam diameter depending on the model used. In this case, a beam diameter $d = 2\omega = 3.5\,\text{mm}$ is needed [9]. After passing through the Beam Shaper, a focusing lens has to be used to generate the Top Hat profile. The resulting beam diameter d', as specified by the manufacturer, is approximately:

$$d' \approx \frac{2\lambda f}{Md}, \tag{3.1}$$

where λ is the wavelength of the incident laser light, f is the focal length of the focusing lens and M is the magnification of a potential optical setup between the Beam Shaper and the lens. Each setup described in the following uses the Lens Optics Objective with $f_{\text{LO}} = 52.88\,\text{mm}$ instead of a single focusing lens, as well as a glass plate simulating the vacuum chamber window used in the actual experiment. The camera (U3-38J0XLE-M-GL) used to acquire the images has a pixel size of 1.45 µm and is placed behind a microscope objective with a magnification of 43.75. For all calculations in this chapter regarding uniformity and crosstalk, the intensity is uniformly weighted across the entire calculation area. The calculation area for the uniformity is chosen as the region in which the intensity exceeds $90\,\%$ of the maximum of the respective profile. Additionally, for the crosstalk calculation described in Equation 2.17, the measured beam profile is digitally replicated at the eight positions of a 3×3 structure with a pitch of 5.288 µm, corresponding to the atomic pitch.

3.1 Magnification with a Telescope

Prior to evaluating the Beam Shaper for the generation of Top Hat profiles near the resolution limit, the profile produced in the focal plane of the Lens Optics Objective is first measured without any additional magnifying optics. The setup consists of a Beam Expander (Edmund Optics 87-564) behind the Triplet Collimator of the fiber coupler to achieve the necessary beam diameter for the subsequent Beam Shaper. After passing through the Beam Shaper, the laser beam is focused with the Lens Optics Objective and passes through the glass plate before being imaged onto the sensor of the camera.
The Beam Expander is set to a magnification of approximately 1.9 in order for the laser beam to reach its necessary diameter for the Beam Shaper. The waist of the Gaussian beam in both the horizontal (x) and vertical (y) dimensions is nearly identical, with $\omega_x = 1.77(2)\,\text{mm}$ and $\omega_y = 1.76(2)\,\text{mm}$, and both values are within $5\,\%$ of the desired value.
The resulting Top Hat profile in the focal plane of the Lens Optics Objective has a $1/e^2$ diameter of 22.80(6) µm. The expected beam diameter according to Equation 3.1 is calculated with f_{LO}, resulting in $d' = 24\,\mu\text{m}$. The measured beam profile is approximately $5\,\%$ smaller than the value specified by the manufacturer, indicating that Equation 3.1 provides only an approximation of the actual beam diameter. The uncertainties of the beam waists and of all quantities derived

from them in this chapter result from fitting an elliptical Gaussian profile to the measured data. The Top Hat profile consists of a central peak and a bright outer ring, whose two sides have different intensities. Scanning along the optical axis leads to an increase or decrease in the central peak intensity, while the intensity of the outer ring remains approximately constant. Moving the Beam Shaper perpendicular to the beam path results in a lateral shift of the central peak. The intensity variation along the outer ring could not be corrected by adjusting either the Beam Shaper or the Lens Optics Objective.
By adjusting the position of the camera along the optical axis of the laser beam, different planes along the propagation direction can be observed, with only one camera position corresponding to the focal plane. In addition to the Top Hat profile in the focal plane, a second Top Hat profile with a larger diameter and therefore lower intensity can be observed in another plane. This profile is referred to as a first-order Top Hat profile, while the one in the focal plane is of zeroth order [24]. Additionally, ellipticity of the beam profile is observed, which may indicate the presence of astigmatism. For the following measurements near the diffraction limit, only zeroth order Top Hat profiles are relevant to determine the practical limits of the Beam Shaper in a local Raman laser setup.
After the Beam Shaper was successfully tested, confirming that the required beam size and profile quality were achieved, a telescope was added to the setup to magnify the beam before it reaches the Lens Optics Objective. This allows a Top Hat profile with a size close to or below the 10 µm limit specified by the manufacturer to be generated in the focal plane [9]. The previous setup is extended using two achromatic lenses with the focal lengths of $f_1 = 45\,\text{mm}$ and $f_2 = 400\,\text{mm}$. The lenses in the setup are separated by a distance equal to the sum of their respective focal lengths. After the telescope, two mirrors, the Lens Optics Objective, the glass plate and the camera are placed. Figure 12 shows the final setup, including an additional Acousto-Optic Deflector (AOD) that is not used when evaluating the effect of the telescope alone. The incoming beam is magnified through the telescope by a factor of $M = f_2/f_1$, resulting in a beam diameter of $d' = {}^{f_2}/_{f_1} \cdot d = 31.1\,\text{mm}$, which is subsequently focused by the Lens Optics Objective. The expected diameter of the Top Hat profile according to 3.1 is $d' = 2.7\,\mu\text{m}$. For the creation of a Top Hat profile in the focal plane, it is relevant that the beam diameter is smaller than the optics in the beam path. If the aperture of any optical component is too small compared to the shaped beam diameter, the beam profile is spatially filtered before reaching the Lens Optics Objective. Since the focusing process performs a Fourier transform, this spatial filtering results in a distorted profile.
For this reason, two three inch mirrors are placed after the telescope. The limiting optical component in this setup is the tube of the Lens Optics Objective, which has a free aperture of 34 mm. A simulation of a normalized super-Gaussian Top Hat beam profile is performed to estimate the quality of the beam profile after passing through the Lens Optics Objective. The Super-Gaussian has the form:

$$I(r) = \exp\left[-2\left(\frac{r}{\omega}\right)^{2n}\right], \tag{3.2}$$

where the order $n = 2$ is chosen as an example and a diameter of $d' = 2\omega = 2.7\,\mu\text{m}$ in the atomic plane is assumed, as given by Equation 3.1. Starting from an ideal Top Hat beam profile in the atomic plane, the beam profile in front of the Lens Optics Objective is calculated using a Fourier transform. The ideal beam profile in the Fourier plane is then truncated by the aperture and transformed back into the spatial domain. The resulting filtered profile therefore represents the best-case scenario, assuming an ideal initial beam profile without distortions or additional optical losses. The results of this simulation are shown in Figure 8. The aperture cutoff in

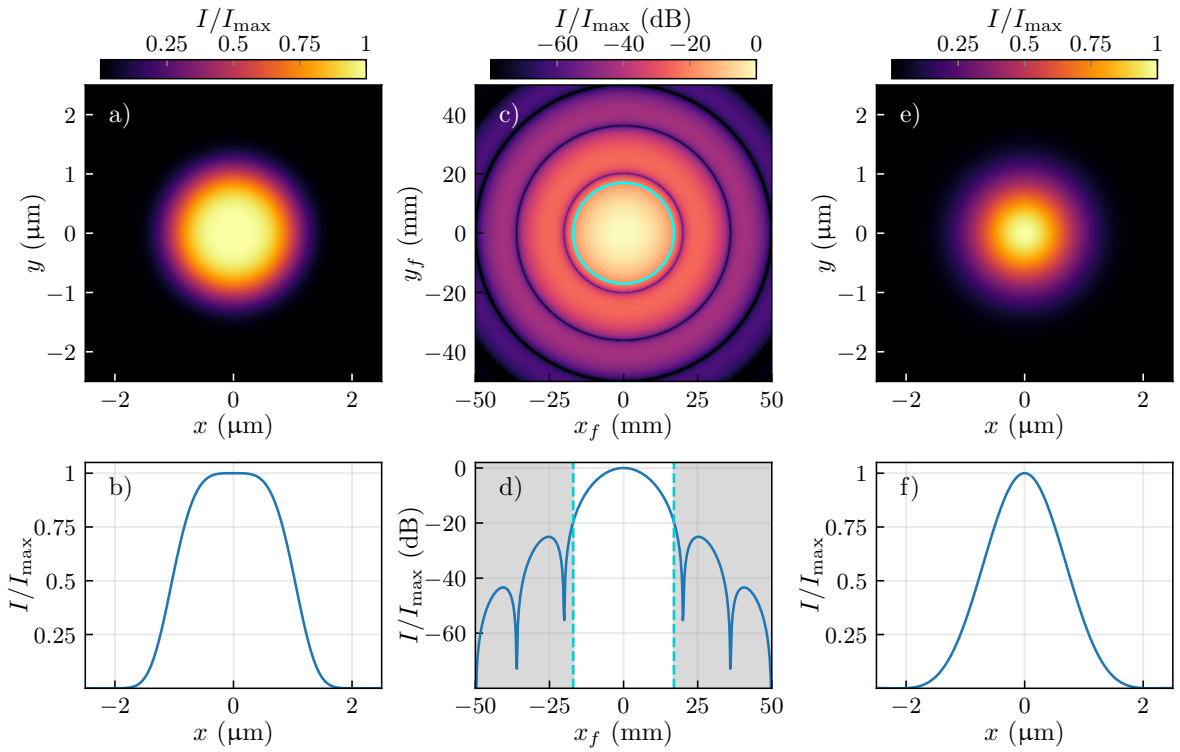


Fig. 8: Simulated aperture cutoff for a Top Hat profile, with all profiles normalized to their respective maxima. a) Top Hat profile to be generated in the atomic plane with a $1/e^2$ diameter of 2.7 µm. b) Cross-section plot through a). c) Transformed profile in the Fourier plane in front of the Lens Optics Objective, with the aperture cutoff indicated in cyan. d) Cross-section plot of the profile in the Fourier plane. e) Back-transformed beam profile after the aperture cutoff, showing the loss of the Top Hat structure. f) Cross-section plot of the back-transformed profile.

the Fourier plane leaves only the central structure of the Fourier image. Consequently, the back-transformed beam profile no longer has the initial Top Hat structure but instead resembles a Gaussian beam with a diameter of approximately 2.50 µm. For a diameter of 4.8 µm, the first secondary maximum is not truncated by the aperture, allowing the back-transformed profile to retain its Top Hat shape. The tube aperture of the Lens Optics Objective, on the other hand, would need a diameter of 60 mm for a Top Hat profile to be created in the focal plane with a diameter of 2.7 µm. The simulation results are confirmed by experimental measurements obtained with the test setup. The resulting image, shown in Figure 9, resembles an Airy profile rather than a Top Hat profile. Although the measured profile diameter of 3.36(6) µm is close to the expected value of 2.56 µm, the Gaussian beam profile predicted by the simulation is not observed experimentally. The loss of the uniform structure of the beam profile can be attributed either to the manufacturer's specified minimum beam diameter of 10 µm for the Beam Shaper [9] or to the limited tube aperture of the Lens Optics Objective, which, as discussed above, acts as a spatial filter. In either case, Top Hat profiles cannot be generated with this combination of lenses and the Beam Shaper used. Therefore, it is concluded that the magnification of the telescope has to be smaller to create a larger beam profile in the focal plane. Additionally, this Top Hat profile has to be smaller than the interatomic distance.

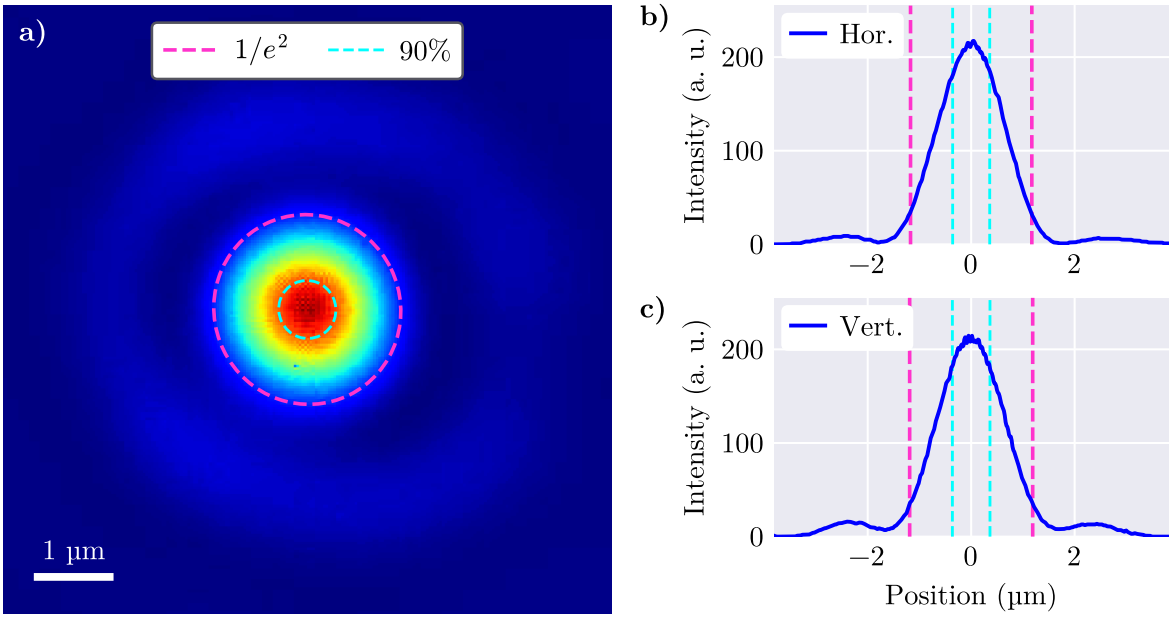


Fig. 9: a) Image of the shaped beam profile with marked intensity levels taken after the Lens Optics Objective and the $45 - 400$ telescope with a $1/e^2$-diameter of approximately 3.36(6) µm. b)+c) Horizontal and vertical cross-section plots with a visible structure of an Airy function.

In the next iteration, the second achromatic lens is replaced with one with a focal length of $f_2 = 300$ mm, while the first lens remains unchanged. The resulting beam diameter is therefore calculated to be $d = {}^{300}/_{45} \cdot 3.5\,\text{mm} = 23.3\,\text{mm}$, which should improve the quality of the focused beam because of less spatial filtering. The expected beam diameter according to Equation 3.1 is 3.6 µm. The resulting image of the beam profile is shown in Figure 10. The beam exhibits an elliptical profile with $1/e^2$-diameters of 5.20(8) µm and 4.42(6) µm along its principal axes and a tilt angle of $52.8(2)^\circ$ relative to the horizontal axis. These values differ by up to $44\,\%$ from the expected values and strongly indicate that the size limit specified by the manufacturer has been reached. The region enclosed by the $90\,\%$ intensity level, indicated in cyan, has diameters of $d_a = 2.26(4)$ µm and $d_b = 1.92(4)$ µm along the principal axes. Within this region, the uniformity of the profile is $U = 3.85(11)\,\%$ and the crosstalk is $\eta = 3.4(12)\,\%$. Also elliptical side structures resembling the secondary maximum of an Airy profile are visible around the central structure, while in the upper right half of the image its intensity increases compared to the remaining parts. The ellipticity of the profile could not be reduced by adjusting the Beam Shaper or the Lens Optics Objective. The telescope itself with its two achromatic lenses was not adjusted and could therefore contribute to the observed deviation in the profile shape due to manual alignment inaccuracies. In particular, the observed ellipticity could potentially be caused by astigmatism introduced by the telescope. This effect cannot be investigated with the Beam Shaper, as only the focal plane can be evaluated. Mounting future setups in a tube system could improve the alignment and, consequently, the quality of the magnification. This was not implemented in the present test setup due to the different lens diameters and the large distances between the lenses.
Although the generated beam profile has an elliptical shape and larger diameters than specified by the manufacturer, the feasibility of single-site addressing with an AOD is investigated next as a proof of concept.

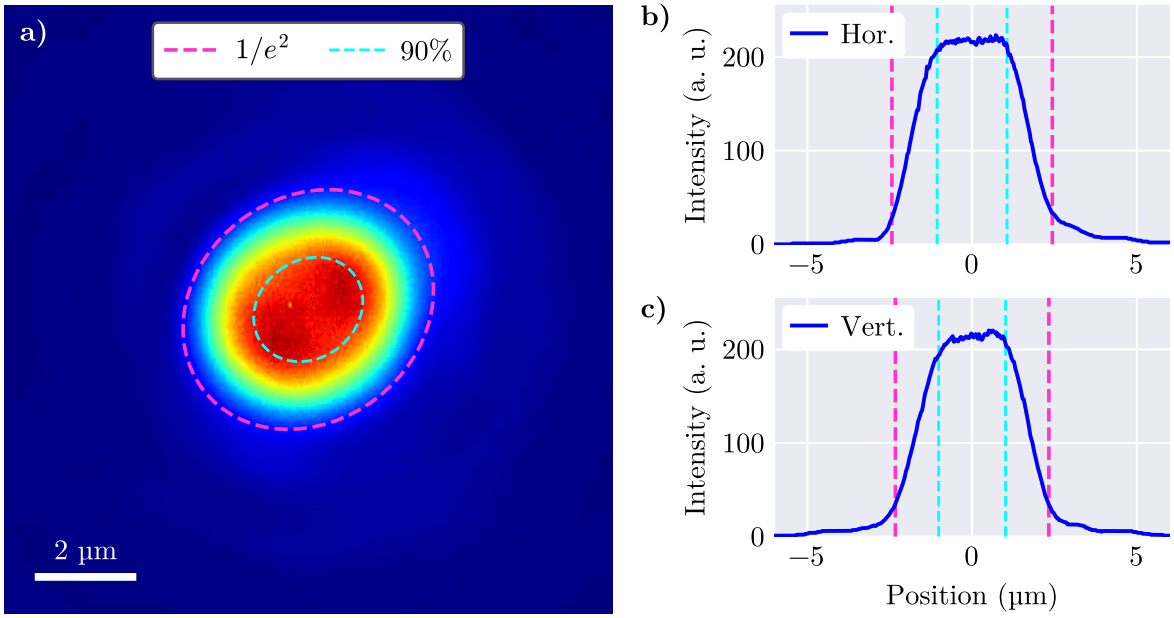


Fig. 10: a) Image of the shaped beam profile with marked intensity levels, taken after the Lens Optics Objective and the $45 - 300$ telescope. The profile is elliptical with diameters of 5.20(8) µm and 4.42(6) µm along both axes. b)+c) Horizontal and vertical cross-section plots.

3.2 Integration of the AODs

To enable single-site addressing with a Top Hat profile, an AOD can be used, as already discussed in section 2.3. For addressing the entire two-dimensional atomic plane, two AODs are arranged perpendicular to each other. The center between the two AODs is positioned at a distance equal to the focal length f_1 from the first achromatic lens of the telescope. The beam propagation behind the AOD is depicted in Figure 11. The angle Θ of the deflected light is transformed into a lateral shift by the first lens f_1. Afterwards, this shift is imaged onto the

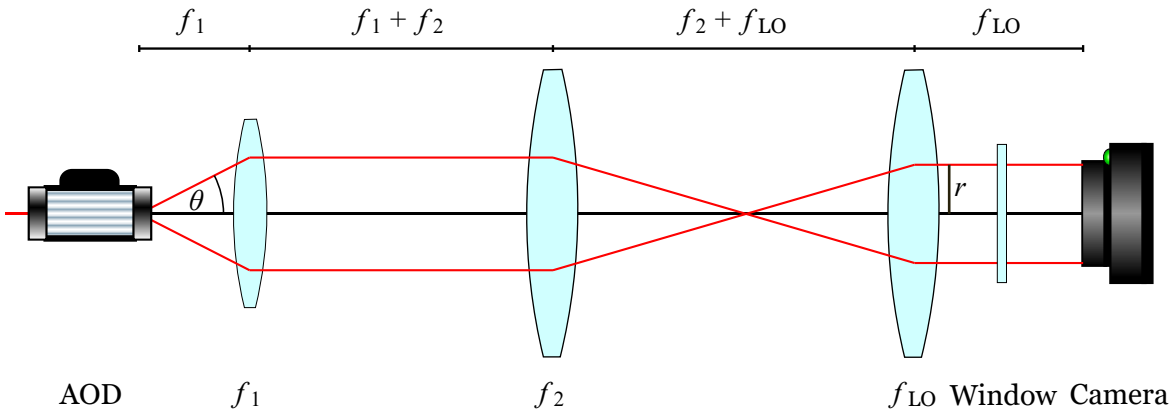


Fig. 11: Schematic drawing of the beam propagation behind the AODs using a telescope and the Lens Optics Objective. For simplicity, only one AOD is drawn. The angle Θ of the deflected light is transformed into a lateral shift by the first lens f_1 and imaged onto the camera sensor by the second achromatic lens f_2 and the Lens Optics Objective.

camera sensor by the second achromatic lens f_2 and the Lens Optics Objective. The resulting position r on the camera sensor depends on the angle of deflection Θ and can be calculated by geometric considerations with:

$$r = \frac{f_1}{f_2} f_{\mathrm{LO}} \tan \Theta. \tag{3.3}$$

The maximum angle of deflection for the AOD is $\Theta_{\mathrm{max}} = \pm 21.5\,\mathrm{mrad}$ [21]. Using Θ_{max} and the current combination of the achromatic lenses $f_1 = 45\,\mathrm{mm}$ and $f_2 = 300\,\mathrm{mm}$ in Equation 3.3 results in a total addressable region of 341.3 µm × 341.3 µm, which corresponds to a 64×64 grid of atoms with a pitch of 5.288 µm. The beam waist in the atomic plane can be calculated with:

$$\omega = \frac{f_1}{f_2} \frac{\lambda f_{\mathrm{LO}}}{\pi \omega_{\mathrm{in}}}. \tag{3.4}$$

This equation does not take the effect of the Beam Shaper into account. Using the beam diameter instead gives:

$$d' = \frac{2}{\pi} \frac{2\lambda f_{\mathrm{LO}}}{d(f_2/f_1)}. \tag{3.5}$$

Together with the magnification M of the telescope, Equation 3.1 is therefore multiplied by the factor $2/\pi$. Consequently, the Top Hat profile is approximately 1.5 times larger than the corresponding Gaussian beam profile.
A schematic of the adjusted setup with an AOD is shown in Figure 12. For the first test, the driving frequency was set to the central frequency of 102.5 MHz for both the horizontal and vertical beam deflection directions, as specified in the data sheet [21]. At this frequency, the diffracted laser beam experiences no net deflection, since the two AOD crystals are cut to compensate for each other's angular refraction. The resulting image taken with the camera is shown in Figure 13. In the center of the profile, a flat region resembling the previously observed Top Hat structure is visible. Additionally, two local intensity maxima can be observed within the central region. These maxima are positioned symmetrically around the center and are not visible in the corresponding cross-section plots due to their anti-diagonal positioning. Adjusting the Beam Shaper results in the same behavior as observed previously, as moving the Beam Shaper perpendicular to the beam path causes a lateral shift of the central intensity structure. The profile still exhibits an elliptical structure with a tilt angle of $1.9(11)^\circ$ relative to the horizontal axes. The $1/e^2$ diameters along the principal axes are 6.4(2) µm and 5.6(2) µm, respectively, representing an increase compared to the profile measured without the AOD. This could be caused by truncation at the aperture of the AOD housing, resulting in a smaller beam

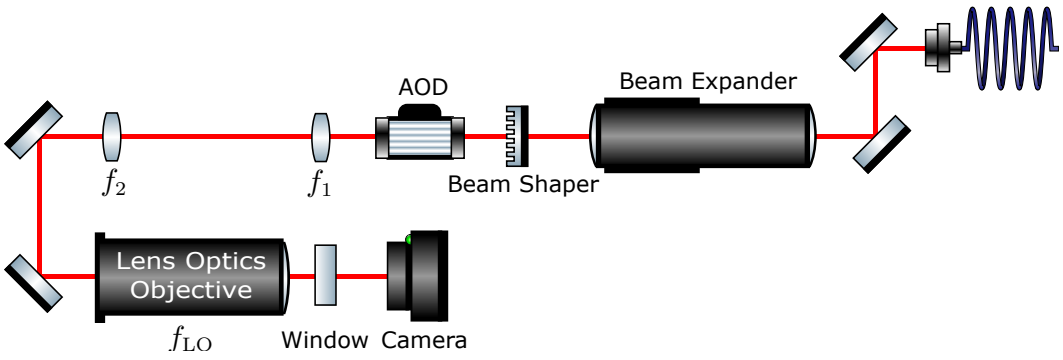


Fig. 12: Schematic setup of the Beam Shaper to generate a Top Hat profile in the focus of a Lens Optics Objective after the deflection with an AOD and the magnification with a $45 - 300$ telescope.

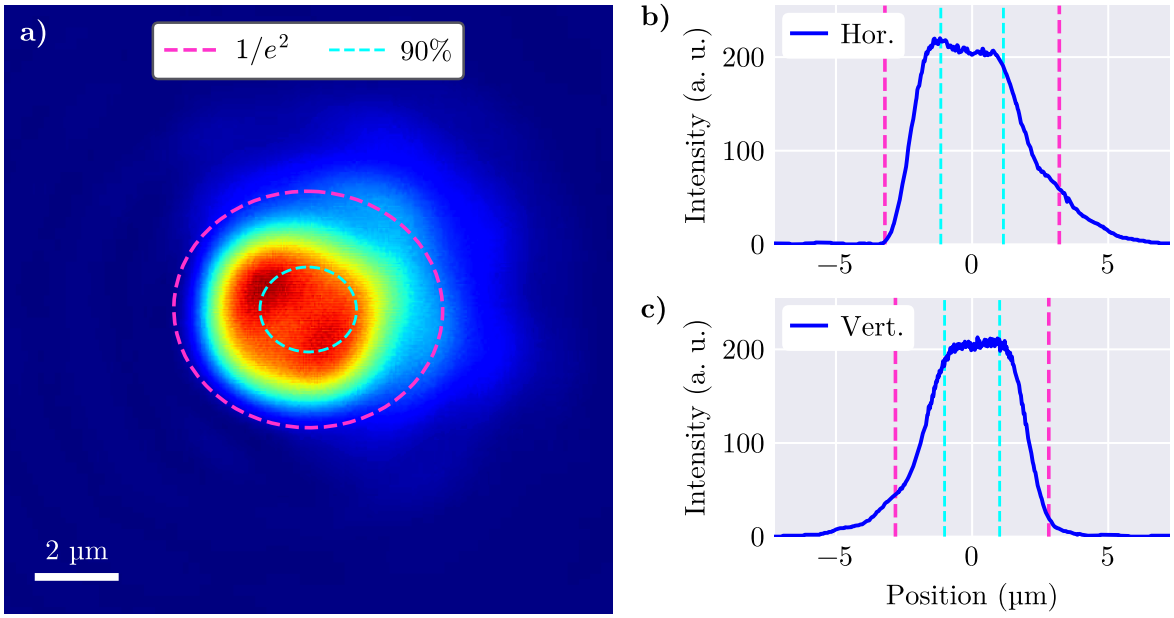


Fig. 13: a) Image of the shaped beam profile deflected with an AOD taken after the Lens Optics Objective and the $45 - 300$ telescope. b)+c) Horizontal and vertical cross-section plots.

size in front of the telescope that is subsequently focused into the atomic plane with a larger waist. The active aperture of the AOD is 7 mm [21]. Thus, a beam with a $1/e^2$ diameter of $d = 3.5$ mm should not be truncated by the aperture. Nevertheless, a cutoff at the housing is visible when observing the beam with an infrared viewer. Other possible causes include optical misalignments or deviations of the actual focal lengths from their specified values at the wavelength used.
The elliptical structure of the beam is now dominated by a distortion along the diagonal axis. Here, the upper right half of the beam shows a large side structure spanning over several micrometers, while the opposite edge shows no such behavior. These distortions are suspected to originate from spatially dependent phase variations introduced by the AOD crystal. As the acoustic wave propagates through the crystal, different parts of the extended Gaussian beam interact with the acoustic wave at different times, potentially introducing a spatially varying phase. Furthermore, the crystal geometry may result in different optical path lengths across the beam, introducing additional phase variations. The resulting beam profile could not be reproduced reliably, even when different optical components were used. Previous attempts to integrate this AOD into the setup of the movable tweezer have already proven difficult due to persistent side structures that could not be eliminated through optical adjustments.
The region enclosed by the 90 % intensity level, indicated in cyan, has diameters of $d_a = 2.30(12)$ µm and $d_b = 2.02(12)$ µm along the principal axes. Within this region, the uniformity of the profile is $U = 4.82(30)$ % and the crosstalk is $\eta = 8.7(19)$ %. Compared to the profile without the AOD, the uniformity value U increases by 25 % and the crosstalk by a factor of three, both due to the dominant distortions.

A proof of concept for generating Top Hat-like profiles with a Beam Shaper at a beam size of approximately 5 µm has been demonstrated, thereby surpassing the manufacturer's specified limit of 10 µm [9]. The size of the resulting profiles are already close to the atomic pitch

of 5.288 µm, with a crosstalk of approximately $3\,\%$. Further reducing the beam size and its resulting crosstalk is prevented by the aperture cutoff at the tube of the Lens Optics Objective. The quality of the beam profile deteriorates after passing through the telescope, as the profile takes on an elliptical shape. After passing through an AOD, the characteristic Top Hat structure can still be observed, while additional distortions further increase the uniformity and the crosstalk of the beam profile. The observed distortions are suspected to originate from the AOD itself, resulting in a beam profile that could not be reproduced reliably, even when different optical components are used.
Accordingly, this approach is rejected, as the resulting intensity profile is unsuitable for single-site addressing of ^{85}Rb atoms. In particular, the crosstalk of approximately $9\,\%$ would significantly compromise the fidelity of single-qubit gate operations.

4 Multi-Tone Simulation for a Top Hat Profile

The second approach for generating uniform intensity profiles uses two perpendicular multi-tone Acousto-Optic Deflectors (AODs) to superimpose multiple deflected laser beams. In the resulting Cartesian coordinate system, each AOD defines one spatial dimension, with the intensity profiles therefore being rectangular and, for the same frequency width in both dimensions, square shaped. Before constructing a new setup with a suitable pair of achromatic lenses with focal lengths f_1 and f_2, as illustrated in Figure 11, theoretical simulations are performed using a Python-based repository [25]. These simulations aim to determine the optimal parameters for generating a uniform intensity profile using two perpendicular multi-tone AODs. The parameters that have to be optimized are:

- The focal lengths of achromatic lenses f_1 and f_2
- The number of tones N_x and N_y in both dimensions of the atomic plane
- The beam waist ω_{in} in front of the achromatic lens f_1, which is then imaged into the atomic plane as ω according to Equation 3.4
- The frequency width W_f of the tones in the AOD, effectively determining the profile width W in the atomic plane according to Equation 3.3
- The amplitudes of each tone $A_{x/y} = \{r_1, r_2, \ldots, r_{N_{x/y}}\}$

Instead of using a Gaussian beam profile as the basis of the multi-tone simulation, an Airy profile is used to evaluate the optical properties of the generated beam profile near the diffraction limit. A first limitation of the free variables is done to limit the number of parameters that have to be optimized. The lenses must be selected to ensure that the entire atomic grid of more than 40×40 atoms can be addressed, according to Equation 3.3. Furthermore, for a new optical setup, a long focal length f_2 is needed for the second achromatic lens due to a long optical path between the Lens Optics Objective and f_2 in the actual experiment. Longer focal lengths are also preferable because they can reduce spherical aberration. The focal lengths of the two achromatic lenses are chosen accordingly as $f_1 = 75\,\text{mm}$ and $f_2 = 750\,\text{mm}$, so a scanning range of $r_{\text{max}} = 227.5\,\mu\text{m}$ or an addressable atomic grid of 43×43 atoms can be generated, according to Equation 3.3. Based on Equation 3.3, the conversion factor c between the frequency width W_f of the AOD and the actual width W of the profile, defined by $W = cW_f$, is $c = 6.316\,\mu\text{m/MHz}$ for this combination of lenses. The beam waist in the atomic plane is on the order of 1 µm, depending on the beam waist in front of the AOD, as given by Equation 3.4. The number of tones for the profile is chosen to be $N_x = 3$ and $N_y = 4$. An optimized set of $N_x = 2$ and $N_y = 3$ tones has an overlap for the spots on its anti-diagonal axis, violating Equation 2.21, resulting in spatial interference and is therefore not suitable for the implementation of a Top Hat profile. Although uniform profiles with $N_x = 4$ and $N_y = 5$ or higher numbers of tones can in principle be generated, they require a larger profile width than the 3×4 configuration, resulting in a larger profile and consequently increased crosstalk. Additionally, the intensity amplitudes of these tones are assumed to be symmetrical around the center and are defined by the ratio of the two outer tones to the inner

ones. With this parametrization, the amplitudes for the tones are given by $A_x = [r_x, 1, r_x]$ and $A_y = [r_y, 1, 1, r_y]$. With these limitations, simulations can be conducted to find suitable parameters for the generation of multi-tone Top Hat profiles that minimize the uniformity U and the crosstalk η. The list of free parameters to be optimized by the simulation is:

- The beam waist ω_{in} in front of the achromatic lens f_1, which is then projected into the atomic plane as ω
- The frequency width W_f of the tones in the AOD, effectively determining the width W of the profile in the atomic plane
- The amplitude ratio of the outer tones to the inner ones in both dimensions r_x and r_y

Since the generated beam profiles have a square structure, the region of interest used to calculate the uniformity and crosstalk is square as well. For the uniformity the region of interest is limited by the profile width W. Larger regions than W would worsen the uniformity for including the falling edge of the profile and are therefore excluded. The region of interest for the crosstalk is limited by a square with the length equal to the pitch of the atomic grid spacing of 5.288 µm. A visualization of these areas is shown in Figure 20.

4.1 Simulation with a Uniform Weighting

The areas defined by the atomic pitch and the tone array are first used as the calculation area within the simulation, which is called the hard metric for further comparison. These calculation areas are chosen on this scale to ensure that the optimized intensity profiles uniformly illuminate the atoms even in the presence of beam-pointing errors between the dipole trap array and the beam profile. This approach thus provides an upper bound for the crosstalk and the uniformity.
A uniform weighting is used for the calculation area, and the values are summed over the corresponding range containing N_h data points:

$$\langle I \rangle_h = \frac{1}{N_h} \sum_{i=1}^{N_h} I_i \quad \Delta I_h = \sqrt{\frac{\sum_{i=1}^{N_h} (I_i - \langle I \rangle_h)^2}{N_h}}. \tag{4.1}$$

To find a suitable set of parameters, the combined score J_h is introduced, which is a weighted sum of uniformity and crosstalk with the weight $\alpha = 0.7$:

$$J_h = \alpha U_h + (1 - \alpha)\eta_h. \tag{4.2}$$

$\alpha = 0.7$ is chosen as a compromise, balancing the importance of uniformity and crosstalk, as $\alpha = 0.9$ underestimates the contribution of crosstalk, whereas $\alpha = 0.5$ assigns it too much weight. For a given set of spot waist and profile width, the amplitude ratios r_x and r_y are optimized to minimize J_h. The waist's upper bound is determined by the aperture of the Lens Optics Objective with 1.7 mm and a magnification of $M = 10$, introduced by the telescope, while a lower bound of 0.8 mm is chosen due to the diffraction limit. The range for the spot waist in the atomic plane is therefore approximately 0.8 µm − 1.7 µm. The range for the frequency width at the AOD is chosen to be 0.25 MHz − 0.45 MHz, which translates to a profile width of 1.58 µm − 2.84 µm. The result of the optimization for J_h is shown in Figure 14. For the overlap of the corner spots according to Equation 2.21, the corresponding area is excluded

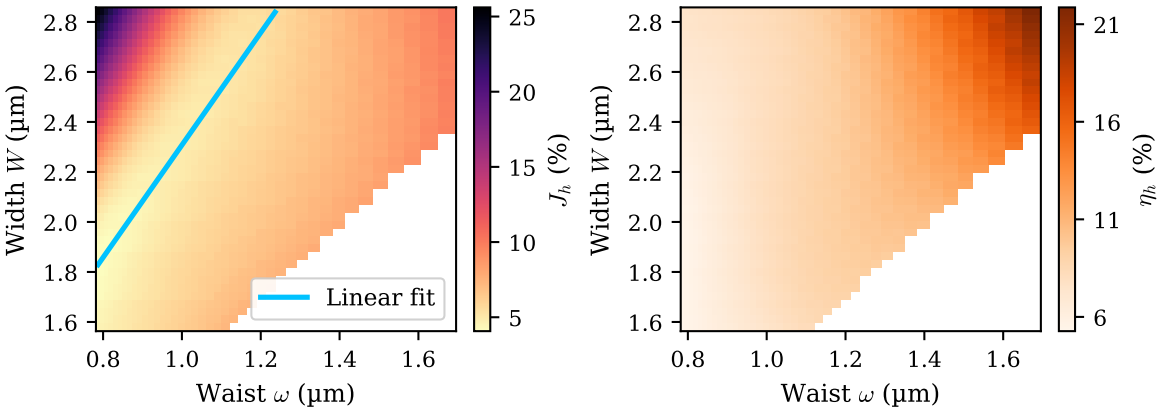


Fig. 14: Left: Two-dimensional plot of the combined score J_h over the spot waist and the profile width, with a linear fit to the path of minimal J_h. Right: Crosstalk η_h plotted for the same variables. For a larger spot waist the crosstalk increases.

from further calculations and left blank due to the expected interference. The crosstalk η_h increases with a rising waist of a single spot, while remaining approximately constant across the profile width. η_h lies in a range of $5\,\%$ to $22\,\%$ and shows no additional side structures. J_h exhibits a minimum region around $5\,\%$, which appears to lie along a straight line for waists between 0.8 µm and 1.3 µm and widths between 1.8 µm and 2.8 µm. In this region, a linear function can be fitted to the data:

$$W(\omega) = a \cdot \omega + b = 2.25(5) \cdot \omega + 0.060(5)\,\mu\text{m}. \tag{4.3}$$

The uncertainties of the fit parameters and the limited resolution of the simulation affect the accuracy of the evaluated quantities. For the hard metric and the potential working points, these quantities are given without explicitly stating their uncertainties. Along the linear fit, J_h lies between $4.1\,\%$ and $5.3\,\%$. This linear relationship between profile width and spot waist indicates that wider beam profiles require larger individual spots to maintain a low uniformity. An ideal model following this description would have a zero constant term. A potential working point can be found for $W = 2.34\,\mu\text{m}$ (0.37 MHz), with a corresponding waist of $\omega = 1.01\,\mu\text{m}$ and the amplitude ratios $r_x = 0.97$ and $r_y = 1.13$. The uniformity of this configuration is $U_h = 3.47\,\%$ and the crosstalk is $\eta_h = 8.17\,\%$. The crosstalk, defined for the eight nearest neighboring profiles, represents an upper bound. The crosstalk decreases with fewer neighboring profiles, with each individual profile contributing approximately $1\,\% - 2\,\%$ to η_h. A major flaw of using a uniform weighting is that it does not take the actual positions of the atoms into account. In this model, the probability to find the atom in the upper right edge of the area is the same as in its center. Using large calculation areas relative to the size of a single spot in the hard metric prevents resolving the crosstalk contributions from the side structures of neighboring Airy profiles. The use of the pitch square always includes multiple secondary maxima and minima within the area used for the crosstalk calculation, revealing no information for a specific set of desirable parameters to actually reduce the crosstalk for each individual atom. To integrate a spatially resolved crosstalk into the simulation, a different simulation approach is used.

4.2 Atom-Weighted Simulation

To take the actual atom position into account, the spatial probability density $|\Psi|^2$ is used to calculate the crosstalk and the uniformity:

$$\langle I \rangle_w = \sum_i I_i \cdot |\Psi|^2 \quad \Delta I_w = \sqrt{\sum_i (I_i - \langle I \rangle_w)^2 \cdot |\Psi|^2}. \tag{4.4}$$

The spatial probability density is parametrized with the atom temperature $T = 17(2)$ µK and a radial trap frequency $\nu_r = 60.4(24)$ kHz [3]. With these parameters, the σ-width is given as $\sigma = 108(8)$ nm and $|\Psi|^2$ can be parametrized according to Equation 2.11.
The Airy profiles used for the individual spots exhibit secondary maxima and minima that can influence the calculated crosstalk but are not resolved using the hard metric. Restricting the effective calculation area according to the spatial probability density of the atoms allows side structures to be resolved more accurately, thereby better minimizing the crosstalk of the generated beam profile visible to the atom.
A simulation using [25] is conducted with the atom-weighted calculation metric. With the same parameter range for the profile width and the spot waist, an optimization is performed for the weighted sum J_w with $\alpha = 0.7$:

$$J_w = \alpha U_w + (1 - \alpha)\eta_w, \tag{4.5}$$

The results are shown in Figure 15. For the overlap of the corner spots, according to Equation 2.21, the corresponding area is left blank due to the expected interference. J_w lies between $0.5\,\%$ to $3.3\,\%$. No linear fit can be found for its minimum, since several minimum regions for J_w form vertical stripes across the profile width W. These stripes result from the crosstalk η_w and are caused by secondary maxima of the Airy profiles. Increasing the waist of the single Airy spots shifts the positions of the secondary maxima and minima of each spot in the same direction relative to the center. This results in a stripe-shaped minimum region for the crosstalk η_w, as the secondary intensity minima of numerous spots in neighboring profiles coincide with the atomic position in the center. The first stripe is caused by the minima of vertically and horizontally neighboring profile spots with $\omega \approx 0.85$ µm.The second stripe is caused by the

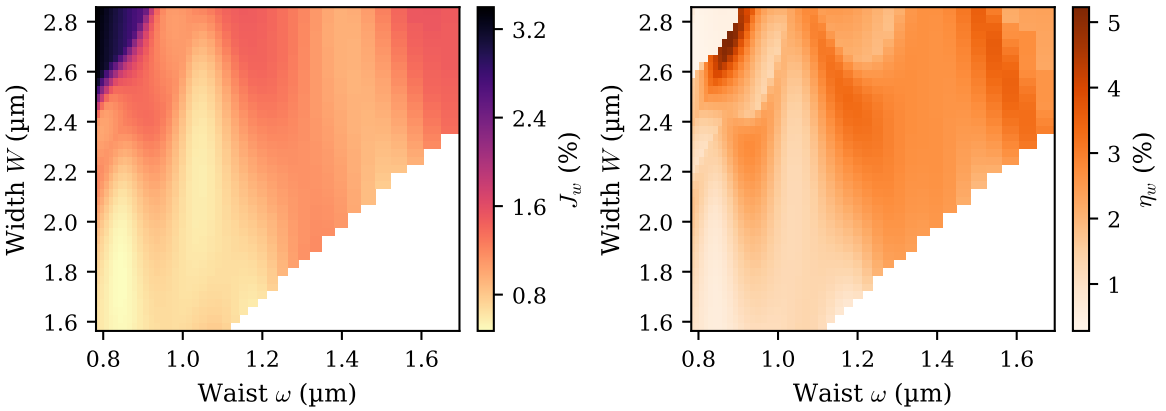


Fig. 15: Left: Two-dimensional plot of the combined score J_w over the spot waist and the profile width. No linear relationship can be found for J_w. Right: Crosstalk η_w plotted for the same variables. Vertical stripes corresponding to the local minima of η_w are visible.

minima of diagonally neighboring profile spots with $\omega \approx 1.1\,\mu m$. Along one stripe, the crosstalk stays almost constant. Increasing the profile width moves the closest spots of neighboring beam profiles closer to the central profile and the atom, while increasing the distance for the outer ones, resulting in a net change of zero for the crosstalk. For larger spot waists, no stripes are visible. The crosstalk the atom experiences depends mostly on the waist of the central structure of the Airy profile, since this waist determines the pattern of local minima and maxima experienced by the atom. η_w depicts discontinuities up to $5\,\%$ for large profile widths. In contrast to the hard metric, this approach is found to fulfill its purpose of resolving the crosstalk η_w caused by the side structures of the Airy profiles. The working point identified for the hard metric at a profile width of $W = 2.34\,\mu m$ is also promising when evaluated using the atom-weighted metric. The corresponding waist is $\omega = 1.05\,\mu m$, with amplitude ratios of $r_x = 1.007$ and $r_y = 0.944$. For these parameters, the atom-weighted uniformity is $U_w = 0.272(3)\,\%$, and the crosstalk is $\eta_w = 1.36(3)\,\%$. The optimization for each parameter set yields cases in which the crosstalk and the atom-weighted uniformity are minimized for amplitude ratios close to the imposed optimization limits ($r_{x,y} \in [0.1, 10]$). Such amplitude ratios are no longer a realization of a 3×4 profile but rather of, for example, a 2×2 profile, violating a basic assumption of the simulation.
An example of such a local minimum is the case $r_x = 2$ and $r_y = 0.18$ with a waist of $\omega = 1.11\,\mu m$ and a width of $W = 2.46\,\mu m$. The uniformity in this case is $U_w = 2.3(1)\,\%$ and the crosstalk is $\eta_w = 2.4(1)\,\%$. At the same time, the uniformity of $U_h > 33\,\%$ calculated with the hard metric reveals a beam profile far from a Top Hat, showing uniform intensity only in the central region experienced by the atom. The resulting beam profile in this case is shown in Figure 16. The power density at the atom's location can vary significantly depending on the beam pointing of multi-tone beam profile and dipole trap relative to each other. Thus, such a case loses the advantage of a uniform beam profile.
The observed discontinuities in the plot of η_w in Figure 15 are found to reveal more such impractical optimization results in which the optimized amplitudes reach the optimization limits.

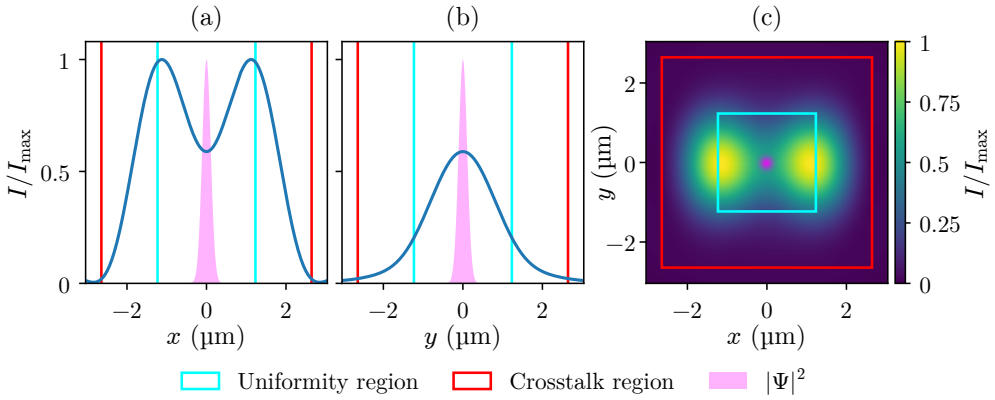


Fig. 16: (a) Horizontal cross-section plot of the uneven beam profile, showing a uniform area for the atom at the minimum between the outer spots. (b) Vertical cross-section plot of the uneven beam profile, revealing a small even region visible to the atom. (c) Two-dimensional plot of the beam profile from the weighted optimization simulation. No Top Hat-like structure is visible.

4.3 Combination of Both Simulations

The first approach, using the hard metric to find parameters for large, uniform intensity profiles, does not resolve the crosstalk contributions of secondary maxima of neighboring Airy profiles. The atom-weighted approach provides this missing information but can favor solutions that do not preserve the desired large, uniform intensity profile. To account for both requirements, the two approaches are combined to identify a set of parameters that balances the profile's uniformity and crosstalk on both scales.
The uniformity and the crosstalk differ by up to an order of magnitude between the two calculation metrics, as previously determined for the potential working point. Therefore, the mean value alone is not a suitable way of combining them. The combined crosstalk η_c and uniformity U_c, minimized during the optimization of the four parameters, consist of the respective mean values of the previous two metrics and an additional penalty term. This term is defined as the absolute difference between the two metrics, weighted by a factor of $\chi = 0.75$:

$$U_c = \frac{1}{2}(U_w + U_h) + \chi|U_w - U_h|, \tag{4.6}$$

$$\eta_c = \frac{1}{2}(\eta_w + \eta_h) + \chi|\eta_w - \eta_h|. \tag{4.7}$$

The weighting factor $\chi = 0.75$ is chosen as a compromise between assigning greater weight to the penalty term and preventing it from dominating the overall metric. This ensures that when the two metrics evolve in opposite directions, with one improving while the other deteriorates, the combined metric still increases if the deterioration outweighs the improvement. The combined score J_c for the optimization is given by a weighted sum with the previous choice of $\alpha = 0.7$:

$$J_c = \alpha U_c + (1 - \alpha)\eta_c. \tag{4.8}$$

For each pair of profile width and spot waist, the amplitudes are optimized to minimize J_c. The results of the optimization are shown in Figure 17. Based on Equation 2.21, the region

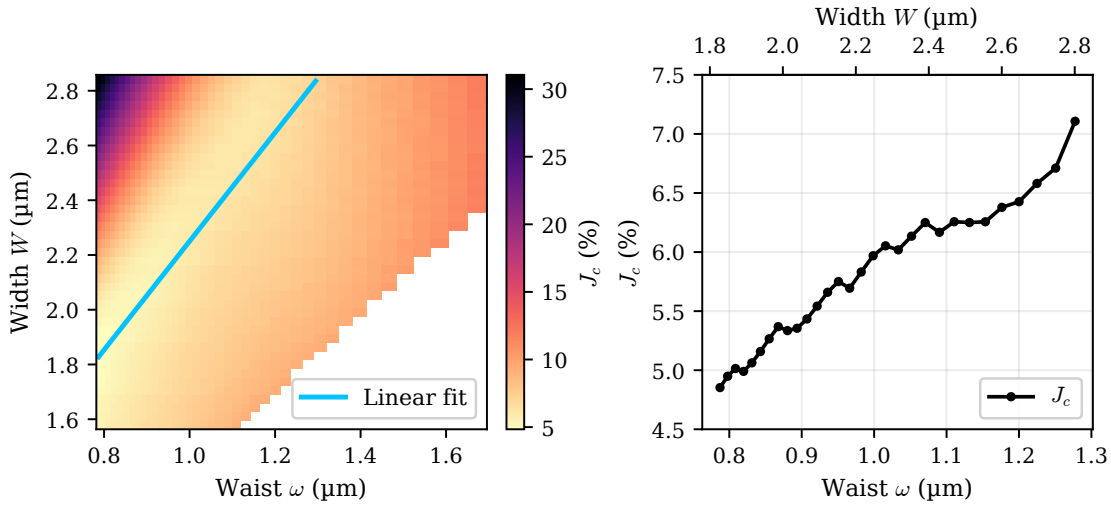


Fig. 17: Left: Plot of the combined score J_c as a function of the spot waist and profile width, with a linear fit along the path of minimal J_c. The excluded region of overlapping spots is left blank. Right: Cross-section plot of J_c along the linear fit.

in which the diagonal spots overlap is left blank for J_c and excluded from further evaluation. Based on the results of this optimization, a linear function can be fitted to the path of minimal J_c:

$$W(\omega) = a \cdot \omega + b = 1.98(3) \cdot \omega + 0.27(3)\,\mu\text{m}. \tag{4.9}$$

Compared to the hard metric, the slope of the linear fit decreases by 12 %, while the constant term increases by more than a factor of four. Along the minimum path itself, J_c lies between 4.5 % and 7.5 %. The curve of J_c shows a zigzag pattern. This pattern may arise from the finite simulation resolution, as the linear fit is evaluated at discrete grid points, leading to jumps between neighboring points.
An overview of the optimized uniformities U_h and U_w, crosstalk values η_h and η_w and their cross-section plots along the linear fit of J_c is shown in Figure 18. For both calculation metrics, the uniformity increases with a larger profile width at a fixed spot waist. This is attributed to the larger spacing between individual spots, which leads to local intensity minima within the central region experienced by the atom. A reduced profile width at a fixed spot waist results in Gaussian-like profiles. Both the weighted uniformity U_w and the hard uniformity U_h show a

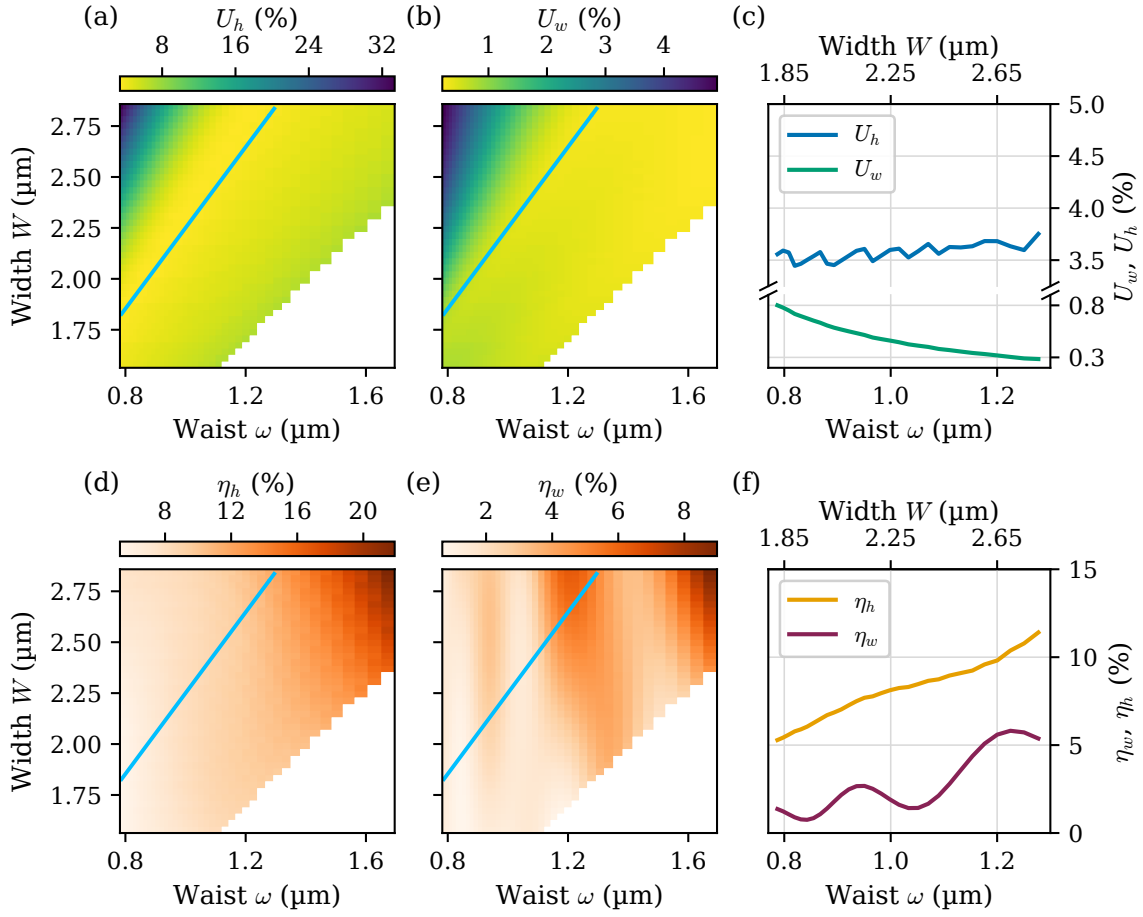


Fig. 18: (a)+(b) Two dimensional plots of the uniformities U_h and U_w, with the linear fit of J_c. Both plots show a minimum path following the linear fit. (c) Cross-section of U_h and U_w along the linear fit. U_h is approximately constant, while U_w decreases. (d)+(e) Two dimensional plots of the crosstalk values η_h and η_w, with the linear fit of J_c. (f) Cross-section of η_h and η_w along the linear fit. η_w depicts a stripe structure, while η_h increases with a rising waist.

minimum path along the linear fit of J_c given by Equation 4.9, as indicated in the plot. For the combined approach, this linear relationship between profile width and spot waist confirms that wider beam profiles require larger individual spots to maintain low uniformity. A second linear path of minimal U_w can be found near the blank overlap region. No linear fit is conducted for this region, since these data points are excluded from the evaluation. Along the linear fit, the hard uniformity remains approximately constant at $U_h \approx 3.6\,\%$ but depicts a zigzag pattern. In contrast, the weighted uniformity U_w decreases monotonically without exhibiting a zigzag structure. The minimum of U_w is reached at the maximum simulated profile width of $W = 2.80\,\mu\text{m}$. The resulting profile, consisting of relatively large individual spots, exhibits only small intensity variations across the region visible to the atoms, thereby minimizing U_w. The uniformity alone is not sufficient to identify a suitable set of parameters for the creation of a multi-tone profile, as the crosstalk must also be considered.

The hard crosstalk η_h for the combined calculation metric depicts the same structure as for the use of just the hard calculation metric. With increasing spot waist, η_h rises and ranges from $5\,\%$ to $22\,\%$. The weighted crosstalk η_w ranges from $0.7\,\%$ to $9\,\%$ and shows a stripe-shaped structure of local minima, which arise from the secondary maxima and minima of the Airy profiles. The first stripe is caused by the minima of vertically and horizontally neighboring profile spots, while the second stripe along the waist axis is caused by the diagonally neighboring profile spots. For higher-order minima, the overlap between the minima of different spots increases, and they start to reach the center simultaneously, causing the stripe to flatten out and bend. The observed discontinuities in Figure 15 do not occur for η_w calculated with the combined metric. The absence of discontinuities demonstrates the advantage of the combined calculation metric, which generates large, uniform beam profiles while resolving the crosstalk experienced by the atom due to secondary maxima of the Airy spots. The difference in the effective calculation area of the two crosstalk metrics results in an offset between the two crosstalk values. Along the linear fit, η_h increases monotonically, showing small bends at the positions of local maxima and minima of η_w. The local extrema of the Airy spots, visible in η_w, cannot be resolved in η_h due to the large area over which it is calculated. Apart from this oscillating pattern, η_w increases along the linear fit of J_c.

In Figure 19, an overview of the optimized amplitude ratios r_x and r_y, together with their

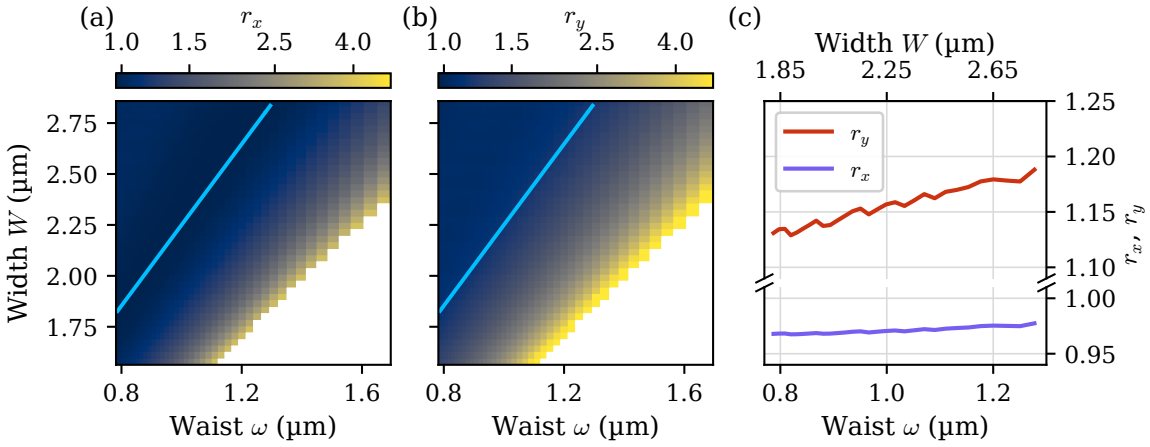


Fig. 19: (a)+(b) Two-dimensional plots of the amplitude ratios r_x and r_y, with the linear fit of J_c. Near the linear fit, both amplitude ratios are approximately one. (c) Cross-section of r_x and r_y along the linear fit. r_x remains approximately constant, while r_y shows a slight linear increase.

cross-sections along the linear fit of J_c, is shown. Both amplitude ratios increase up to $r_{x,y} > 4$ near the excluded evaluation area, indicating this region belongs to a uniform 2×2 multi-tone profile or other configurations. Apart from this region, both amplitude ratios remain close to 1.0 across the entire range of spot waists and profile widths, with $r_y > r_x$. Along the linear fit of J_c, r_y increases from 1.13 to 1.19, exhibiting the previously observed zigzag pattern, while r_x remains nearly constant at $r_x \approx 0.97$.
A suitable working point is again found for $W = 2.34\,\mu\text{m}$, with a corresponding waist of $1.04\,\mu\text{m}$ according to the linear fit, which translates to $\omega_{\text{in}} = 1.28\,\text{mm}$ in front of the AOD. The amplitude ratios are $r_x = 0.97$ and $r_y = 1.16$. With the hard metric, the uniformity is $U_h = 3.56\,\%$ and the crosstalk is $\eta_h = 8.40\,\%$. In the atom-weighted case, the uniformity is $U_w = 0.42(1)\,\%$ with a crosstalk of $\eta_w = 1.42(3)\,\%$. The uncertainties of the weighted values result from the uncertainties of the temperature T and the trap frequency ν_r. Compared to the working point discussed above for the hard and the atom-weighted metrics, the crosstalk and uniformity have slightly increased ($\sim 0.2\,\%$) within the margin of the uncertainty. A visualization of the resulting multi-tone Top Hat profile for these parameters is shown in Figure 20. The horizontal cross-section of the beam profile exhibits an almost flat intensity profile up to the edge of the uniformity region, while the vertical cross section reveals a small dip in the central intensity. In both cases, the width of the spatial probability density is much smaller than the beam profile and its substructure, resulting in the optimized low uniformity U_w.

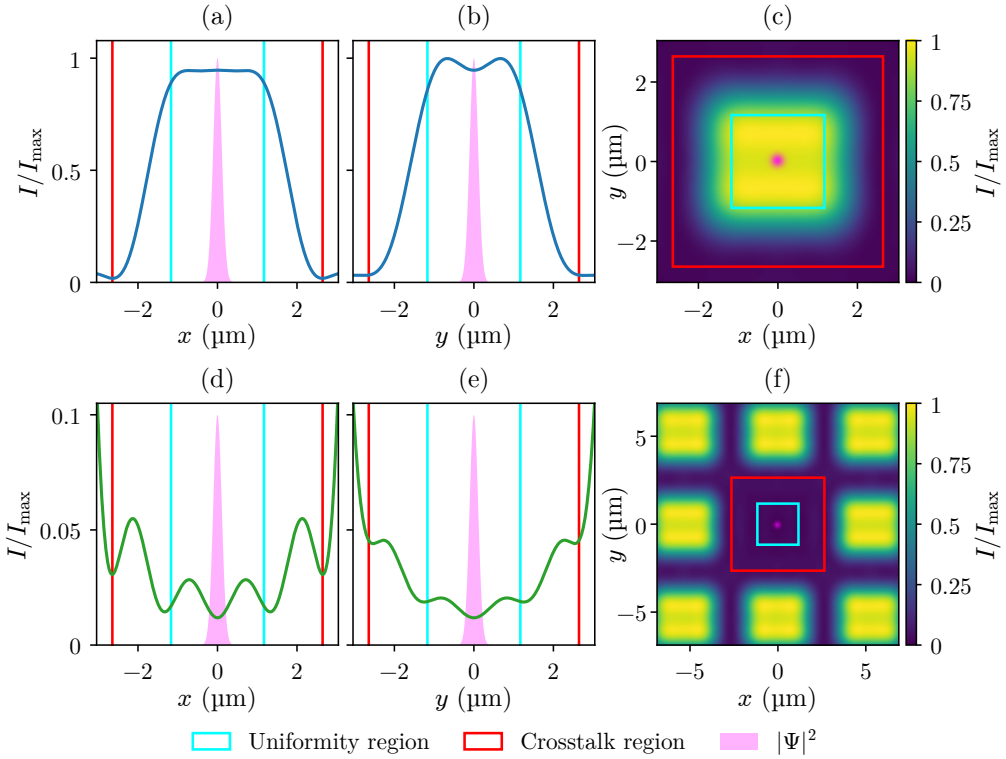


Fig. 20: (a)+(b) Horizontal and vertical cross-section plots of the optimized Top Hat beam profile for the chosen working point, showing a uniform area for the atom in the center. (c) Two-dimensional plot of the optimized Top Hat beam profile and the regions of evaluation. (d)+(e) Horizontal and vertical cross-section plots of the light of neighboring profiles illuminating the center. (f) Two-dimensional plot of the eight neighboring profiles, with the regions of evaluation.

Cross-section plots of the light from the eight neighboring beam profiles illuminating the center are shown in Figure 20 (d) and (e). The intensity, responsible for the crosstalk, is normalized to the maximum intensity of the central beam profile. In the center, the neighboring light has a global minimum of approximately $1.5\,\%$ for both directions. From the center outward, the intensity increases up to $5\,\%$ of the maximum intensity at the edges of the crosstalk region. In both cases, the intensity depicts a series of local minima and maxima. For the horizontal direction, shown in (a), the ratio of maximum and minimum is more pronounced. Beyond the crosstalk region, the rising intensity indicates the neighboring beam profile.
Using 4×5 tones instead would require a much larger profile width for a uniform profile with a fixed waist of $\omega = 1\,\mu\text{m}$, increasing the atom-weighted crosstalk. For $\omega = 1\,\mu\text{m}$ and a profile width of $W = 3.16\,\mu\text{m}$, the atom-weighted crosstalk would be $\eta_w = 3.0\,\%$, roughly twice the value at the working point of the 3×4 profile, significantly distorting single-qubit gate operations. Instead, the spot waist would have to be reduced to 0.65 µm at a constant profile width of $W = 2.34\,\mu\text{m}$ to generate a uniform beam profile with uniformity and crosstalk values comparable to those of the 3×4 working point. Such a small spot waist would require an incoming beam waist of $\omega_{\text{in}} = 2.06\,\text{mm}$ in front of the AOD. After the telescope, the magnified beam would be truncated at the tube aperture of the LENS-Optics objective, making this spot waist impractical for the intended setup.

4.3.1 Beam Pointing of the Top Hat Profile Relative to the Dipole Trap

If the dipole trap, and thus the atom, is displaced from the center of the beam profile, for example due to beam-pointing errors caused by long optical paths of the laser systems used, the local intensity and the crosstalk experienced by the atom can differ from their values at the center. This can lead to variations in the excitation probability and reduce the fidelity of the gate operations. The main advantage of a Top Hat profile is its low uniformity, which allows for small relative displacements between the dipole trap and the center of the Top Hat beam profile without significantly compromising the fidelity of gate operations.
To evaluate the advantage for the optimized multi-tone profile, the atom's position relative to the Top Hat beam profile is varied, and the resulting changes in the uniformity and the crosstalk are calculated. Only the atom-weighted case is evaluated, since changes of the uniformity and the crosstalk for the hard metric are mainly geometric. The atom is shifted outward from the center up to the outermost spot in the horizontal, vertical and diagonal directions.
The resulting changes in uniformity and crosstalk are shown in Figure 21. The uniformity stays close to $1\,\%$ over the inner two-thirds of the array in all three directions. When the atom reaches the outermost spot, the falling edges of the profile are included in the calculation, and U_w rises to $3.8\,\%$ horizontally and $6.9\,\%$ vertically. The two axes differ because the profile consists of different numbers of tones per axis. The 3×4 grid has a constant spacing in both directions but is positioned such that the horizontal axis through its center runs between the spots rather than through them. This means the atom is moved horizontally within an intensity valley while passing through the spot centers vertically. Additionally, the amplitudes for r_x and r_y differ, which causes further differences in the uniformity. Along the diagonal axis, U_w remains the lowest of the three directions, staying below $2.2\,\%$ until close to the corner, and reaches a local minimum of $0.81\,\%$ at $r = 0.94\,\mu\text{m}$. At the corner of the profile, at $r = 1.65\,\mu\text{m}$, U_w reaches its maximum of $9.0\,\%$, since two falling edges are included in the calculation of the standard deviation. In all cases, the uniformity stays below $5\,\%$ for displacements up to 1 µm from the center. For the expected maximum displacement from the profile center, corresponding to $3\sigma \approx 0.3\,\mu\text{m}$, the uniformity can be considered stable.

The crosstalk increases in all three directions and reaches a maximum of $3.25\,\%$ at $r = 0.7\,\mu\text{m}$ horizontally, $2.38\,\%$ at $r = 0.85\,\mu\text{m}$ vertically and $3.51\,\%$ at $r = 0.94\,\mu\text{m}$ diagonally. The maximum of η_w is caused by the secondary maxima of the neighboring profiles. Once the atom is shifted further, the crosstalk decreases again toward the edge of the beam profile.

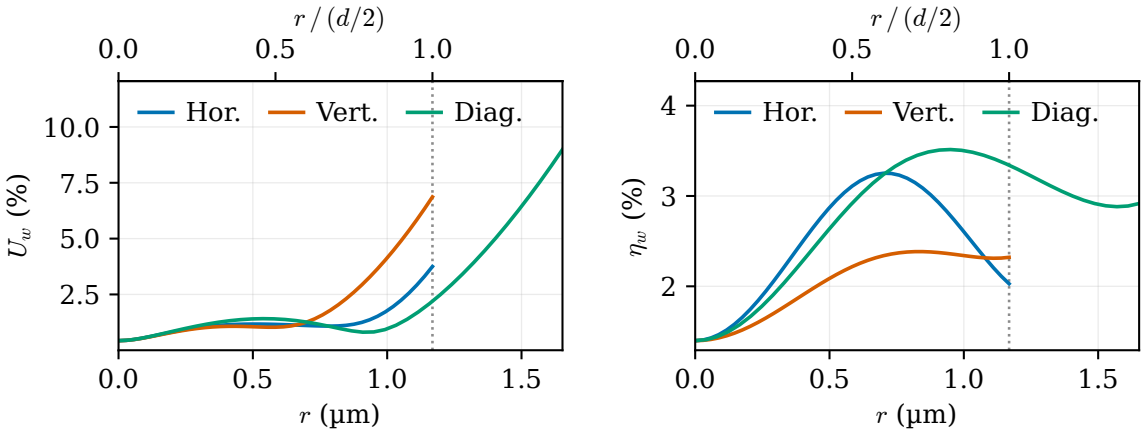


Fig. 21: Calculated uniformity U_w and crosstalk η_w for the atom displaced from the profile center along the horizontal, the vertical and the diagonal directions up to the position of the outermost spot.

4.4 Comparison with a Single Spot

Another method for single-site addressing of ^{85}Rb is the use of a single Airy profile. This has the advantage of leaving just the initial beam waist as the only parameter for the optimization while eliminating interference concerns. The waist, as the only free parameter of the Airy profile, is optimized in a simulation [25]. The hard and the atom-weighted calculation metrics are used to give an estimation of a good set of potential waists. The calculation area of the hard metric defined by the tone square has to be replaced with a fixed area, since only one tone is used per profile. The area chosen for this is a circle with a radius of 1 µm. Since the tube of the Lens Optics Objective has a diameter of $d = 34\,\text{mm}$, the maximum initial waist is set to $\omega_i = 1.7\,\text{mm}$, which translates to 0.8 µm within the atomic plane. The maximum value for the waist within the atomic plane is limited by the pitch of the atomic grid and is therefore set to 2.5 µm, which translates back to an initial waist of $\omega_{\text{in}} = 0.54\,\text{mm}$. For a better comparison, two calculation areas are used for the calculation of the crosstalk of the hard metric. The first, η_h, uses the same definition as before with a pitch-sized square, while the second, $\eta_{h,\text{circ}}$, uses the same circular area with a radius of 1 µm as the uniformity. The uniformity and the crosstalk of the hard calculation metric for different waists of a single Airy profile are shown in Figure 22.

The uniformity U_h decreases with increasing waist ω, as the falling edges of the broader beam profile move outside the calculation area. Consequently, the standard deviation decreases, resulting in a lower uniformity. On the other hand, the crosstalk increases in both cases with a rising waist, since neighboring secondary maxima extend into the calculation area. The pitch square crosstalk η_h is more than three times larger than the circular crosstalk $\eta_{h,\text{circ}}$ due to the greater area of included neighboring light. The smaller the calculation area, the better the

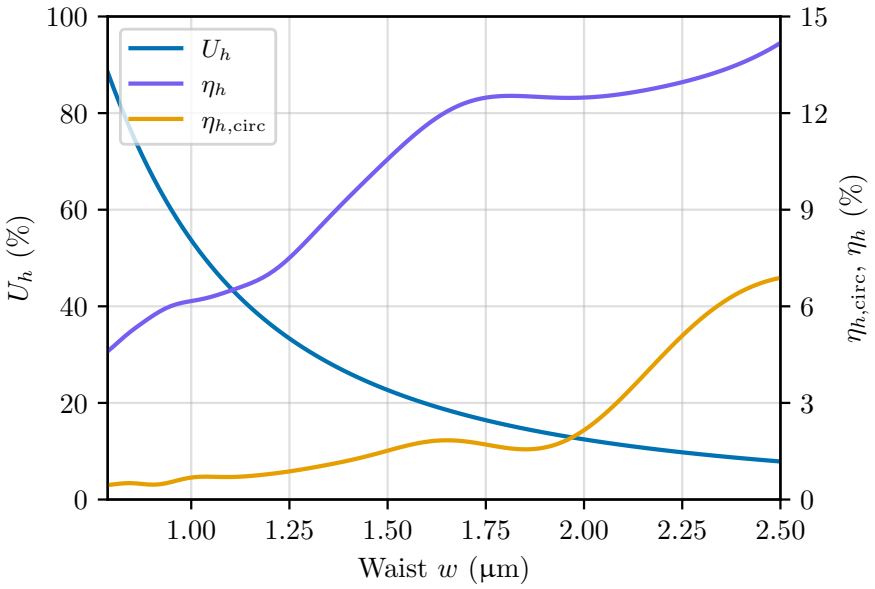


Fig. 22: Uniformity U_h and crosstalk η_h calculated with the hard metric as a function of the waist of a single Airy profile. For the crosstalk, the two definitions with the pitch square and the circle with a radius of 1 µm are used.

side structures of the Airy profiles can be resolved. Both crosstalk curves show a plateau from 1.75 µm to 2 µm, where the uniformity has already fallen below 20 %. The crosstalk $\eta_{h,\mathrm{circ}}$ exhibits a local minimum at a waist of approximately $\omega = 1.9$ µm. After the plateau, both crosstalk values increase, reaching up to $\eta_h = 14$ % and $\eta_{h,\mathrm{circ}} = 7$ %. A suitable working point can be found for $\omega = 1.9$ µm. With the hard metric, the uniformity is $U_h = 13.8$ %, and the

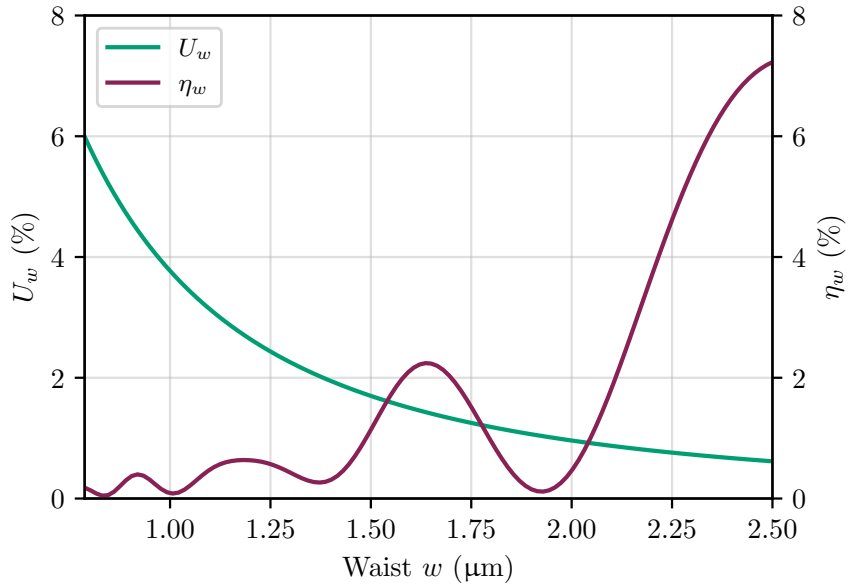


Fig. 23: Uniformity U_w and crosstalk η_w weighted with the spatial probability density of the atom as a function of the waists of a single Airy profile. For $\omega = 1.9$ µm, a suitable working point for experimental use is found.

crosstalk values for both calculation areas are $\eta_h = 12.5\,\%$ and $\eta_{h,\mathrm{circ}} = 1.62\,\%$.
The uniformity and the crosstalk weighted with the spatial probability density of the atom are shown in Figure 23. Starting from $6.7\,\%$, the uniformity U_w decreases with increasing waist in the atomic plane and remains below $5\,\%$ over almost the entire range. The crosstalk shows the oscillating pattern already observed for the multi-tone approach in Figure 18, which results from the minima and maxima of the neighboring Airy profiles. For the entire scanned range of the waist in the atomic plane, the crosstalk stays below $8\,\%$. Furthermore, a crosstalk η_w larger than $2.5\,\%$ is only reached for waists above 2.1 µm. The working point $\omega = 1.9\,\mu\mathrm{m}$, selected based on the hard metric, also provides suitable values for the weighted case, with a uniformity of $U_w = 1.06(17)\,\%$ and a crosstalk of $\eta_w = 0.15(5)\,\%$, with it being minimized due to the side structures of neighboring Airy profiles. The other local minima of the crosstalk are discarded as potential working points due to their higher uniformities. Therefore, the last minimum in the specified range is selected, as this waist exhibits the lowest uniformity.
A visualization of the Airy profile for the chosen working point is shown in Figure 24. With this choice for the waist, the secondary maxima of the Airy profile lie outside the calculation area of the crosstalk (red) and therefore contribute light to the neighboring sites, increasing the hard crosstalk η_h. The atom in the profile center receives almost no light from neighboring profiles, since the global intensity minimum coincides with the maximum of the spatial probability density.

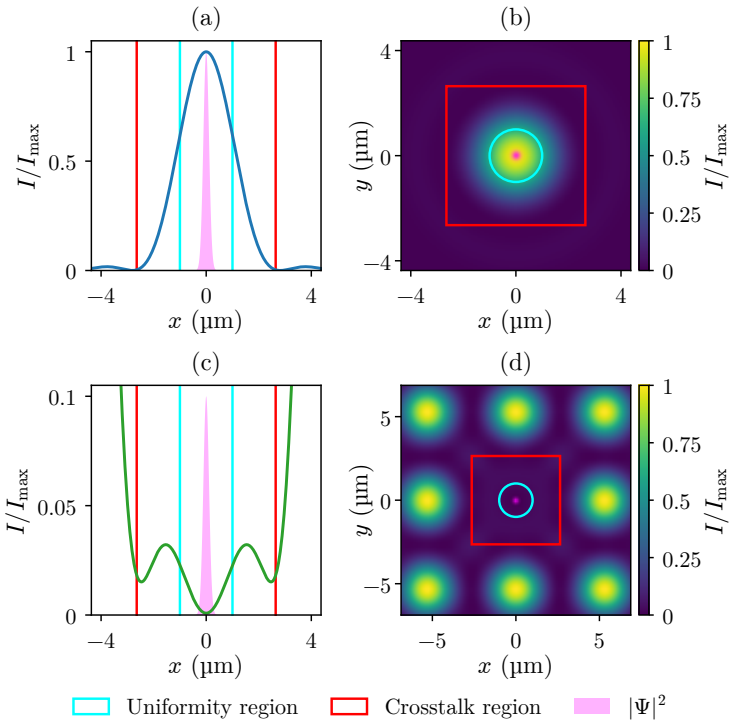


Fig. 24: The calculation area of the uniformity in the plot is given by a circle with a radius of 1 µm. For the crosstalk, the calculation area shown is the pitch square. (a) Cross-section of an Airy profile with $\omega = 1.9\,\mu\mathrm{m}$, together with the areas used for the calculation of uniformity and crosstalk. (b) Two-dimensional plot of one Airy profile with the calculation areas. (c) Cross-section for light of eight neighboring profiles illuminating the central region. (d) Two-dimensional visualization of the eight neighboring Airy profiles responsible for the crosstalk.

4.4.1 Beam Pointing of the Single Spot Relative to the Dipole Trap

A major disadvantage of Airy spots is the non-uniform intensity distribution along the flanks of the central maximum. Consequently, even small displacements of the dipole trap, and thus the atom, relative to the center of the Airy profile can alter the local intensity illuminating the atom. This can reduce the fidelity of the performed gate operations. A displacement can be taken into account in the simulation of a single Airy spot by moving the atom within the profile and calculating the uniformity and the crosstalk at each position, as previously done for the multi-tone profile. Only the atom-weighted case is considered, since changes in the uniformity and the crosstalk with respect to the atom position are of interest.
For an Airy profile on a Cartesian grid, the uniformity is identical for vertical, horizontal and diagonal displacements due to the rotational symmetry of the profile. Along the diagonal of the spot grid, the crosstalk differs slightly at larger waists because of the Cartesian arrangement of the neighboring spots. Since this difference is small, only one displacement direction is considered in the following To evaluate the effect of atomic displacements, simulations are performed for several waists of the Airy profile, resulting in two-dimensional maps of the uniformity U_w and the crosstalk η_w as functions of the displacement r and the spot waist ω. The waist is varied from 0.75 µm to 2.5 µm, while the atom is displaced outwards by up to 2.5 µm from the center. In Figure 25 the results of the simulation are shown. For displacements smaller than the waist, the uniformity U_w increases monotonically for all waist values, reaching values up to $U_w = 10\,\%$. For displacements larger than the waist, two lines of local maxima and one line of local minima are visible. These lines are caused by the secondary maxima and minima of the Airy profile. When the atom is placed near a secondary minimum, the mean intensity averaged over its spatial probability density is very low, such that even small intensity variations result in a high uniformity. Near a secondary maximum, the opposite effect occurs, leading to local minima in the uniformity.
For a fixed displacement, the crosstalk η_w shows an oscillating pattern of minima and maxima as a function of the waist, which results from the secondary maxima and minima of the Airy profile, as already observed in Figure 23. For larger displacements, this effect becomes less pronounced and occurs for different waists, as the light from other neighboring Airy spots starts to illuminate the atom. The crosstalk reaches values of $\eta_w = 10\,\%$ and above for displacements smaller than the waist of the Airy profile. For larger displacements, the crosstalk exhibits lines

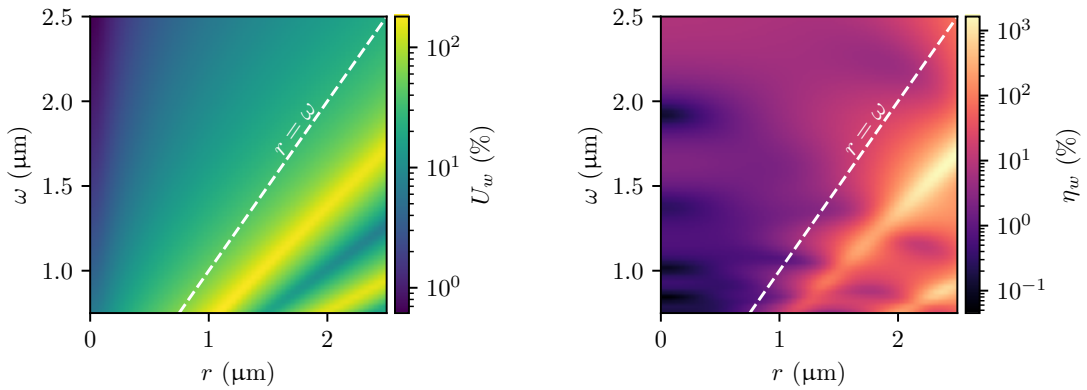


Fig. 25: Two-dimensional plots of the uniformity U_w and the crosstalk η_w as functions of the displacement r of the atom from the center and the waist ω of the Airy profile. The line $r = \omega$ is marked in white.

of local maxima resulting from the secondary minima of the Airy profile. On these lines of local maxima, the intensity of the central spot is very low, while the eight neighboring spots still illuminate the atom. To further assess the suitability of the previously introduced working point, the changes in uniformity and crosstalk caused by atomic displacement are investigated for this case. For the selected waist of $\omega = 1.9\,\mu\text{m}$, the corresponding cross-sections of Figure 25 are evaluated and the changes of uniformity and crosstalk are shown in Figure 26. The atomic displacement increases both the uniformity and crosstalk, which reach up to $27.8\,\%$ and $19.8\,\%$, respectively. For shifts of up to 0.5 µm, the crosstalk remains below $1\,\%$, while the uniformity deviation exceeds the $5\,\%$ mark. At the displacement of 1 µm previously considered for the multi-tone profile, the Airy profile exhibits a uniformity of $U_w = 10.8(12)\,\%$ and a crosstalk of $\eta_w = 3.7(4)\,\%$. Therefore, a limit of 0.5 µm for the misalignment of the trap array relative to the Raman tweezer array can be defined.

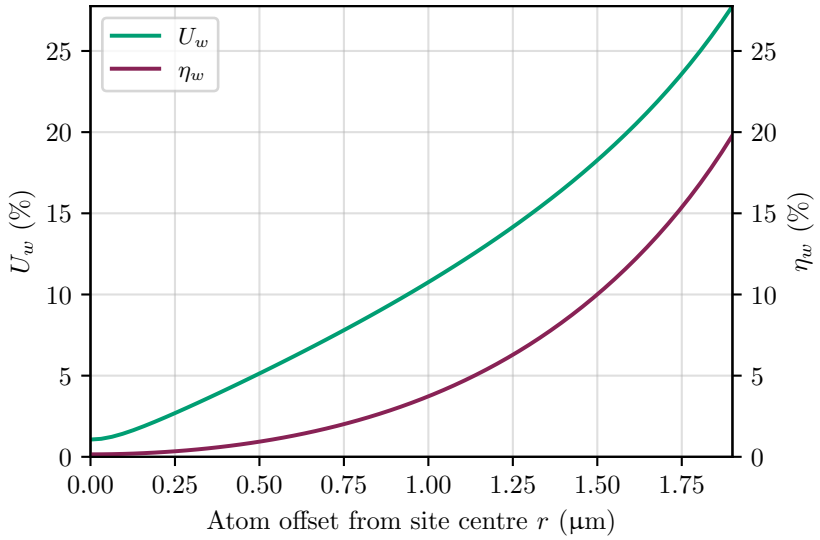


Fig. 26: Uniformity U_w and crosstalk η_w for an atom displaced vertically from the center up to the waist $r = \omega = 1.9\,\mu\text{m}$ of the Airy profile.

4.5 Beating and Crest Factor

When light fields with similar frequencies interfere, the resulting intensity exhibits temporal oscillations at frequencies determined by the differences between the individual field frequencies. This phenomenon is referred to as beating. For the chosen working point of the multi-tone profile, the frequency differences between the individual spots generated by the AODs lie in the kilohertz range. The resulting beating is simulated to evaluate its impact on the intensity at the atom position. The time-dependent uniformity, as a direct way to measure the impact of the resulting intensity variations, determines the temporal stability of the multi-tone Top Hat profile and its suitability for quantum gate operations.
For the 3×4 multi-tone profile, the tone frequencies are determined by the frequency width of 0.37 MHz and the frequency offset of 100 MHz of the two perpendicular AODs. The resulting tone frequencies are:

$$f_{x,i} \in \{100.000,\ 100.185,\ 100.370\}\,\text{MHz}, \tag{4.10}$$

$$f_{y,j} \in \{100.000,\ 100.123,\ 100.247,\ 100.370\}\,\text{MHz}. \tag{4.11}$$

Accordingly, the frequency spacings are :

$$\Delta f_x = \frac{W_f}{2} = 185\,\text{kHz}, \qquad \Delta f_y = \frac{W_f}{3} = 123.3\,\text{kHz}. \tag{4.12}$$

The greatest common divisor of the frequency spacings, which determines the fundamental beating frequency and its corresponding period, can be calculated with the Least Common Multiple (LCM) of the maximum spot indices and is given by:

$$f_0 = \frac{W_f}{\text{LCM}(2,3)} = \frac{W_f}{6} = 61.67\,\text{kHz}, \qquad T_0 = \frac{1}{f_0} = 16.22\,\mu\text{s}, \tag{4.13}$$

so that $\Delta f_x = 3f_0$ and $\Delta f_y = 2f_0$ hold. Using the conversion factor $c = 6.316\,\mu\text{m/MHz}$, the spots have pitches of $p_x = 1.17\,\mu\text{m}$ and $p_y = 0.78\,\mu\text{m}$ on the rectangular lattice with the full width of $W = 2.34\,\mu\text{m}$. The order $k_{i,j}$ and the frequency $f_{i,j}$ of a spot (i,j), with $0 \le i \le 2$ and $0 \le j \le 3$, are given by:

$$k_{i,j} = 3i + 2j, \qquad f_{i,j} = 200\,\text{MHz} + k_{i,j} \cdot f_0. \tag{4.14}$$

The resulting spot orders on the 3×4 grid range from $k_{0,0} = 0$ to $k_{2,3} = 12$ and form the pattern shown in Table 4.1. The twelve spots of the 3×4 grid fall into eleven spot groups $\mathcal{S}_m = \{\,(i,j) : k_{i,j} = m\,\}$. With $z_{(i,j)} = g_{(i,j)}\, e^{i\phi_{(i,j)}}$, the group fields $H_m(\mathbf{r}) = \sum_{(i,j)\in\mathcal{S}_m} z_{(i,j)}$ are given by:

$$\begin{aligned}
&H_0 = z_{(0,0)}, && H_2 = z_{(0,1)}, && H_3 = z_{(1,0)}, && H_4 = z_{(0,2)},\\
&H_5 = z_{(1,1)}, && H_6 = z_{(0,3)} + z_{(2,0)}, && H_7 = z_{(1,2)}, && H_8 = z_{(2,1)},\\
&H_9 = z_{(1,3)}, && H_{(10)} = z_{(2,2)}, && H_{12} = z_{(2,3)}, && H_1 = H_{11} = 0.
\end{aligned} \tag{4.15}$$

The group field H_6 contains the two spots at opposite corners, $(0,3)$ and $(2,0)$, which are 3.305 µm apart. The orders $k = 1$ and $k = 11$ do not occur, so that $H_1 = H_{11} = 0$.
The time average contains, in addition to the incoherent sum, the term for the stationary interference of the corner pair, as already discussed in chapter 2 with Equation 2.37:

$$\langle I \rangle_t = \sum_{(i,j)} g_{(i,j)}^2 + 2\, g_{(0,3)}\, g_{(2,0)} \cos(\phi_{x,0} + \phi_{y,3} - \phi_{x,2} - \phi_{y,0})\,. \tag{4.16}$$

Tab. 4.1: Spot orders $k_{i,j}$ of the 3×4 grid created with two perpendicular AODs, with $f_0 = 61.67\,\text{kHz}$ and a frequency width of 0.37 MHz.

	$i=0$	$i=1$	$i=2$
$j=3$	**6**	9	12
$j=2$	4	7	10
$j=1$	2	5	8
$j=0$	0	3	**6**

Here, the tone phases determine the stationary interference pattern, which vanishes for $\Delta\phi = \phi_{(0,3)} - \phi_{(2,0)} = (\phi_{x,0} + \phi_{y,3}) - (\phi_{x,2} + \phi_{y,0}) = \pm\pi/2$.
A pair of spots separated by $(\Delta i, \Delta j)$ has a beating order d defined as the difference between their spot orders:

$$d = 3\,\Delta i + 2\,\Delta j, \qquad |\Delta i| \leq 2, \quad |\Delta j| \leq 3. \tag{4.17}$$

Their corresponding distance in the focal plane is given by $\sqrt{(\Delta i\, p_x)^2 + (\Delta j\, p_y)^2}$. The cross term is weighted by the spatial overlap $g_s(\mathbf{r})g_{s'}(\mathbf{r})$ of the spots, so closely spaced pairs have a larger contribution on the beating.
An analysis of the temporal variation of the intensity for the given parameters reveals rapid oscillations with a beating period of $T_0 = 16.22\,\mu\text{s}$. Over the beating period, the intensity varies across the entire profile, leaving the center, and thus the atom, dark for several microseconds. The uniform intensity of the generated multi-tone profile is achieved in the time average and not on timescales comparable to the beating period. Since quantum gate operations require light pulses of a suitable duration, typically on the order of microseconds, to perform the desired quantum state transformations, these temporal intensity variations can compromise the fidelity of the gate operations. This contradicts the whole purpose of the generated Top Hat profile. If a uniform mean intensity is achieved only for long pulse durations, the multi-tone Top Hat profile offers no advantage over a single Airy profile. The seven phases of the beam profile, as remaining degrees of freedom, have to be chosen in such a way that the created profile illuminates the atom in the center for a given pulse duration as uniformly as possible. Randomly chosen phases can leave the center dark for several microseconds and introduce a gradient in the intensity. Reliable gate operations for atoms with spatial thermal fluctuations would be distorted, so the phases have to be chosen accordingly. A point-symmetric intensity profile, $I(-\mathbf{r}, t) = I(\mathbf{r}, t)$ for all t, promotes a more uniform illumination of the central atom, since the intensity gradient at the center vanishes at all times. As shown in section A.1, the condition for a point-symmetric intensity profile is:

$$\phi_{x,i} + \phi_{x,N_x-1-i} = c_x \;\; \forall i, \qquad \phi_{y,j} + \phi_{y,N_y-1-j} = c_y \;\; \forall j, \tag{4.18}$$

where $c_{x,y}$ are phase constants that must be chosen accordingly. A secondary condition for the optimization of the phases can be found using the Crest factor C [26]. The Radio Frequency (RF) multi-tone signal $s(t)$, which creates the acoustic wave inside the acousto-optic crystal and thus the multi-tone profile, can be characterized by the Crest factor, defined as:

$$C = \frac{\max_t |s(t)|}{\text{rms}_t[s(t)]} \propto \sqrt{\frac{P_{\text{peak}}}{P_{\text{avg}}}}. \tag{4.19}$$

Since the peak RF power P_{peak} that can be applied to the AOD is limited, a high Crest factor limits the achievable mean RF power, thereby reducing the diffraction efficiency of each spot. The

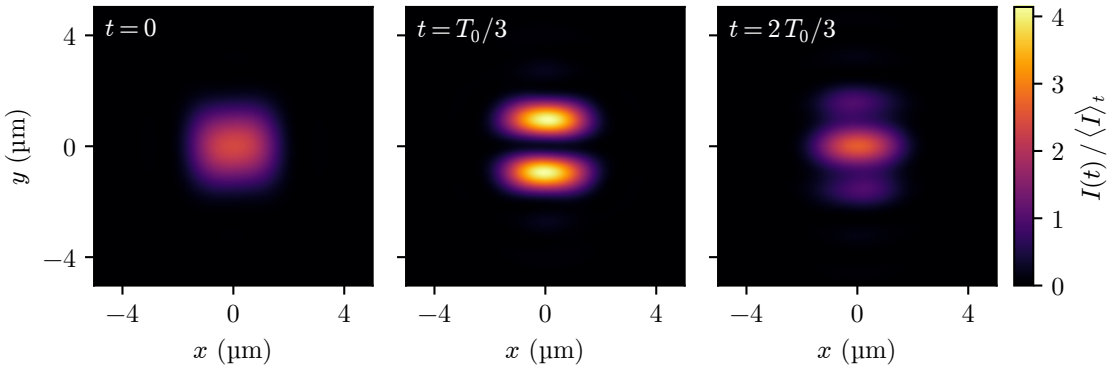


Fig. 27: Oscillating intensity of a 3×4 multi-tone beam profile with a beating period of $T_0 = 16.22\,\mu s$, shown for $t \in \{0, {}^{T_0}/_3, {}^{2T_0}/_3\}$ and normalized to the time average of the atom-weighted intensity $\langle I \rangle_t$.

goal is therefore to find a set of phases that satisfies the symmetry condition while minimizing the Crest factors of the RF signals. As shown in section A.2, the Crest factor for $N_x = 3$ cannot be minimized through phase selection while satisfying the symmetry condition. In contrast, for $N_y = 4$, such a minimization is possible. The optimization of the phases yields:

$$\phi_x = [0°, 12°, 24°], \quad \phi_y = [0°, 98°, 16°, 114°].$$

With this choice of phases, the Crest factors are $C_x \approx 2.449$ and $C_y \approx 2.196$. The condition for a point-symmetric intensity profile is fulfilled with the phase constants $c_x = 24°$ and $c_y = 114°$, while the stationary interference term in Equation 4.16 vanishes, since $\Delta\phi = 90°$. A visualization of the intensity fluctuations over a full beating period time is shown in Figure 27 for the chosen working point and the optimized phases.

Table 4.2 lists all beating orders and the corresponding spot pairs for the 3×4 multi-tone profile with the optimized phases. The last column gives the contribution of each beating order, evaluated as a Root Mean Square (RMS) over the spot square of 2.34 µm × 2.34 µm. With $\sigma_d = \sqrt{2|D_d|^2}$ and $\langle I \rangle_t = D_0$, it is given by $U_{t,d} = \left\langle \sigma_d^2 / \langle I \rangle_t^2 \right\rangle_R^{1/2}$, so that it is defined over the same region R as the spatial uniformity U_h. The beating order $d = 0$ has to be excluded from the evaluation of the beating because its constant intensity contributes to the mean intensity but has no oscillating component. Furthermore, the stationary interference contribution of the corner pair to $d = 0$ vanishes for the chosen phases, according to Equation 4.16. The beating order $d = 3$ has the strongest contribution of $51.9\,\%$, produced by the nearest neighbors along x, followed by $d = 2$ with a contribution of $40.4\,\%$, produced by the nearest neighbors along y. The nearest diagonal neighbors, with the beating orders $d = 1$ and 5, contribute comparably to the beating, with around $20\,\%$ each. In general, the higher the beating order, the smaller its contribution, since the spatial overlap of the spots decreases. The order $d = 12$ is produced by a single pair of corner spots and has the lowest contribution to the beating. Its magnitude remains unchanged regardless of the choice of phases, since destructive interference with another pair of spots is not possible. The total time-dependent uniformity is then given by:

$$U_t = \left\langle U_t(\mathbf{r})^2 \right\rangle_R^{1/2} = \left\langle \frac{\sigma_I^2}{\langle I \rangle_t^2} \right\rangle_R^{1/2} = \left(\sum_{d>0} \left\langle \frac{\sigma_d^2}{\langle I \rangle_t^2} \right\rangle_R \right)^{1/2} = \left(\sum_{d>0} \left(U_{t,d} \right)^2 \right)^{1/2} = 76.7\,\%. \tag{4.20}$$

The beating frequencies f_0 and $2f_0$ are close to the radial trap frequency $\nu_r = 60.4\,\text{kHz}$ [3] and twice the trap frequency $2\nu_r$, respectively. $f_0 = 61.67\,\text{kHz}$ lies 1.27 kHz above ν_r, while

Tab. 4.2: Beating orders for a 3×4 multi-tone profile at the working point with $f_0 = 61.67$ kHz, together with the distances of the corresponding spot pairs and their contribution to the time-dependent uniformity, calculated as an RMS over the spot square R, for the tone phases $\phi_x = [0°, 12°, 24°]$ and $\phi_y = [0°, 98°, 16°, 114°]$ used in this work. The beating order $d = 11$ does not occur.

d	f_{beat}(kHz)	number of pairs	$(\Delta i, \Delta j)$	distance (µm)	$U_{t,d}$ (%)
0	0	1	$(2, -3)$	3.31	–
1	61.7	10	$(1, -1)$, $(-1, 2)$	1.40, 1.95	20.0
2	123.3	11	$(0, 1)$, $(2, -2)$	0.78, 2.81	40.4
3	185.0	10	$(1, 0)$, $(-1, 3)$	1.17, 2.61	51.9
4	246.7	9	$(0, 2)$, $(2, -1)$	1.56, 2.46	20.4
5	308.3	6	$(1, 1)$	1.40	18.8
6	370.0	7	$(2, 0)$, $(0, 3)$	2.34, 2.34	16.1
7	431.7	4	$(1, 2)$	1.95	6.9
8	493.3	3	$(2, 1)$	2.46	3.2
9	555.0	2	$(1, 3)$	2.61	6.2
10	616.7	2	$(2, 2)$	2.81	5.4
12	740.0	1	$(2, 3)$	3.31	1.8

$2f_0 = 123.34$ kHz lies 2.56 kHz above $2\nu_r = 120.8$ kHz. Intensity modulations at frequencies close to these characteristic trap frequencies can cause heating, which may lead to atom loss and reduced fidelity of qubit gate operations [27]. Consequently, long laser pulses spanning multiple beating periods may pose a higher risk of heating. For pulse durations much shorter than the beating period, the spectral broadening associated with the finite pulse duration can reduce the risk of resonant heating. For example, a pulse duration of $T_p = 1$ µs corresponds to a spectral width of approximately $\delta\nu = 1$ MHz, which is about 17 times the trap frequency ν_r. The high time-dependent uniformity $U_t = 76.7$ % confirms the temporal non-uniformity of the generated multi-tone profile observed in Figure 27. To reliably avoid heating, short pulses are required. On these timescales, the intensity profile does not yet reach a Top Hat shape. A uniform intensity profile can only be achieved for long pulse durations with $t \gg T_p$, at the risk of atom loss due to heating.

4.6 Pulse Area with Beating

For Rabi oscillations or, more general, the excitation of atoms between two states, the Bloch sphere picture introduced in chapter 2 can be used. For a light pulse with a time-dependent intensity, the angle θ, also known as the pulse area, which characterizes the population transfer between the two states, is given by:

$$\theta(\mathbf{r}, t_0, T_p) = \int_{t_0}^{t_0+T_p} \Omega(\mathbf{r}, t)\, \mathrm{d}t, \tag{4.21}$$

where t_0 the start time of the pulse and T_P is the pulse duration. For a two-photon process, the total intensity I is split between the two beams with the weight β, such that their respective intensities are βI and $(1-\beta)I$. These two intensities drive transitions between $|F{=}2, m\rangle$ and $|F{=}3, m\rangle$ via the $5P$ manifold of ^{85}Rb [2]. The single-photon Rabi frequency $\Omega_{q,i}$ of beam q coupling to an intermediate state $|i\rangle$ is given by $\Omega_{q,i} = d_i E_q/\hbar$. Here, d_i is the transition dipole matrix element between the ground state $|g\rangle$ and the intermediate state $|i\rangle$. Using the relation $I_q = \frac{1}{2}\varepsilon_0 c E_q^2$ [28], the squared Rabi frequency can be written as:

$$\Omega_{q,i}^2 = \frac{2\, d_i^2}{\varepsilon_0 c\, \hbar^2}\, I_q \equiv d_i^2\, \kappa\, I_q, \qquad \kappa = \frac{2}{\varepsilon_0 c\, \hbar^2}. \tag{4.22}$$

For large detuning, the intermediate states are eliminated adiabatically, as already discussed in subsection 2.2.1. Each of these intermediate states contributes to the Rabi frequency of the two-photon process:

$$\Omega = \sum_i \frac{\Omega_{1,i}\, \Omega_{2,i}}{2\Delta_i} = \underbrace{\kappa\sqrt{\beta(1-\beta)} \sum_i \frac{d_{2i}\, d_{3i}}{2\Delta_i}}_{A_{\text{Rabi}}}\, I = A_{\text{Rabi}} I. \tag{4.23}$$

Here, the contributions from all relevant intermediate states are incorporated into the Rabi constant A_{Rabi}. Thus, the effective Rabi frequency is directly proportional to the laser intensity I. An intermediate state is, for example, given by a hyperfine level of the $5P$ manifold:

$$|i\rangle = |5P_{J'}, F', m'\rangle\,, \tag{4.24}$$

with $J' = \frac{1}{2}$ and $F' = 2, 3$ for the D_1 line, and with $J' = \frac{3}{2}$ and $F' = 1, 2, 3, 4$ for the D_2 line [2]. $d_{2i} = \langle i|\hat{d}|F{=}2, m\rangle$ and $d_{3i} = \langle i|\hat{d}|F{=}3, m\rangle$ are the dipole matrix elements, and Δ_i is the detuning of the laser beams from the specific intermediate state $|i\rangle$. Δ is the global detuning of both laser beams relative to the center frequency of the D_1 line of ^{85}Rb [2]. The detuning Δ_i can be calculated from the center frequency $\nu_{J'} \in \{\nu_{D_1}, \nu_{D_2}\}$ of the corresponding line and the hyperfine shift $\Delta_{\text{hfs}}(J', F')$ of each intermediate energy level relative to this center frequency:

$$\Delta_i = 2\pi \left[\Delta + (\nu_{D_1} - \nu_{J'}) - \Delta_{\text{hfs}}(J', F')\right]. \tag{4.25}$$

The calculations above are performed with a Python-based simulation [25] using atomic data from the Alkali-Rydberg-Calculator (ARC) package [29]. The resulting Rabi constant is then calculated for a detuning of $\Delta = 50\,\text{GHz}$:

$$A_{\text{Rabi}} = 11.66\, \frac{\text{rad}\,\text{m}^2}{\text{W}\,\text{s}}. \tag{4.26}$$

With the previously chosen set of phases, a point-symmetric beam profile can be generated, resulting in a point-symmetric spatial distribution of the pulse area. Using the proportionality between the Rabi frequency and the intensity, the pulse area can be expressed as:

$$\theta(\mathbf{r}, t_0, T_p) = \int_{t_0}^{t_0+T_p} \Omega(\mathbf{r}, t)\,\mathrm{d}t = A_{\text{Rabi}} \int_{t_0}^{t_0+T_p} I(\mathbf{r}, t)\,\mathrm{d}t. \tag{4.27}$$

Up to this point, the simulated beam profiles are normalized and do not include physical units. To obtain the physical intensity distribution, a scaling factor s is introduced, which relates the normalized profile to the total power P of the time-averaged Top Hat profile:

$$P = \int \langle I(\mathbf{r}, t)\rangle_t, \mathrm{d}^2 r = s \int \langle \tilde{I}(\mathbf{r}, t)\rangle_t, \mathrm{d}^2 r \equiv s\tilde{P}_{\text{prof}}, \qquad \Longrightarrow \qquad s = \frac{P}{\tilde{P}_{\text{prof}}}. \tag{4.28}$$

Here, $\tilde{P}_{\text{prof}}$ represents the spatially integrated normalized intensity profile. With the scaling factor $s = P/\tilde{P}_{\text{prof}}$, the intensity can be expressed as $I(\mathbf{r}, t) = s\tilde{I}(\mathbf{r}, t)$, and the pulse area accordingly becomes:

$$\theta(\mathbf{r}, t_0, T_p) = A_{\text{Rabi}}\, s \int_{t_0}^{t_0+T_p} \tilde{I}(\mathbf{r}, t)\,\mathrm{d}t = A_{\text{Rabi}}\, \frac{P}{\tilde{P}_{\text{prof}}} \int_{t_0}^{t_0+T_p} \tilde{I}(\mathbf{r}, t)\,\mathrm{d}t. \tag{4.29}$$

With the definition of $\tilde{P}_{\text{prof}}$, the pulse area is given by:

$$\boxed{\theta(\mathbf{r}, t_0, T_p) = A_{\text{Rabi}}\, P\, \frac{\displaystyle\int_{t_0}^{t_0+T_p} \tilde{I}(\mathbf{r}, t)\,\mathrm{d}t}{\displaystyle\int \langle \tilde{I}(\mathbf{r}, t)\rangle_t\, \mathrm{d}^2 r}}. \tag{4.30}$$

The time integral is the only part that still depends on the beating. When the Fourier series $\tilde{I}(\mathbf{r}, t) = \sum_d D_d(\mathbf{r})\, e^{\mathrm{i}2\pi d f_0 t}$ is inserted into Equation 4.30, each order $d \neq 0$ contributes the time integral:

$$\int_{t_0}^{t_0+T_p} e^{i2\pi d f_0 t}\,\mathrm{d}t = \frac{e^{i2\pi d f_0 (t_0+T_p)} - e^{i2\pi d f_0 t_0}}{i\,2\pi d f_0}, \tag{4.31}$$

while the order $d = 0$ simply yields T_p. Extracting the phase factor $e^{i2\pi d f_0 (t_0+T_p/2)}$ and using $e^{ix} - e^{-ix} = 2i\sin(x)$ [30] gives:

$$e^{i2\pi d f_0 (t_0+T_p/2)}\, \frac{e^{i\pi d f_0 T_p} - e^{-i\pi d f_0 T_p}}{i\,2\pi d f_0} = e^{i2\pi d f_0 (t_0+T_p/2)}\, \frac{\sin(\pi d f_0 T_p)}{\pi d f_0}. \tag{4.32}$$

Using the normalized sinc function $\mathrm{sinc}(z) = \sin(\pi z)/(\pi z)$ [30], the time integral becomes:

$$\int_{t_0}^{t_0+T_p} \tilde{I}(\mathbf{r}, t)\,\mathrm{d}t = T_p \sum_d D_d(\mathbf{r})\ \mathrm{sinc}(d f_0 T_p)\, e^{i2\pi d f_0 (t_0+T_p/2)}. \tag{4.33}$$

The equation for the pulse area used in the simulation is then given by:

$$\theta(\mathbf{r}, t_0, T_p) = \frac{A_{\text{Rabi}}\, P\, T_p}{\tilde{P}_{\text{prof}}} \sum_d D_d(\mathbf{r})\ \mathrm{sinc}(d f_0 T_p)\, e^{i2\pi d f_0 (t_0+T_p/2)}. \tag{4.34}$$

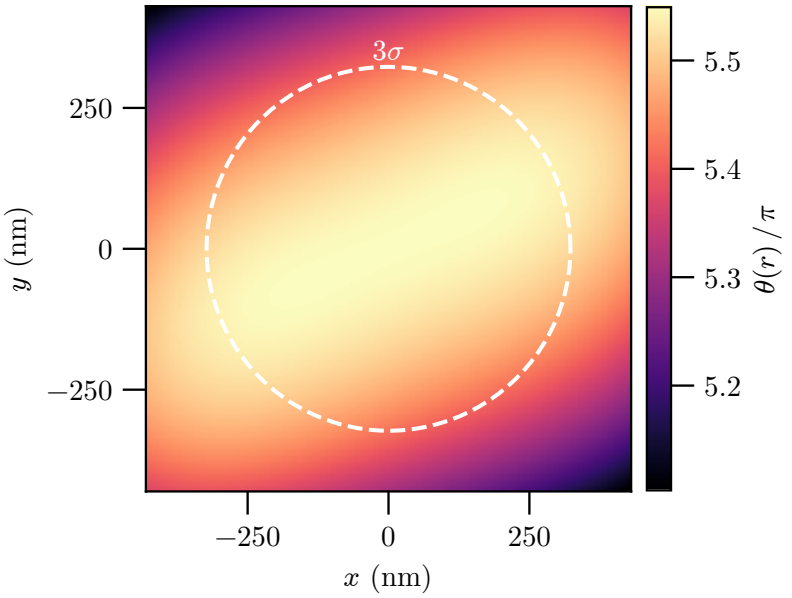


Fig. 28: Simulated pulse area of a multi-tone Top Hat profile with $A_{\text{Rabi}} = 11.66 \frac{\text{rad m}^2}{\text{W s}}$, $P = 10\,\mu\text{W}$, $T_p = 1\,\mu\text{s}$ and $t_0 = 15.4\,\mu\text{s}$. Only the central region is shown in which the atom has the highest spatial probability density, marked by the 3σ region.

$\tilde{P}_{\text{prof}}$ is calculated in section A.3. The simulation for the pulse area of the 3×4 multi-tone Top Hat profile is conducted for a profile power of $P = 10\,\mu\text{W}$, a pulse duration of $T_p = 1\,\mu\text{s}$ and a starting time of $t_0 = 15.4\,\mu\text{s}$. The results are shown in Figure 28 and are focused around the spatial probability density of the atom within the 3σ region.
The maximum pulse area in the center of the profile is $\theta_{\text{max}} = 5.55\pi$, while the minimum value within the 3σ region is $\theta_{\text{min}} = 5.40\pi$. This corresponds to a nearly uniform pulse area with a maximum relative deviation of 2.8%. Other starting times t_0 result in higher spatial deviations of the pulse area, as discussed in the following.
The laser pulse can be generated in an experiment using an Acousto-Optic Modulator (AOM) as an optical switch in the beam path. When the AOM is triggered correctly, the light is diffracted and reaches the atoms. The timing of the trigger is essential for achieving the desired pulse area. The atom-weighted intensity of the multi-tone profile with $P = 10\,\mu\text{W}$, normalized to its time average, is shown in Figure 29. The intensity exhibits three maxima above the time average, with the global maximum reaching $I_{\text{max}} = 2.6 \cdot \langle I \rangle_t$. Additionally, four local minima occur at which $I(t)/\langle I \rangle_t < 1$. At $t = 14\,\mu\text{s}$, a global minimum is reached, and the atom is not illuminated. The starting time $t_0 = 15.4\,\mu\text{s}$ for a light pulse with a duration of $T_p = 1\,\mu\text{s}$ is marked in red. With this intensity variation, the mean pulse area can be calculated by averaging the resulting pulse area over the atom's spatial probability density for different starting times t_0, as shown in Figure 30. The pulse area and the intensity exhibit the same behavior, as described by Equation 4.30, but are shifted in time relative to each other due to the integration over the pulse duration $T_p = 1\,\mu\text{s}$. The global intensity minimum results in the minimum pulse area at a starting time of $t_0 = 13.5\,\mu\text{s}$. A global maximum in the pulse area is visible at $t_0 = 10.2\,\mu\text{s}$, with $\theta = 6.8\pi$. In this case, the spatial deviation of the pulse area exceeds 20% and is thus substantially higher than for $t_0 = 15.4\,\mu\text{s}$ in Figure 28, so this starting time is not preferred.

This confirms the previous choice of $t_0 = 15.4\,\mu s$ as the preferred starting time, based on the spatial uniformity of the pulse area. A timing error of $1\,\mu s$ in the AOM trigger can cause a deviation of up to $65\,\%$ in the mean pulse area $\langle\theta\rangle$. For this set of optimized parameters, even small trigger errors can distort gate operations. In general, a compromise has to be found for the starting time t_0 between a low sensitivity of the mean pulse area to trigger errors and a spatially uniform pulse area in the center of the profile. The length of the light pulses therefore cannot be chosen arbitrarily, since for longer pulses it must be ensured that the atom is still illuminated. Otherwise, extending the pulse into a dark phase of the beating does not increase the accumulated pulse area. This imposes a restriction on Rabi oscillations and gate operations on certain timescales. A possible next step in a future iteration of the simulation would be to jointly optimize the integration time window and its starting time t_0. The optimization would aim to achieve a spatially uniform pulse area not only for the full integration window but also for shorter integration durations within this window, while ensuring that the atom remains sufficiently illuminated. This would provide greater flexibility in the choice of pulse durations T_p. Trigger errors for pulses shorter than the optimized integration window would only result in small changes in the mean pulse area, while maintaining its spatial uniformity. This could make the approach more robust for single-qubit gate operations.

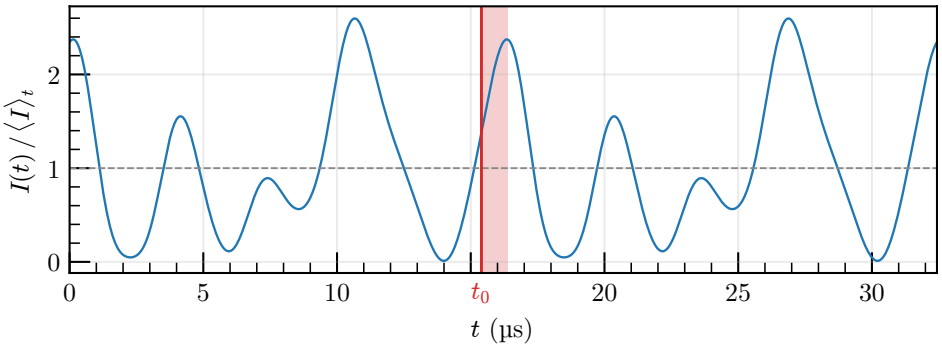


Fig. 29: Intensity $I(t)$ of a 3×4 multi-tone profile at $P = 10\,\mu W$, shown over two beating periods, normalized to its time average and averaged over the spatial probability density of the atom. The red line marks the starting time $t_0 = 15.4\,\mu s$ of the pulse, the shaded area its duration $T_p = 1\,\mu s$.

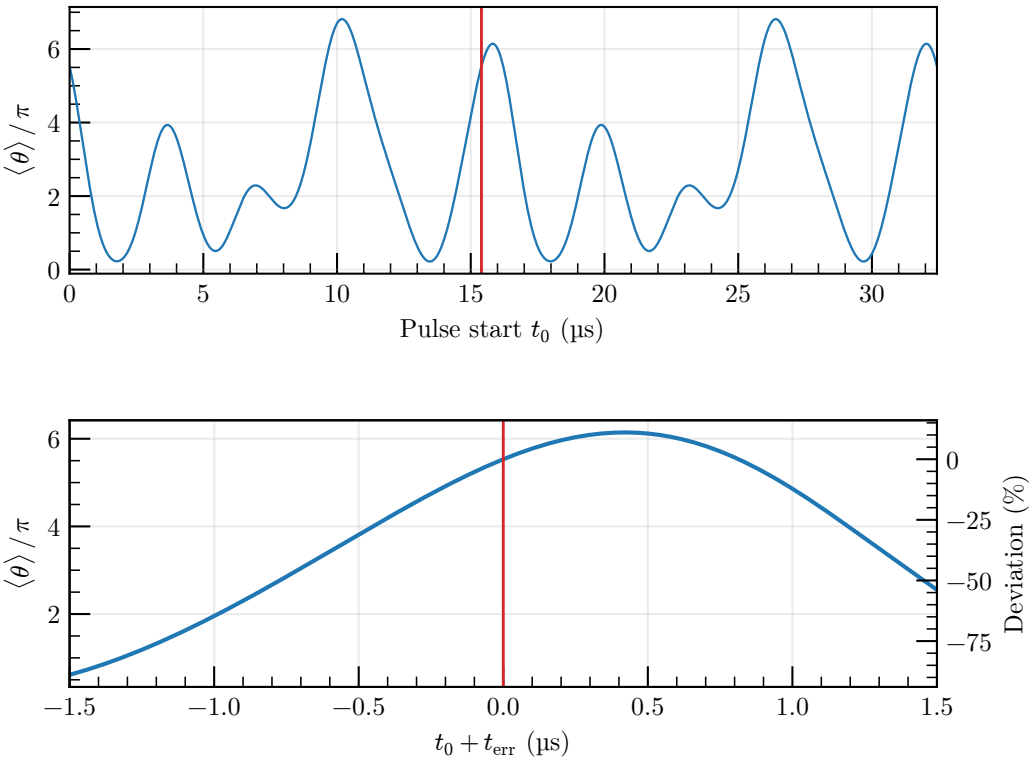


Fig. 30: Pulse area of a $T_p = 1\,\mu s$ pulse for the 3×4 multi-tone profile at $P = 10\,\mu W$, averaged over the spatial probability density of the atom. Top: Pulse area $\langle \theta \rangle / \pi$ as a function of the starting time t_0 over two beating periods, with $t_0 = 15.4\,\mu s$ marked in red. Bottom: Enlarged section around $t_0 = 15.4\,\mu s$, showing the deviation of the pulse area caused by an error in the trigger time. For $t_{\mathrm{err}} = -1\,\mu s$, the pulse area deviates by up to 65 %.

4.7 Validation with the Test Setup

To verify the simulation results, the test setup previously shown in Figure 12 is used without the Beam Shaper. A disadvantage of this setup is the beam waist in the focal plane behind the Lens Optics Objective. According to Equation 3.4, the expected beam waist is $\omega = 1.15\,\mu m$. The measured waist of a single spot in the focal plane is $\omega = 1.8\,\mu m$. This deviation may be caused by optical misalignment, as already discussed in chapter 3. Despite this deviation, the setup is sufficient for proof-of-concept measurements.
Using the same number of tones in both dimensions leads to stationary spatial interference along the anti-diagonals. Therefore, an $N \times (N+1)$ configuration with $N = 3$ tones was used throughout the time-independent simulations and the beating analysis, as discussed in chapter 2. This limitation on the choice of the number of tones is validated by taking an image of an $N \times N$ multi-tone beam profile. This image is then compared with a simulation of the coherent superposition of the corresponding spots, conducted with a Python-based analysis routine [25]. The image of a 6×6 multi-tone profile with a frequency width of 0.5 MHz is shown on the left of Figure 31. The symmetric structure of the Top Hat profile is no longer visible in the measured image. Instead, a bright diagonal stripe and an additional side structure below appear. The measured profile shares some features with the simulation shown in Figure 31 on the right, which also predicts a prominent diagonal stripe resulting from the spatial interference of the individual spots. The two profiles differ in the visibility of the central square structure. While the simulation retains the square form at the four corners, with the central region dominated by the diagonal stripe, this structure is no longer visible in the measurement. The additional side structure observed in the measurement is not reproduced by the simulation and may therefore originate from effects caused by the AOD. This result confirms the necessity to choose different numbers of tones for generating uniform intensity profiles in a multi-tone scheme. Otherwise, spatial interference would distort the resulting profile, and the advantage of a Top Hat profile would be lost.

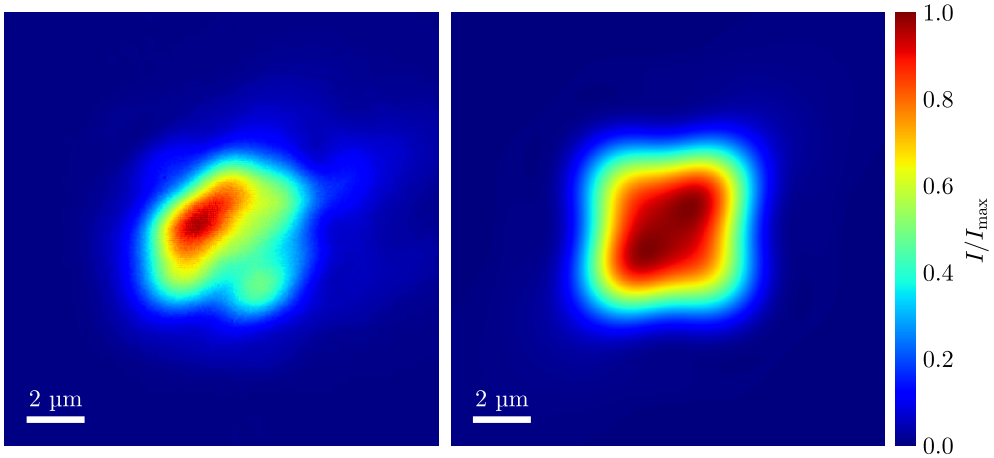


Fig. 31: Left: Image of a 6×6 multi-tone beam profile with $W_f = 0.5$ MHz, created with the test setup. A bright diagonal stripe and a small side structure are visible. Right: Simulated image with the same parameters. A tilted square with a dominant diagonal stripe in its center is visible.

To verify whether beating occurs in the generated multi-tone Top Hat profiles, a test image is taken with a camera. The image represents a temporal average of the intensity over the exposure time T:

$$\langle I(\mathbf{r},t)\rangle = \frac{1}{T}\int_0^T I(\mathbf{r},t)dt. \tag{4.35}$$

To resolve the time-dependent intensity variations caused by the beating and obtain high contrast within the image, the exposure time of the camera must be shorter than the beating period to avoid averaging over multiple beating periods. Additionally, the camera itself requires a global shutter so that all pixels are illuminated by the same period sequence of the beating. With a rolling shutter and long readout times, different rows of the image would be recorded at different phases of the beating, so the image would contain no distinct information about the beating. The camera used (CM3-U3-13S2M-CS) has a minimal exposure time of 45.6 µs and the required global shutter. For the test image, the two perpendicular AODs are used with an achromatic lens with $f_1 = 45$ mm, as shown in Figure 32.
The optimized multi-tone profile has a beating period of $T_0 = 16.22$ µs and therefore cannot be resolved with the camera. Since no camera with a shorter exposure time is available, the beam profile has to be adapted. The parameters of the profile are chosen such that the beating period of $T_0 = 100$ µs is more than twice the minimum exposure time of the camera. For this purpose, a profile with $N_x = 13$ and $N_y = 14$ tones and a frequency width of $W_f = 1.56$ MHz is created. The results of the simulation and the corresponding measurement are shown in Figure 33. The simulation on the left shows four bright diagonal stripes extending across the entire profile. Small-scale substructures between the bright stripes are also visible and may originate from the side structures of the individual Airy profiles. The Peak-to-Valley (PV) value is calculated within the region enclosed by the 70 % intensity level of the profile as:

$$PV = \frac{I_{\text{max}} - I_{\text{min}}}{\langle I\rangle}. \tag{4.36}$$

The obtained PV value in the simulation is $PV_{\text{sim}} = 68.5\,\%$. The measurement on the right of Figure 33 exhibits the same diagonal stripes as the simulation, although their positions are shifted. This difference arises because the image was taken manually rather than with a synchronized trigger, as would be provided, for example, by an AOM. Additionally, a grid of bright and dark substructures is visible across the profile. This grid is much more pronounced in the measurement than in the simulation and consists of 13×14 bright spots, which can be identified as the individual spots forming the profile. The PV value of the image is $PV_{\text{im}} = 59.6\%$. The difference of almost 10 % between simulation and measurement is considered acceptable given the limitations of the measurement method. The camera does not provide a direct and exact conversion from recorded pixel values to optical intensity. Imaging aberrations and

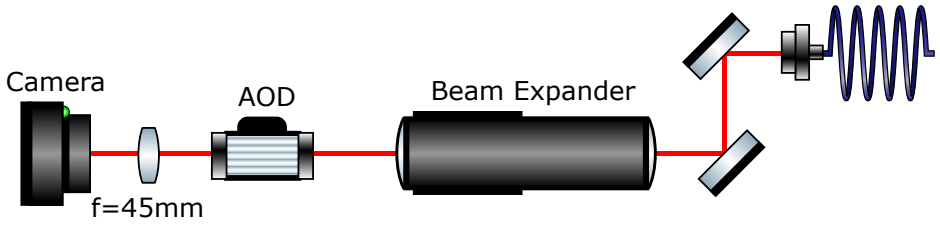


Fig. 32: Schematic of the optical setup used for measuring the beating of the multi-tone Top Hat profiles.

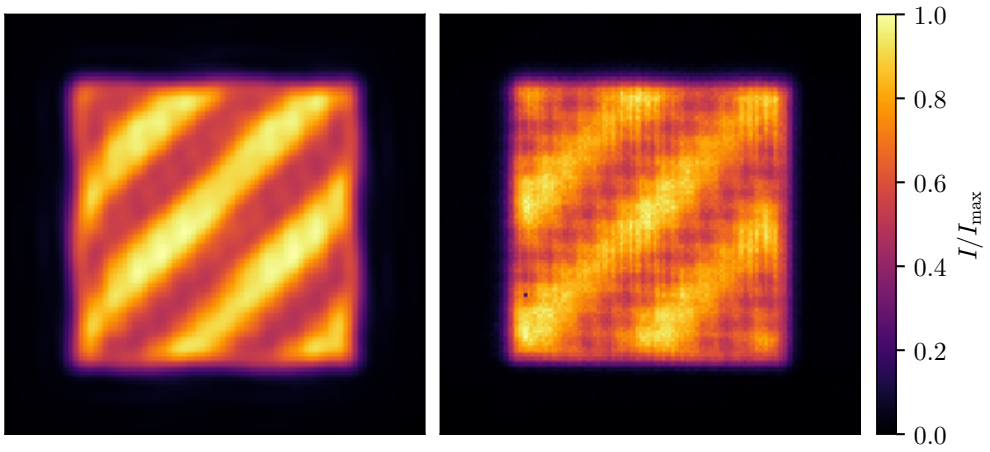


Fig. 33: Left: Simulated image of the beating of a 13×14 multi-tone profile with a frequency width of 1.56 MHz for an exposure time of 45.6 µs. Dominant diagonal stripes are visible. Right: Measured image with the same parameters. The measurement also shows diagonal stripes and is in good agreement with the simulation.

sensor nonlinearities can contribute to deviations between the measured intensity distribution and the simulated profile. The overall agreement between the measured and simulated profiles provides strong validation of the simulation results for the temporal beating. To evaluate the beating of a 3×4 multi-tone profile with the optimized parameters of $W_f = 0.37$ MHz with the chosen phases $\phi_x = [0°, 12°, 24°]$ and $\phi_y = [0°, 98°, 16°, 114°]$, the frequency spectrum is measured with a photodiode (DET10A/M). Since the future setup has not yet been built, the setup used for the beating image in Figure 32 is used again. The photodiode is placed in the focal plane of the lens so that the whole beam profile is measured. Additionally, a simulation with a Python-based analysis routine [25] of the beating orders is conducted that differs from the one shown in Table 4.2. This difference results from the measurement method itself, since the photodiode measures the fluctuation of the total power by integrating the intensity over the whole detector area. From this signal, the variation of the different frequency components can be determined. For an ideal measurement in the focal plane of the lens, the contribution of each beating order is calculated as:

$$U_{t,d}{}^{\text{sim}} = \frac{\sqrt{2}|\int D_d \, \mathrm{d}A|}{\int D_0 \, \mathrm{d}A}. \tag{4.37}$$

Intensity fluctuations at different positions can cancel each other out. When one region becomes brighter while another becomes darker, the total power stays the same, and the photodiode measures no change. For the whole detector area, this results in lower contributions of the different beating orders compared to the simulation conducted for Table 4.2 based on Equation 2.40. The latter approach calculates the intensity variations for each position in the profile and then the squared values are averaged over the whole profile. With this calculation method, every intensity variation has a positive contribution and does not average out as is the case in Equation 4.37. To measure the beating, this evaluation method could be implemented using a camera with an exposure time significantly shorter than the beating period to acquire a time-resolved sequence of spatially resolved beam profile images.
To determine the parameters of the intensity profile used in the simulation, the beam profile is measured with a camera (CM3-U3-13S2M-CS). A two-dimensional Gaussian function with

a waist of $\omega = 6.4\,\mu m$ is fitted to the data and subsequently used for the simulation. For the measurement, the mean voltage signal of $\bar{U} = 4.90\,mV$ is recorded and the peak amplitude $\hat{U}_d$ of the beating order is determined from the Fourier Transform. The contribution of each beating order is then calculated as:

$$U_{t,d}{}^{\text{meas}} = \hat{U}_d/(\sqrt{2}\,\bar{U}). \tag{4.38}$$

The resulting spectrum of the beating orders is shown in Figure 34 together with the simulated spectrum. The beating orders $d = 2$ and $d = 3$, which result from vertically and horizontally adjacent spots, have the largest contributions to the beating in both the simulation and the measurement, and the two agree well with each other. For $d = 1$ and $d = 5$ resulting from diagonally adjacent spots, this also holds true. The main difference between measurement and simulation is found in the orders $d = 4$ and $d = 6$, resulting from the spot pairs separated by two sites vertically or horizontally. The measured contributions are significantly higher than the simulated ones, which can be explained with a higher overlap of the corresponding spots due to side structures that are not included in a simple Gaussian model for the spots in the simulation. For higher orders $d > 6$, caused by more distant spot pairs, the simulation shows little to no beating at these frequencies. In contrast, the measurement consists of frequency components up to $d = 12$. Deviations between simulation and measurement may also arise from differences in the size of the individual spot. The simulation assumes ideal Gaussian profiles with a diameter of 12.8 µm in the focal plane of the lens. If the photodiode is not positioned exactly in the focal plane, the actual beam diameter at the detector may differ from the simulated value. Additionally, misalignment may cause partial clipping of the beam profile at the active area of the photodiode, leading to further differences between the measured and simulated results. By combining the contributions of all beating orders, the time-dependent uniformity of the entire beam profile can be calculated. The resulting values, $U_t^{\text{meas}} = 35.3\,\%$ and $U_t^{\text{sim}} = 36.5\%$, show good agreement between measurement and simulation, supporting the validity of the simulation model. With the spot diameter and the number of tones, this measurement method provides information about the spatial intensity variations without requiring camera-based imaging of the beam profile.

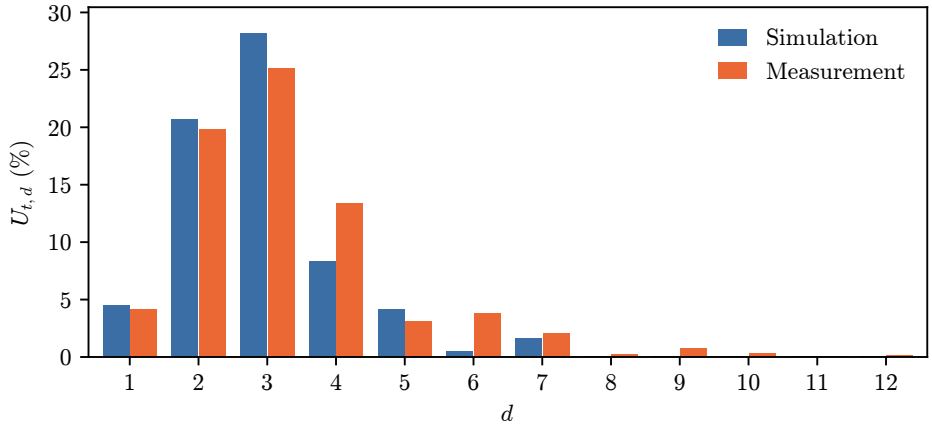


Fig. 34: Beating orders d of a 3×4 profile with $W_f = 0.37\,MHz$ and a beating frequency of $f_0 = 61.67\,kHz$, compared with the simulation. Measured with a photodiode in the focal plane of a lens with $f = 45\,mm$. Up to $d = 7$, the simulated spectrum contains beating frequencies, whereas the measured spectrum consists of beating frequencies up to $d = 12$.

4.8 Conclusion

Based on simulations with two different calculation metrics and their combination, a comparison of the time-independent multi-tone approach and the single Airy spot is shown in Table 4.3. The working points differ in terms of the beam waist within the atomic plane with 1.04 µm and 1.9 µm. Switching between the two approaches requires no changes to the optical hardware. Only an adjustment of the beam waist using the Beam Expander in front of the two perpendicular AODs, as in the test setup shown in Figure 12, and a switch between single-tone and multi-tone RF signals are necessary. A direct comparison of the hard

Tab. 4.3: Comparison of the selected working points for the multi-tone and single Airy spot approaches, including the parameters used to generate each profile and the resulting uniformity and crosstalk, as well as their sensitivity to relative displacements between the beam profile and the dipole trap.

Parameter	Multi-tone	Single Airy spot
Beam waist ω	1.04 µm	1.9 µm
Profile width W	2.34 µm	—
Amplitude ratio r_x	0.97	—
Amplitude ratio r_y	1.16	—
Hard uniformity U_h	$3.56\,\%$	$13.8\,\%$
Hard crosstalk η_h	$8.40\,\%$	$12.5\,\%$
Weighted uniformity U_w	$0.42(1)\,\%$	$1.06(17)\,\%$
1 µm shift U_w	$4.1(3)\,\%$	$10.8(12)\,\%$
Weighted crosstalk η_w	$1.42(3)\,\%$	$0.15(5)\,\%$
1 µm shift η_w	$3.5(1)\,\%$	$3.7(4)\,\%$

metrics with regard to the uniformity is not suitable here, since for the single Airy spot a circle with a radius of 1 µm is used, while for the multi-tone profile the calculation area is given by the spot square of 2.34 µm width. For the crosstalk η_h, on the other hand, both calculation areas are identical. In the atom-weighted case, the multi-tone approach yields a crosstalk of $1.42(3)\,\%$, almost ten times as high as that of the single Airy spot with $0.15(5)\,\%$. Since this result is based on the spatial probability density of the atom rather than purely geometric considerations, it is a clear advantage of the single spot over the multi-tone profile. The atom-weighted uniformities are of the same order of magnitude, although the single Airy spot profile exhibits a more than two times higher uniformity of $1.06(17)\,\%$ compared to $0.42(1)\,\%$ for the multi-tone profile. A displacement of the dipole trap relative to the multi-tone profile causes a much smaller increase in the uniformity and the crosstalk, as intended by its design. For displacements of $r = 1$ µm, the uniformity of the multi-tone profile increases to $4.1(3)\,\%$, while that of the single Airy profile reaches $10.8(12)\,\%$. For the crosstalk, the values of the two approaches are almost identical, at $3.5(1)\,\%$ and $3.7(4)\,\%$. Except for the atom-weighted crosstalk, these values are either of the same order of magnitude or favor the a multi-tone profile.

For the selected working point of the multi-tone profile, a simulation of the beating for the superimposing of laser beams was conducted and partially validated by measurements of the frequency spectrum and the spatial intensity distribution. The beating has a period of 16.22 µs, and the intensity shows variations of up to four times its time average. The whole profile has a time-dependent uniformity of $U_t = 76.7\,\%$ inside its spot square, which contradicts the purpose of generating uniform intensity profiles with a multi-tone AOD on timescales comparable to the beating period. For illumination times longer than the beating period, as required for applications such as optical pumping, the time-averaged profile takes on the desired Top Hat shape. With an appropriate choice of phases for each tone, the profile can be symmetrical with regard to its center, allowing the atom to be illuminated more uniformly on shorter timescales. To use multi-tone profiles on timescales on the order of the beating period, a synchronized trigger is required. With a correctly timed trigger, the spatial distribution of the pulse area is uniform in the center, around the assumed position of the atom. This requires stable beam pointing for the dipole trap relative to the profile center. Additionally, the beating limits the choice of pulse durations, since extending a pulse into a dark phase of the beating does not increase the pulse area. This imposes a restriction on Rabi oscillations and gate operations on certain timescales. The use of a single Airy profile remains a viable option, since it is not subject to stationary interference or beating. Even on short timescales, the atom can be illuminated without a synchronized trigger, and the length of light pulses can be chosen as required for the intended Rabi oscillations. Even though the uniformity of an Airy profile is worse than that of the time-averaged multi-tone profile and the Airy profile can only be approximated as uniform close to its center, the single Airy profile is the preferred choice for single-site addressing. With its lower atom-weighted crosstalk and the absence of stationary spatial interference or beating, the single Airy profile offers advantages that outweigh its disadvantages compared to a multi-tone Top Hat profile for our approach to single-site addressing on a micrometer scale.

5 Planning of an Optical Setup

The next step is the construction of a modular Raman laser board that integrates the existing global addressing scheme described in [7] in a more compact form, alongside a new setup for single-site addressing, which has not previously been available in the Quantum Information Processing System (QUIPS)-C experiment.
Combining both setups into one will make the whole system more efficient by reducing the lengths of optical paths. Spatial restrictions are set due to the already placed optical setups on the experiment table. The whole setup is placed on an aluminum board, which allows for construction work to be performed in a different laboratory. The dimensions of the board are 750 mm $\times$ 850 mm.
A schematic of a possible new optical setup is shown in Figure 35. The laser light is provided by the Raman laser system developed in [5] and integrated into the experimental setup in [6]. With this system, the necessary frequency shift is generated for the two-photon Raman transition. In both cases, the light is coupled out of the fibers and filtered to ensure the desired polarization and wavelength. A small fraction of the light is then directed onto a photodiode for intensity stabilization and monitoring. The global addressing part of the setup is a rearranged version of the system described in [7], aiming to uniformly address the entire atomic grid of more than 40×40 atoms simultaneously while reducing the optical path length between the setup and the vacuum chamber. The laser beam for the global addressing passes through a Beam Expander and a diffractive Beam Shaper. It is then magnified by a telescope to achieve the required beam diameter before being focused by the Lens Optics Objective, thereby generating a square Top Hat profile in the atomic plane.
The laser light for the single-site addressing passes through a Beam Expander. To improve the imaging quality, a 1:1 telescope is planned to be placed between the two perpendicular Acousto-Optic Deflectors (AODs) that follow. This allows the first lens of the subsequent telescope, which determines the scanning range according to Equation 3.3, to be positioned at a distance equal to its focal length $f_1 = 75$ mm from the second AOD. In a future setup, alternative 1:1 imaging systems consisting of multiple lenses, similar to the Lens Optics Objective, could be considered to reduce spherical aberration. The telescope behind the AODs is designed with focal lengths of $f_1 = 75$ mm and $f_2 = 750$ mm, resulting in a scanning range of $r_{\text{max}} = 227.5$ µm, as already discussed in chapter 4. All lenses are positioned such that the distance between adjacent lenses equals the sum of their respective focal lengths. Although alternative focal lengths could provide a larger scanning range according to Equation 3.3, spatial constraints must be considered, as they may prevent the lenses from being positioned at the required distances. After passing through the telescope, the light is combined with the laser beam used for global addressing using a Polarizing Beam Splitter (PBS). A camera at the remaining port of the PBS allows the beam quality of both setups to be monitored. The combined beam passes through two Dichroic Mirrors (DMs), which are used to couple light from a Rydberg laser system into the vacuum chamber and to enable camera-based readout of light propagating in the opposite direction. The light then passes through a $\lambda/4$ plate before being imaged through the glass window into the vacuum chamber by the Lens Optics Objective. As required for the two-photon process, the light inside the vacuum chamber is thus circularly polarized.

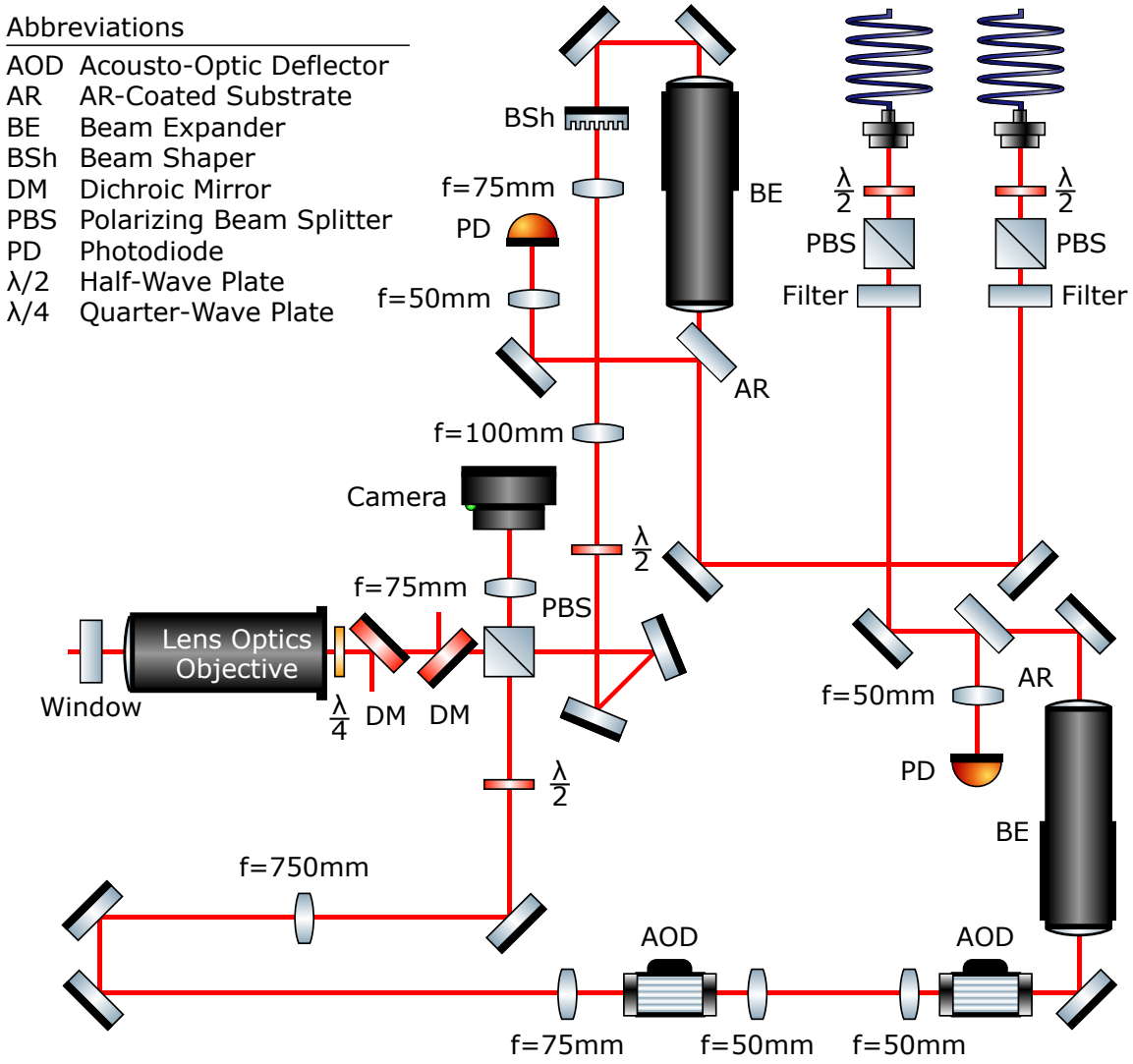


Fig. 35: Schematic of a possible new optical setup for the beam shaping of the Raman laser beams. It consists of a global addressing scheme using a Beam Shaper and a part for single-site addressing with two perpendicular AODs. The laser beams are provided by the same laser board with two fiber couplers and are each shaped for their respective purpose and superimposed before reaching the vacuum chamber, represented by the glass window.

An advantage of the single-site addressing setup is the flexibility to switch between the different addressing methods with minimal additional effort. Switching from multi-tone Top Hat profiles to single Airy profiles requires only adjusting the beam waist using the variable Beam Expander and driving each AOD with a single-tone signal.

6 Conclusion and Outlook

In this work, different approaches for single-site addressing of ^{85}Rb with a uniform intensity profile have been examined. Initially, the use of a Beam Shaper in a test setup yielded promising results with diameters below 10 µm. The diameter of these profiles is larger than expected, as the beam size approaches the diffraction limit, where the empirical relation for the profile size provided by the manufacturer is no longer applicable. Additionally, the recorded images show an ellipticity of the beam profile that could not be eliminated by optical adjustments. Using an Acousto-Optic Deflector (AOD) together with a Beam Shaper distorts the beam profile even further. In combination with an increased beam waist due to possible misalignments, the resulting profile is not reproducible and in conclusion not suitable for single-site addressing.
The approach of superimposing multiple deflected spots using two perpendicular AODs to generate Top Hat profiles was also investigated. The frequency shift imposed on the spots by the AODs made it necessary to use $N \times (N+1)$ spots to avoid stationary interference. Time-independent simulations were conducted for a profile consisting of 3×4 spots. The parameters that minimize the uniformity and the crosstalk were found to follow a linear relation between the spot waist ω and the width W of the profile. The amplitude ratios $r_{x,y}$ of the outer tones to the inner ones were determined by optimization in both dimensions. The determined optimal parameters are $N_x = 3$, $N_y = 4$, $\omega = 1.04\,\mu\text{m}$, $W = 2.34\,\mu\text{m}$ $(0.37\,\text{MHz})$, $r_x = 0.97$ and $r_y = 1.16$. Afterwards, the beating of the overlapping spots was simulated for these parameters. The beating results in strong temporal intensity fluctuations across the profile, which can negate the advantages of the intended Top Hat profile for short pulse durations. For a beating period of $T_0 = 16.22\,\mu\text{s}$, a time-dependent uniformity of $U_t = 76.7\%$ was found. For pulse durations much longer than T_0, on the other hand, the rapid temporal variations are averaged out, causing the measured intensity profile to approach the desired Top Hat structure. This could be beneficial for the longer addressing times required for optical pumping, provided that the beating frequencies close to the trap frequency do not cause heating. With a point-symmetric intensity profile, achieved through an appropriate choice of phases, a uniform pulse area for pulse durations of 1 µs remains feasible with a synchronized trigger. A possible next step in a future iteration of the simulation would be the optimization of a uniform pulse area for pulse durations other than 1 µs, coupled to the starting time t_0, to reduce a potential trigger error on the pulse area. For the desired multi-tone profile to be created in the atomic plane, the focal plane of the laser light must coincide precisely with it, and accurate positioning is required to ensure precise addressing of each site. Achieving this alignment poses a challenge for the experiment and future setups. To avoid stationary interference and beating, the use of a single Airy spot with $\omega = 1.9\,\mu\text{m}$ is preferred. Compared to a multi-tone Top Hat profile, this approach yields a higher hard crosstalk and a higher atom-weighted uniformity value, while the crosstalk weighted with the spatial probability density of the atom is lower for a single Airy profile. Taking these results and the remaining challenges in generating Top Hat profiles near the diffraction limit into account, the next step is to build an optical setup for single-site addressing. The existing global addressing scheme is to be combined with a local addressing setup consisting of two perpendicular AODs, providing the flexibility to switch between the different addressing methods.

Appendix

The following notations are used for the performed calculations:

N_x, N_y	number of tones per axis, tone indices n, m
W_f	frequency span per axis, tone spacing $\Delta f = W_f/(N-1)$
φ_n, φ_m	tone phases
a_n, b_m	tone intensity weights, $a_s = a_n b_m$, spot field amplitude $A_s = \sqrt{a_n b_m}$
$u(r)$	focal mode, normalized to $u(0) = 1$
f_0, T_0	fundamental beat frequency and period
k_s	frequency order of spot s, H_m group field of order m
D_d	Fourier coefficient of the intensity, beat order d
C	Crest factor of the Radio Frequency (RF) signal per axis

The amplitude ratios r_x and r_y are ratios of the optical intensity. Accordingly, the intensity weights of the outer tones are $r_{x,y}$, while the spot field and RF amplitudes are given by their square roots:

$$(a_n) = [\, r_x, 1, \ldots, 1, r_x \,], \qquad (b_m) = [\, r_y, 1, \ldots, 1, r_y \,], \qquad \varrho_{x,y} := \sqrt{r_{x,y}}. \tag{A.1}$$

A.1 The Centering Criterion

To better illuminate the center of the multi-tone profile, the intensity profile has to be point-symmetric:

$$I(-\mathbf{r}, t) = I(\mathbf{r}, t) \quad \forall\, t. \tag{A.2}$$

In terms of the electric field this is realized by:

$$E(-\mathbf{r}, t) = e^{i\alpha(t)} E^*(\mathbf{r}, t), \tag{A.3}$$

with one and the same $\alpha(t)$ for the whole field. Each spot s of Equation 2.30 can be decomposed into the two Acousto-Optic Deflector (AOD) tone indices (n, m):

$$E(\mathbf{r}, t) = \sum_{n,m} \sqrt{a_n b_m}\, u(\mathbf{r} - \mathbf{c}_{nm})\, e^{i2\pi(f_n + g_m)t}\, e^{i(\varphi_n + \psi_m)}. \tag{A.4}$$

The frequency for a spot created by one AOD is given by $f_n = f_0 + W_f \frac{n}{N-1}$, and increases linearly with n. Here, W_f is the frequency width. The profile itself already has four symmetries by construction, with the mirror index $\bar{n} = N - 1 - n$:

$$\text{The spot grid is centered:} \qquad \mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{c}_{nm}, \tag{A.5}$$

$$\text{The spot profile is radially symmetric:} \qquad u(-\mathbf{r}) = u(\mathbf{r}), \tag{A.6}$$

$$\text{The amplitudes are symmetric:} \qquad a_{\bar{n}} = a_n, \qquad b_{\bar{m}} = b_m, \tag{A.7}$$

$$f_n \text{ increases linearly:} \qquad f_{\bar{n}} + f_n = F, \qquad F = 2f_0 + W_f. \tag{A.8}$$

An equal frequency condition holds for g_m. Evaluating $E(-\mathbf{r}, t)$ and relabeling the sum by the mirror index results in:

$$E(-\mathbf{r}, t) = \sum_{\bar{n},\bar{m}} \sqrt{a_{\bar{n}} b_{\bar{m}}}\, u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}})\, e^{i2\pi(f_{\bar{n}}+g_{\bar{m}})t}\, e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})}.$$

Since $n, m \to \bar{n}, \bar{m}$ is a bijection of $\{0, \ldots, N-1\}$ onto itself, renaming the summation indices n, m does not change the sum. Now the transformation for each component of the field can be performed. The single spot field does not change using its symmetries:

$$u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}}) = u\big(-\mathbf{r} + \mathbf{c}_{nm}\big) = u\big(-(\mathbf{r} - \mathbf{c}_{nm})\big) = u(\mathbf{r} - \mathbf{c}_{nm}).$$

The amplitudes remain unchanged:

$$\sqrt{a_n b_m} = \sqrt{a_{\bar{n}} b_{\bar{m}}}.$$

The linearity of the frequencies together with the constant frequency offset F results in an additional time-dependent phase for all spots:

$$e^{i2\pi(f_{\bar{n}}+g_{\bar{m}})t} = e^{i2\pi(F+G)t} \cdot e^{-i2\pi(f_n+g_m)t}.$$

After identifying all symmetries the field takes the form:

$$E(-\mathbf{r}, t) = e^{i2\pi(F+G)t} \sum_{n,m} \sqrt{a_n b_m} \cdot u(\mathbf{r} - \mathbf{c}_{nm}) \cdot\ e^{-i2\pi(f_n+g_m)t} e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})}.$$

For a direct comparison the complex conjugate field is calculated:

$$E^*(\mathbf{r}, t) = \sum_{n,m} \sqrt{a_n b_m} \cdot u(\mathbf{r} - \mathbf{c}_{nm}) \cdot e^{-i2\pi(f_n+g_m)t} e^{-i(\varphi_n+\psi_m)}. \tag{A.9}$$

Both sums run over the same basis functions with the same real coefficients and differ only by a phase per term. They are proportional, and hence Equation A.3 holds for $\alpha(t) = [2\pi(F + G)t + \Theta]$, if and only if $e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})} = e^{i\Theta} e^{-i(\varphi_n+\psi_m)}$ for all n, m with a single constant Θ:

$$(\varphi_n + \varphi_{N_x-1-n}) + \big(\psi_m + \psi_{N_y-1-m}\big) = \Theta.$$

Since the first summand only depends on n and the second only on m, the sum is constant for every combination if and only if each summand is constant, resulting in the condition for a point-symmetric beam profile with $c_x, c_y \in \mathbb{R}$:

$$\boxed{\varphi_n + \varphi_{N_x-1-n} = c_x \ \ \forall n, \qquad \psi_m + \psi_{N_y-1-m} = c_y \ \ \forall m.} \tag{A.10}$$

A.2 Crest Factor

The Crest factor of an RF signal $s(t)$ is calculated as the ratio of the maximum signal to its Root Mean Square (RMS) value:

$$C = \frac{\max_t |s(t)|}{\mathrm{rms}_t[s(t)]}. \tag{A.11}$$

The RF signal in a multi-tone setup is given by N different superimposed oscillations with RF amplitude $v_n := \sqrt{a_n}$, frequency f_n and phase φ_n:

$$s(t) = \sum_{n=0}^{N-1} v_n \cos\left(2\pi f_n t + \varphi_n\right) = \mathrm{Re}[e^{i2\pi f_c t} A(t)]. \tag{A.12}$$

$A(t)$ can be expressed through a center frequency f_c:

$$A(t) = \sum_n v_n e^{i(2\pi(f_n - f_c)t + \varphi_n)} = |A(t)| e^{i\Theta(t)}. \tag{A.13}$$

So the RF signal takes the following form:

$$s(t) = \mathrm{Re}[|A(t)| e^{i[2\pi f_c t + \Theta(t)]}] = |A(t)| \cos\left(2\pi f_c t + \Theta(t)\right). \tag{A.14}$$

The mean squared value is therefore given by:

$$\langle s(t)^2 \rangle = \frac{1}{2} \langle |A(t)|^2 \rangle, \tag{A.15}$$

which leads to an RMS value of:

$$\mathrm{rms}_t[s(t)] = \sqrt{\langle s(t)^2 \rangle} = \sqrt{\frac{1}{2} \langle |A(t)|^2 \rangle}. \tag{A.16}$$

As a result, the Crest factor is given by:

$$C = \frac{\max_t |s(t)|}{\mathrm{rms}_t[s(t)]} = \sqrt{2} \frac{\max_t |A(t)|}{\mathrm{rms}_t[A(t)]}. \tag{A.17}$$

For the RMS value, the time average of each cross term between two different frequencies vanishes, since the complex exponential averages out over one period:

$$\langle |A(t)|^2 \rangle = \sum_n v_n^2 + \sum_{n \neq n'} v_n v_{n'} \langle e^{i[2\pi(f_n - f_{n'})t + (\varphi_n - \varphi_{n'})]} \rangle = \sum_n v_n^2, \tag{A.18}$$

$$\mathrm{rms}_t[A(t)] = \sqrt{\langle |A(t)|^2 \rangle} = \sqrt{\sum_n v_n^2}. \tag{A.19}$$

The lower bound for the Crest factor is found for a single tone in which maximum and RMS value coincide. The upper limit is calculated using the triangle inequality on $\max_t |A(t)|$:

$$\sqrt{2} \leq C \leq \sqrt{2} \frac{\sum_n v_n}{\sqrt{\sum_n v_n^2}} \xrightarrow{v_n \equiv 1} \sqrt{2N}. \tag{A.20}$$

For calculating the Crest factor for the cases $N = 3$ and $N = 4$, the curvature of the phases is defined as an auxiliary variable:

$$\xi = \varphi_1 - \frac{\varphi_0 + \varphi_2}{2}. \tag{A.21}$$

The Crest factor can be used to calculate the peak RF power if the average power P is given according to its definition in Equation A.11:

$$P_{\mathrm{peak}} = C^2 P_{\mathrm{avg}}. \tag{A.22}$$

A power amplifier reaches compression and clips the instantaneous peak voltage. A condition to avoid compression can be formulated with $P_{\text{peak}} \leq P_{1\text{dB}}$, which results in:

$$P_{\text{avg}} \leq \frac{P_{1\text{dB}}}{C^2}. \tag{A.23}$$

With a 1 dB compression point of 40 dBm $\approx$ 10 W, the resulting average power is given for the two Crest factors:

	C	C^2	$P_{\text{avg}} \leq P_{1\text{dB}}/C^2$
x axis, $N_x = 3$	2.4494	6.00	1.67 W
y axis, $N_y = 4$, $\varphi_1 = 98°$	2.1963	4.82	2.07 W

A.2.1 $N_x = 3$ and $N_y = 4$ Tones

The amplitudes of the RF signal for three tones are given by:

$$v_0 = v_2 = \varrho_x := \sqrt{r_x}, \quad v_1 = 1. \tag{A.24}$$

For three tones, the center frequency can be chosen as the second one $f_c = \bar{f} = f_1$. Together with the tone difference $\Delta f_x = f_n - f_c$, $A(t)$ can be expressed as:

$$A(t) = \varrho\, e^{\mathrm{i}(\varphi_0 - 2\pi\Delta f_x t)} + e^{\mathrm{i}\varphi_1} + \varrho\, e^{\mathrm{i}(\varphi_2 + 2\pi\Delta f_x t)}. \tag{A.25}$$

Using the phase curvature ξ from Equation A.21, the phases are reparametrized as:

$$\varphi_0 - \varphi_1 = -\xi + \delta, \qquad \varphi_2 - \varphi_1 = -\xi - \delta, \qquad \delta = \tfrac{1}{2}(\varphi_0 - \varphi_2), \tag{A.26}$$

the amplitude takes the following form:

$$A(t)\, e^{-\mathrm{i}\varphi_1} = \varrho\, e^{\mathrm{i}(-\xi+\delta-2\pi\Delta f_x t)} + 1 + \varrho\, e^{\mathrm{i}(-\xi-\delta+2\pi\Delta f_x t)} \tag{A.27}$$

$$= \varrho\, e^{-\mathrm{i}\xi}\left[e^{\mathrm{i}(\delta-2\pi\Delta f_x t)} + e^{-\mathrm{i}(\delta-2\pi\Delta f_x t)}\right] + 1 \tag{A.28}$$

$$= 2\varrho\, e^{-\mathrm{i}\xi}\cos\vartheta + 1, \qquad \vartheta = 2\pi\Delta f_x\,(t_\delta - t), \quad t_\delta = \frac{\delta}{2\pi\Delta f_x}. \tag{A.29}$$

The squared absolute value eliminates the remaining complex phase, resulting in:

$$|A(t)|^2 = 4\varrho^2\cos^2\vartheta + 4\varrho\cos\xi\,\cos\vartheta + 1. \tag{A.30}$$

The maximum lies at $\cos\vartheta = \text{sign}(\cos\xi)$:

$$\max_t |A(t)| = \sqrt{4\varrho^2 + 4\varrho\,|\cos\xi| + 1}, \tag{A.31}$$

$$\text{rms}_t|A(t)| = \sqrt{2\varrho^2 + 1}, \qquad C_x(\xi) = \sqrt{2}\,\frac{\sqrt{4\varrho^2 + 4\varrho\,|\cos\xi| + 1}}{\sqrt{2\varrho^2 + 1}}. \tag{A.32}$$

For a point symmetric beam profile, $\xi = 0$ holds, and the maximum becomes:

$$\max_t |A(t)| = 2\varrho + 1 = \textstyle\sum_n v_n. \tag{A.33}$$

Under the symmetry condition for three tones, the Crest factor is independent of the tone phases and always takes its maximum value depending only on the amplitudes v_n. For $\varrho = \sqrt{0.97}$, the Crest factor is given by:

$$C_x = \sqrt{2}\frac{\sqrt{0.97}+1+\sqrt{0.97}}{\sqrt{0.97+1+0.97}} \approx 2.4494. \tag{A.34}$$

For calculating the Crest factor for four tones, the same approach as before is used. The envelope frequency $f_{\text{env}} = \Delta f_y$ is calculated with the frequency width W_f as $f_{\text{env}} = {}^{W_f}/_{N_y-1} = \frac{0.370\,\text{MHz}}{3} = 0.1233\,\text{MHz}$. Together with the center frequency f_c, the following frequency parametrization is used with the frequency index n and phases φ_n:

$$f_n = f_0 + n f_{\text{env}}, \quad f_c = f_0 + \tfrac{3}{2} f_{\text{env}}, \qquad f_n - f_{\text{c}} = \left(n - \tfrac{3}{2}\right) f_{\text{env}} =: \hat{n}\, f_{\text{env}}. \tag{A.35}$$

With $\hat{n} \in \left\{-\frac{3}{2}, -\frac{1}{2}, +\frac{1}{2}, +\frac{3}{2}\right\}$, the exponent can be further simplified:

$$u := \pi f_{\text{env}} t \quad \Longrightarrow \quad 2\pi\hat{n} f_{\text{env}} t = 2\hat{n}u \in \{-3u,\, -u,\, +u,\, +3u\}, \tag{A.36}$$

and $t \in [0, 1/f_{\text{env}})$ corresponds to $u \in [0, \pi)$. A reparameterization of the phases is possible due to the symmetry condition:

$$\chi_n := \varphi_n - \tfrac{c_y}{2} \quad \Longrightarrow \quad \chi_{\text{a}} := \chi_3 = -\chi_0, \qquad \chi_{\text{i}} := \chi_2 = -\chi_1, \tag{A.37}$$

the parameters necessary for calculating the Crest factor are, with $b := \varphi_1$:

n	$\hat{n}$	v_n	φ_n	χ_n
0	$-\frac{3}{2}$	ϱ	0	$-c/2 = -\chi_{\text{a}}$
1	$-\frac{1}{2}$	1	b	$b - c/2 = -\chi_{\text{i}}$
2	$+\frac{1}{2}$	1	$c - b$	$c/2 - b = \chi_{\text{i}}$
3	$+\frac{3}{2}$	ϱ	c	$c/2 = \chi_{\text{a}}$

The optimization of the tone phases with regard to the Crest factor in chapter 4 results in $\varphi_n = [0°, 98°, 16°, 114°]$, so the reparameterization is $\chi_a = 57°$ and $\chi_i = -41°$, with $c = 114°$ and $b = 98°$. Each term of the envelope can be written as:

$$v_n e^{\mathrm{i}(2\hat{n}u+\varphi_n)} = e^{\mathrm{i}c/2}\, v_n e^{\mathrm{i}(2\hat{n}u+\chi_n)}, \tag{A.38}$$

$$A(t)\, e^{-\mathrm{i}c/2} = \varrho_y\, e^{\mathrm{i}(-3u-\chi_{\text{a}})} + e^{\mathrm{i}(-u-\chi_{\text{i}})} + e^{\mathrm{i}(+u+\chi_{\text{i}})} + \varrho_y\, e^{\mathrm{i}(+3u+\chi_{\text{a}})}. \tag{A.39}$$

Together with a trigonometric identity ($e^{\mathrm{i}z} + e^{-\mathrm{i}z} = 2\cos z$):

$$\varrho_y\left[e^{-\mathrm{i}(3u+\chi_{\text{a}})} + e^{+\mathrm{i}(3u+\chi_{\text{a}})}\right] = 2\varrho_y \cos(3u+\chi_{\text{a}}) \tag{A.40}$$

$$e^{-\mathrm{i}(u+\chi_{\text{i}})} + e^{+\mathrm{i}(u+\chi_{\text{i}})} = 2\cos(u+\chi_{\text{i}}), \tag{A.41}$$

the envelope becomes real:

$$A(t)\, e^{-\mathrm{i}c/2} = 2\varrho_y \cos(3u+\chi_{\text{a}}) + 2\cos(u+\chi_{\text{i}}) \quad \in \mathbb{R}. \tag{A.42}$$

Substituting $u' := u + \chi_{\text{i}}$:

$$u + \chi_{\text{i}} = u', \tag{A.43}$$

$$3u + \chi_{\text{a}} = 3u' - 3\chi_{\text{i}} + \chi_{\text{a}} = 3u' + \gamma, \tag{A.44}$$

yields:

$$A(t)\,e^{-\mathrm{i}c/2} = 2\varrho_y \cos(3u' + \gamma) + 2\cos(u'). \tag{A.45}$$

Since both cosine terms change sign under $u' \to u' + \pi$:

$$u' \to u' + \pi : \quad \cos u' \to -\cos u', \quad \cos(3u' + \gamma) \to -\cos(3u' + \gamma), \tag{A.46}$$

$|A(t)|$ has a period of π in u'. For the working point holds:

$$\gamma = \chi_\mathrm{a} - 3\chi_\mathrm{i} = 3b - c = 57^\circ - 3 \cdot (-41^\circ) = 180^\circ, \tag{A.47}$$

which simplifies the envelope to:

$$A(t)\,e^{-\mathrm{i}c/2} = 2\cos u' - 2\varrho\cos(3u'). \tag{A.48}$$

To find its maximum for the Crest factor, several trigonometric identities [30] are used:

$$\begin{aligned}
\cos(3u') = \cos(2u' + u') &= \cos(2u')\cos(u') - \sin(2u')\sin(u') && \text{(A.49)}\\
&= (2\cos^2(u') - 1)\cos(u') - 2\sin^2(u')\cos(u') && \text{(A.50)}\\
&= 4\cos^3(u') - 3\cos(u'). && \text{(A.51)}
\end{aligned}$$

With the substitution $p := \cos u' \in [-1, 1]$, the function to be maximized becomes:

$$G(p) := A(t)\,e^{-\mathrm{i}c/2} = 2p - 2\varrho_y(4p^3 - 3p) = (2 + 6\varrho_y)\,p - 8\varrho_y\,p^3, \tag{A.52}$$

since $|A(t)| = |G(p)|$ holds true. Because $G(p)$ is odd and only $|G(p)|$ is of interest, only $p \in [0, 1]$ is considered. The maximum follows from:

$$G'(p) = (2 + 6\varrho_y) - 24\varrho_y\,p^2, \tag{A.53}$$

$$G'(p_*) = 0 : \quad p_*^2 = \frac{2 + 6\varrho_y}{24\varrho_y} = \frac{1 + 3\varrho_y}{12\varrho_y}. \tag{A.54}$$

This root is indeed a maximum and lies within the interval:

$$G''(p) = -48\varrho_y p < 0, \qquad p_* \le 1 \;\Leftrightarrow\; \varrho_y \ge \tfrac{1}{9}. \tag{A.55}$$

The maximum value is then calculated as:

$$G(p_*) = p_*\big[(2 + 6\varrho_y) - 8\varrho_y p_*^2\big]. \tag{A.56}$$

Together with $8\varrho_y p_*^2 = 8\varrho_y \cdot \frac{1+3\varrho_y}{12\varrho_y} = \frac{2}{3}(1 + 3\varrho_y)$ and $(2 + 6\varrho_y) - \frac{2}{3}(1 + 3\varrho_y) = \frac{4}{3}(1 + 3\varrho_y)$, the maximum is:

$$G(p_*) = \tfrac{4}{3}(1 + 3\varrho_y)\,p_* = \tfrac{2}{3}\,\frac{(1 + 3\varrho_y)^{3/2}}{\sqrt{3\varrho_y}}. \tag{A.57}$$

The Crest factor for four tones with the optimized phases is therefore:

$$\boxed{\max_t |A(t)| = \frac{2}{3}\,\frac{(1 + 3\varrho_y)^{3/2}}{\sqrt{3\varrho_y}}, \qquad C_y = \sqrt{2}\,\frac{\frac{2}{3}(1 + 3\varrho_y)^{3/2}/\sqrt{3\varrho_y}}{\sqrt{2\varrho_y^2 + 2}} \approx 2.1963} \tag{A.58}$$

A.3 Power of a Normalized Profile

For calculating the pulse area θ of the multi-tone profile, the power of a normalized profile $\tilde{P}_{\text{prof}}$ has to be calculated. The profile can then be scaled to the desired power P, using the scaling factor s, such that $P = s\tilde{P}_{\text{prof}}$.
A single spot has the form $g_s = A_s\, u(\mathbf{r} - \mathbf{c}_s)$ with the field amplitude $A_s = \sqrt{a_n b_m}$. The intensity weight of a spot is:

$$a_s \equiv A_s^2 = a_n\, b_m. \tag{A.59}$$

The optical power of a single spot can be calculated as

$$\int g_s^2 \,\mathrm{d}^2 r = a_s P_1, \qquad P_1 = \int |u|^2 \mathrm{d}^2 r. \tag{A.60}$$

In the following, the notation $r = |\mathbf{r}|$, $q = |\mathbf{q}|$ and $l = |\mathbf{l}|$ is used. The mode functions (Gaussian and Airy) are radially symmetric, so u is written as a function of the scalar radius, $u(r) \equiv u(|\mathbf{r}|)$, and the same holds for $\tilde{u}(q)$ and for the overlap $O(l)$. Calculating the power of a Gaussian spot with the substitution $x = {}^{2r^2}/_{\omega_0^2}$ results in:

$$P_1^{\mathsf{G}} = \int_0^\infty e^{-2r^2/w_0^2}\, 2\pi r \,\mathrm{d}r = \frac{\pi w_0^2}{2} \int_0^\infty e^{-x}\,\mathrm{d}x = \frac{\pi w_0^2}{2}. \tag{A.61}$$

For one Airy profile, with $x = kr$ and the integral $\int_0^\infty J_\nu^2(x)\,\mathrm{d}x/x = 1/(2\nu)$ [31], the power is:

$$P_1^{\mathsf{A}} = \int_0^\infty \left(\frac{2J_1(kr)}{kr}\right)^2 2\pi r \,\mathrm{d}r = \frac{8\pi}{k^2} \int_0^\infty \frac{J_1^2(x)}{x}\,\mathrm{d}x = \frac{8\pi}{k^2}\cdot\frac{1}{2} = \frac{4\pi}{k^2}, \tag{A.62}$$
$$k = \frac{3.8317}{1.483\, w_0}.$$

The Airy spot carries $4\pi/k^2/(\pi w_0^2/2) = 1.20$ times the power of a Gaussian spot with the same waist w_0, because of its rings. The time average of the coherent sum, as discussed in chapter 2, is given by:

$$\langle \tilde{I}(\mathbf{r},t)\rangle_t = \sum_s g_s^2(\mathbf{r}) + 2 \sum_{\substack{s<s' \\ k_s = k_{s'}}} g_s(\mathbf{r})\; g_{s'}(\mathbf{r}) \cos(\phi_s - \phi_{s'}), \tag{A.63}$$

because a pair of spots with $k_s \neq k_{s'}$ oscillates at $(k_s - k_{s'})f_0$ and averages to zero over T_0, while a pair of spots with $k_s = k_{s'}$ does not oscillate at all. Integrating term by term over the plane gives:

$$\tilde{P}_{\text{prof}} = \sum_s a_s\, P_1 + 2 \sum_{\substack{s<s' \\ k_s = k_{s'}}} A_s A_{s'} \cos(\phi_s - \phi_{s'}) \underbrace{\int u(\mathbf{r} - \mathbf{c}_s) u(\mathbf{r} - \mathbf{c}_{s'})\,\mathrm{d}^2 r}_{O(l_{s,s'})}, \tag{A.64}$$

with $l \equiv l_{s,s'} = |\mathbf{c}_s - \mathbf{c}_{s'}|$ being the distance between the two spots. Calculating the power of the normalized profile reduces to evaluating the power of the single spots and the overlap integral $O(l)$. For Gaussian spots, the overlap can be calculated with the midpoint $\mathbf{m} = \frac{1}{2}(\mathbf{c}_s + \mathbf{c}_{s'})$ and the Gaussian integral [30]:

$$e^{-\frac{(\mathbf{r}-\mathbf{c}_s)^2}{w_0^2}}\, e^{-\frac{(\mathbf{r}-\mathbf{c}_{s'})^2}{w_0^2}} = e^{-\frac{l^2}{2w_0^2}}\, e^{-\frac{2(\mathbf{r}-\mathbf{m})^2}{w_0^2}},$$

$$O^{\mathrm{G}}(l) = \frac{\pi w_0^2}{2}\, e^{-l^2/(2w_0^2)} = P_1^{\mathrm{G}}\, e^{-l^2/(2w_0^2)} = P_1^{\mathrm{G}} u(l). \tag{A.65}$$

For the Airy mode, the integral is calculated in Fourier space. The Fourier transform of a uniformly illuminated circular disk is proportional to the Airy function [31]:

$$\frac{1}{(2\pi)^2}\int_{|\mathbf{q}|<k} e^{i\mathbf{q}\cdot\mathbf{r}}\,\mathrm{d}^2q = \frac{k^2}{4\pi}\,\frac{2J_1(kr)}{kr} = \frac{k^2}{4\pi}\,u(r), \qquad r = |\mathbf{r}|, \tag{A.66}$$

where the left side is an integral over the vector $\mathbf{q}$ and the right side depends only on the length r. Conversely, the Fourier transform of u is:

$$\tilde{u}(q) = \int u(|\mathbf{r}|)\, e^{-i\mathbf{q}\cdot\mathbf{r}}\,\mathrm{d}^2r = \frac{4\pi}{k^2}\,\Theta(k-q)\,, \qquad q = |\mathbf{q}|. \tag{A.67}$$

Θ is the Heaviside step function that has the value one inside the cutoff of the largest possible wave vector $k = 2\pi\,\mathrm{NA}/\lambda$ and zero outside. With the Fourier transform, the shifted Airy spot can be expressed by:

$$u(|\mathbf{r}-\mathbf{l}|) = \frac{1}{(2\pi)^2}\int \tilde{u}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{l}}\mathrm{d}^2q,$$

and the overlap becomes:

$$O^{\mathrm{A}}(l) = \int u(|\mathbf{r}|)\, u(|\mathbf{r}-\mathbf{l}|)\,\mathrm{d}^2r = \frac{1}{(2\pi)^2}\int |\tilde{u}(q)|^2\, e^{i\mathbf{q}\cdot\mathbf{l}}\,\mathrm{d}^2q. \tag{A.68}$$

The result depends only on the distance $l = |\mathbf{l}|$. With $\Theta^2 = \Theta$ applied, the overlap is given by:

$$O^{\mathrm{A}}(l) = \frac{1}{(2\pi)^2}\left(\frac{4\pi}{k^2}\right)^2 \frac{2\pi k J_1(kl)}{l} = \frac{8\pi\, J_1(kl)}{k^3 l} = \frac{4\pi}{k^2}\cdot\frac{2J_1(kl)}{kl} = P_1^{\mathrm{A}}\, u(l). \tag{A.69}$$

So in both cases, the overlap of two spots is calculated as the power of the single spot profile multiplied by the mode function evaluated at their respective distance.
The power of the normalized profile composed of Airy profiles is given by:

$$\tilde{P}_{\mathrm{prof}} = \frac{4\pi}{k^2}\sum_s a_s + \frac{8\pi}{k^2}\sum_{\substack{s<s'\\ k_s=k_{s'}}} A_s A_{s'}\,\cos(\phi_s-\phi_{s'})u(l_{s,s'}). \tag{A.70}$$

For an appropriate choice of phases, for example $\phi_s - \phi_{s'} = \pm\pi/2$ for the corner pair, the interference term vanishes, and only the sum of the single-spot powers remains.

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List of Tools and Aids

Generative AI was used in this thesis as listed below, following the *AI Guide for Students at TU Darmstadt* (2026-09-24). AI tools are not cited as sources. All physical models, parameters and conclusions were specified and checked by the author, who is responsible for them.

Type of application	Own work	Parts of the work affected	AI tools	Remark
Writing and revising simulation code (multi-tone profiles, optimization of amplitudes and tone phases, beating analysis, GUIs)	Physical model, choice of metrics and parameters, specification of every function, tests against analytical limits and measurements	chapter 3, chapter 4, repository `Multitone_Simulation` [25]	Claude (Anthropic), model `claude-opus-5-5`, via Claude Cowork	Aug–Nov 2026
Numerical evaluation and uncertainty estimates	Working points and uncertainty budget defined by the author; all numbers checked against the simulation data	chapter 3, chapter 4, Table 4.1, Table 4.2	as above	
Figures and plots	Content, layout and selection of all figures	Figure 5, Figure 7	as above	

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Darmstadt, 03.11.2026



Daniel Hagn